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Lu et al.

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(54) **COMPOUND FOR TREATING
NEURODEGENERATIVE DISORDERS**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 1032 days.

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A61K 31/517 (2006.01)
A61P 25/00 (2006.01)
C07D 311/20 (2006.01)
C07D 311/22 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 31/353** (2013.01); **A61K 31/517**
(2013.01); **A61P 25/00** (2018.01)

(58) **Field of Classification Search**

CPC **A61K 31/353**; **A61K 31/352**; **A61P 25/00**;
C07D 311/20; **C07D 311/22**

See application file for complete search history.

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(57) **ABSTRACT**

The present invention relates to use of a compound with a
substituted bicyclic structure, a pharmaceutically acceptable
salt, stereoisomer, solvate, polymorph, tautomer, isotopic
compound, metabolite or prodrug thereof in the preparation
of a medicament for preventing or treating polyglutamine
(polyQ) related disorders.

6 Claims, 9 Drawing Sheets

Specification includes a Sequence Listing.

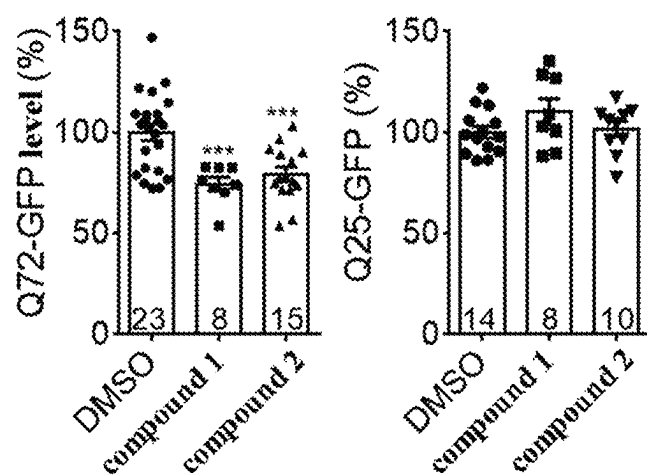


Figure 1

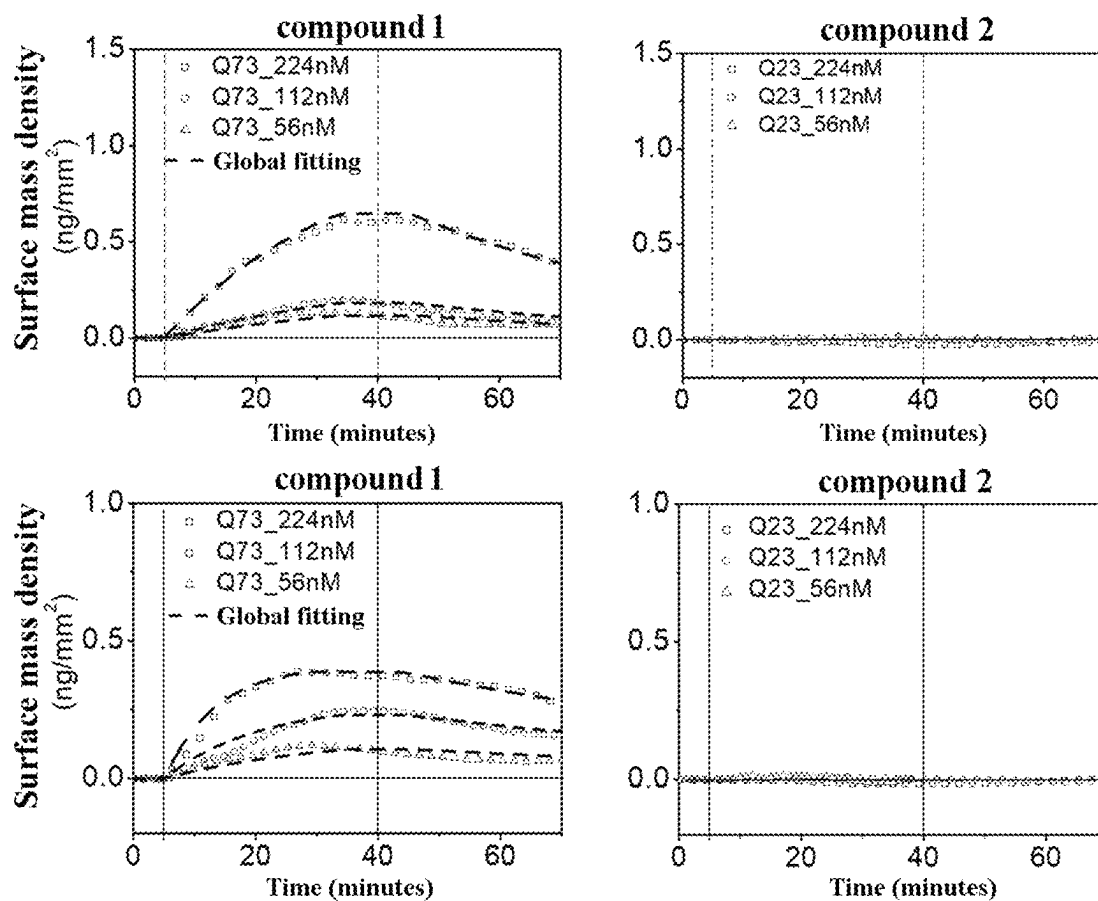


Figure 2

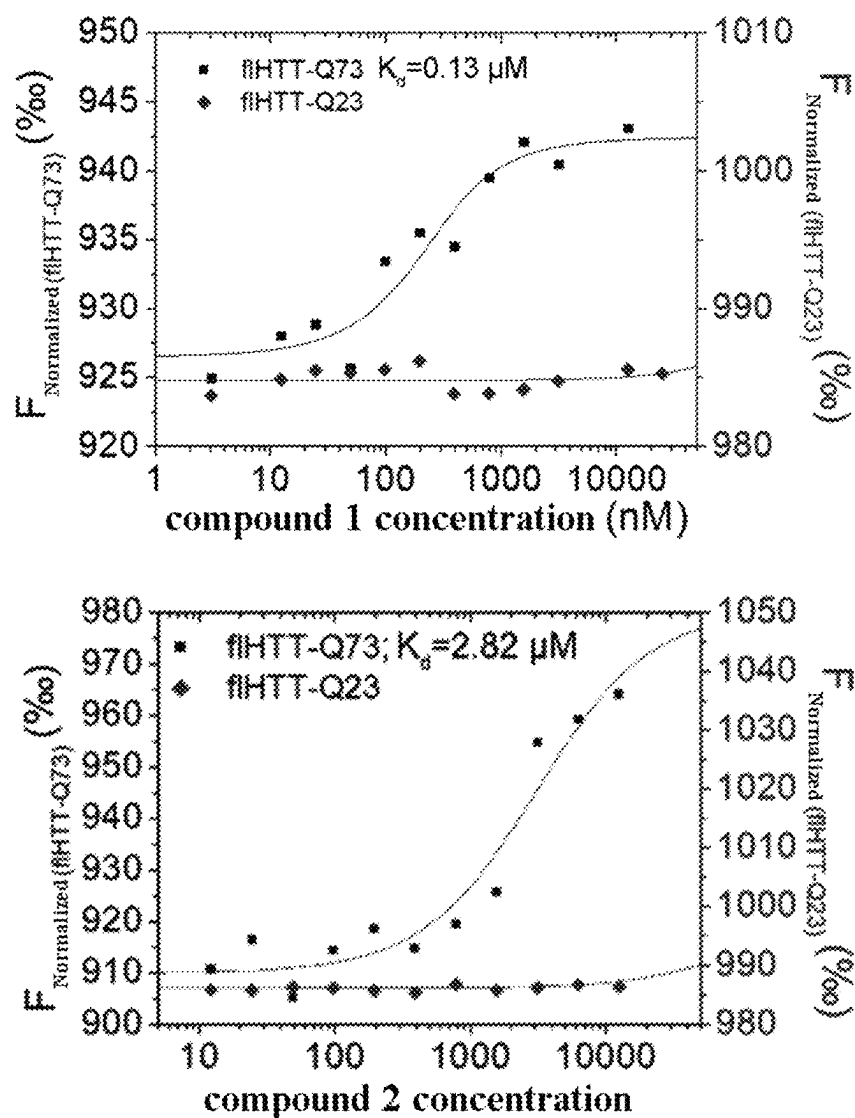


Figure 3

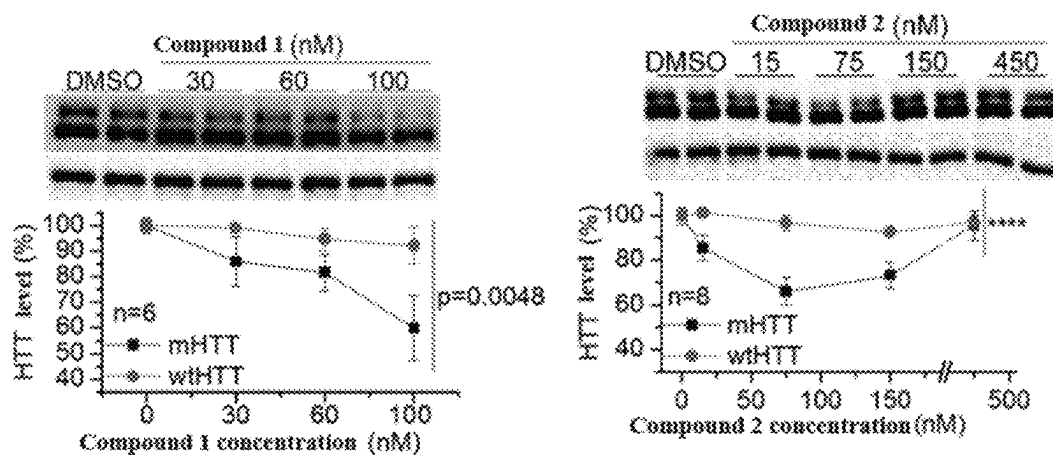


Figure 4

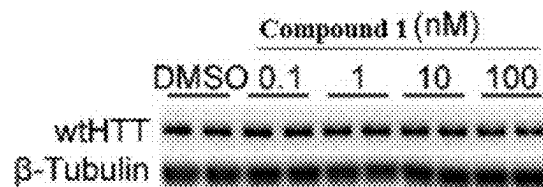


Figure 5

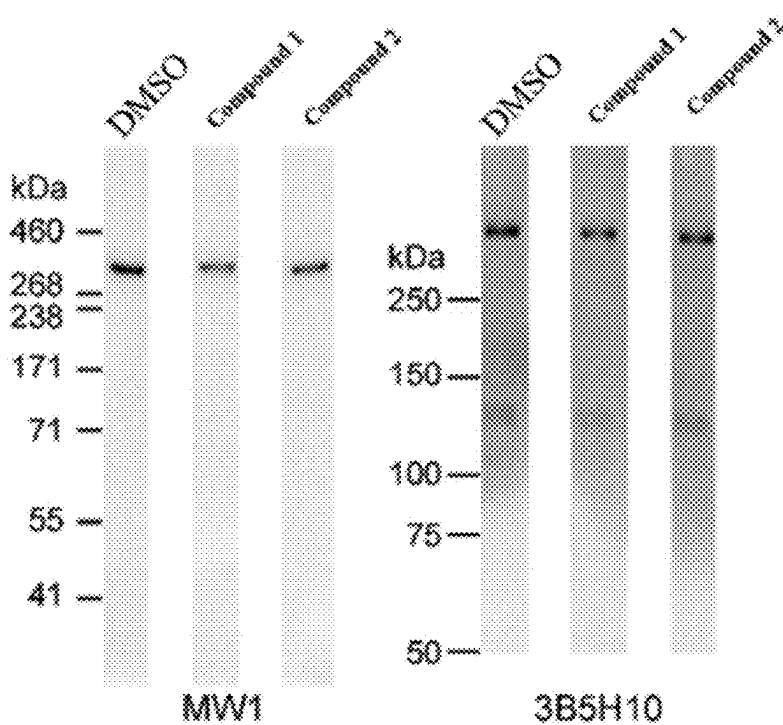


Figure 6

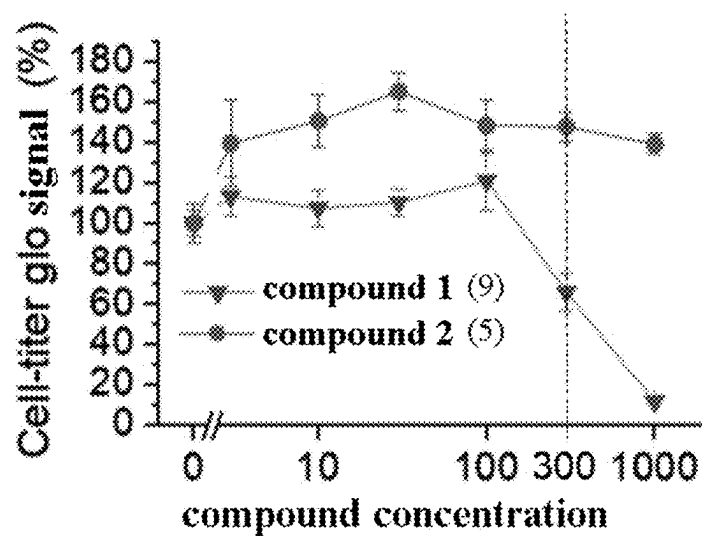


Figure 7

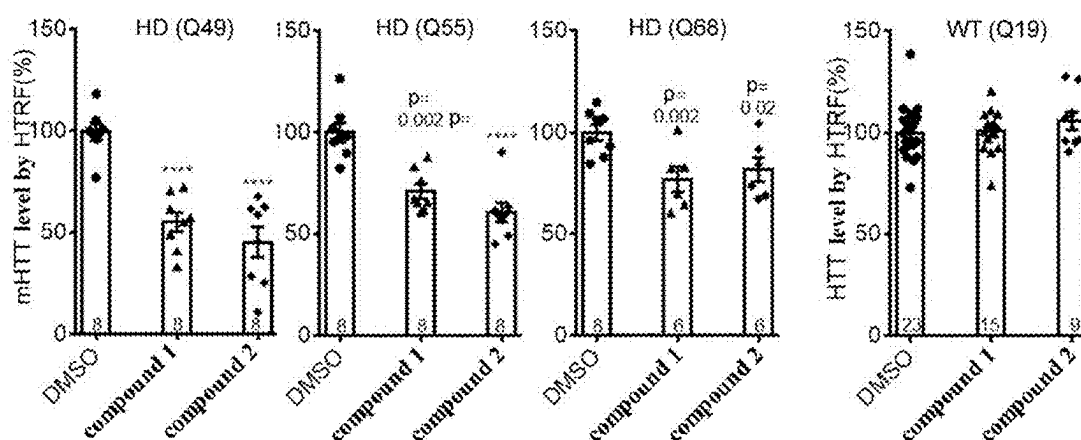


Figure 8

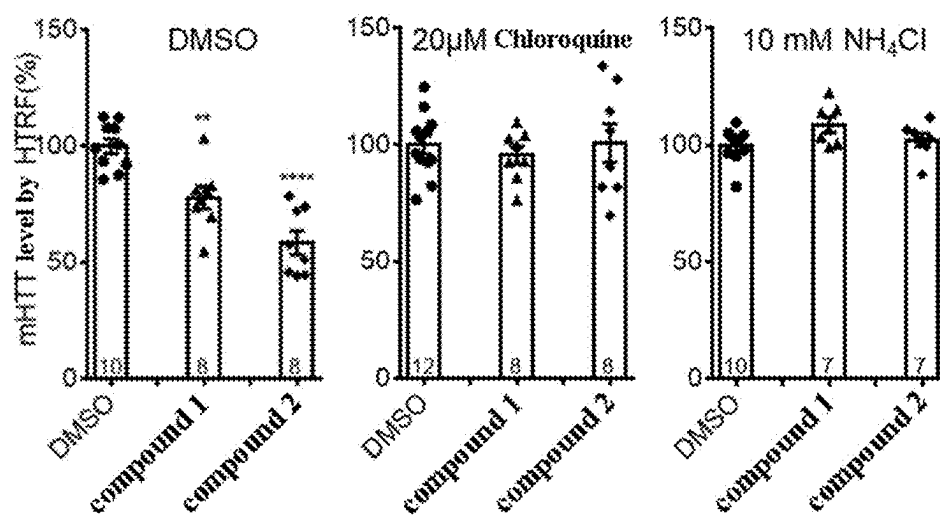


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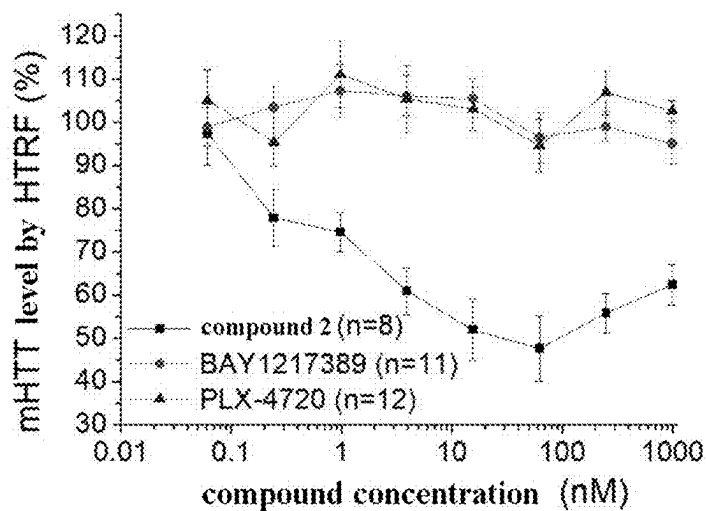


Figure 10

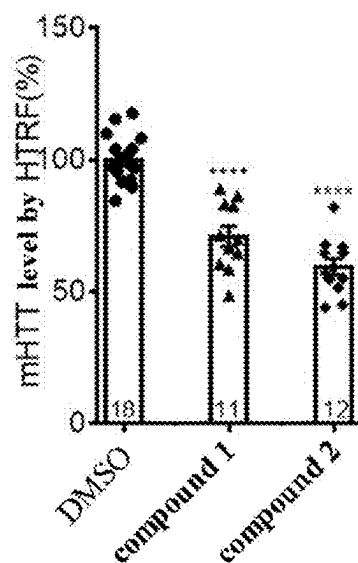


Figure 11

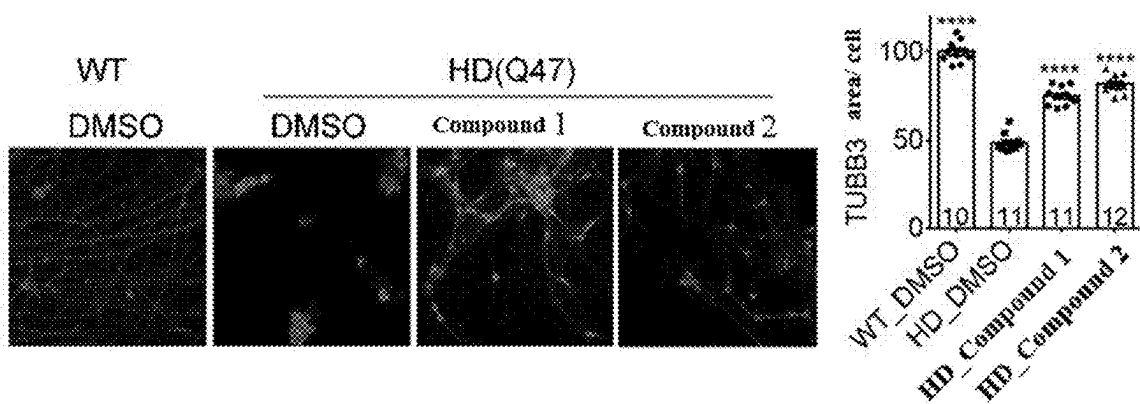


Figure 12

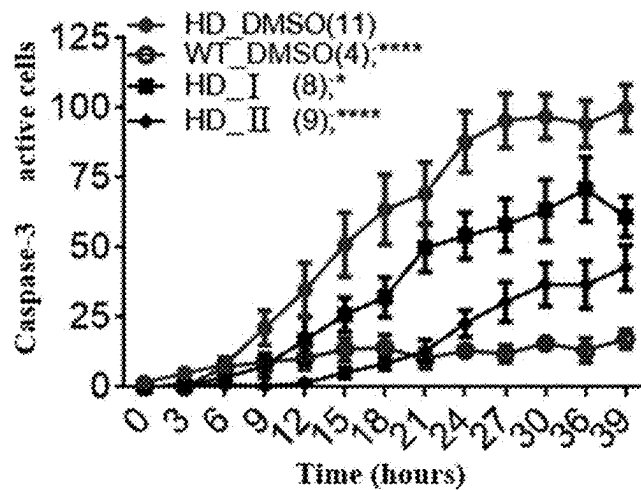


Figure 13

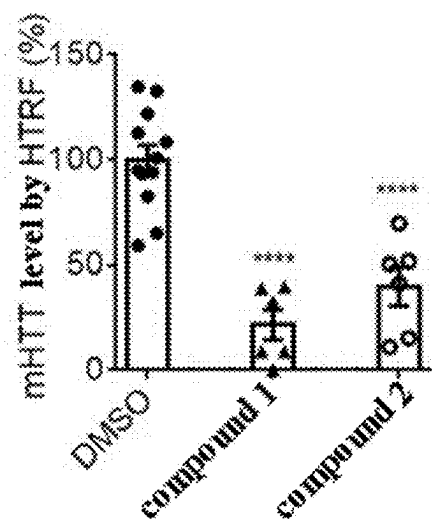
HTT-Q128 *Drosophila* heads

Figure 14

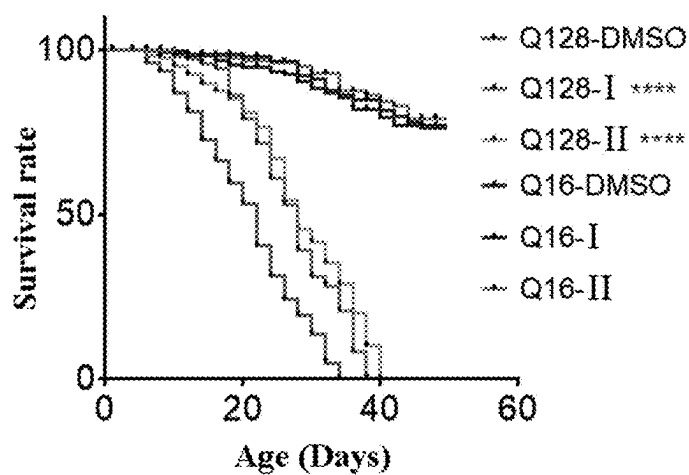


Figure 15

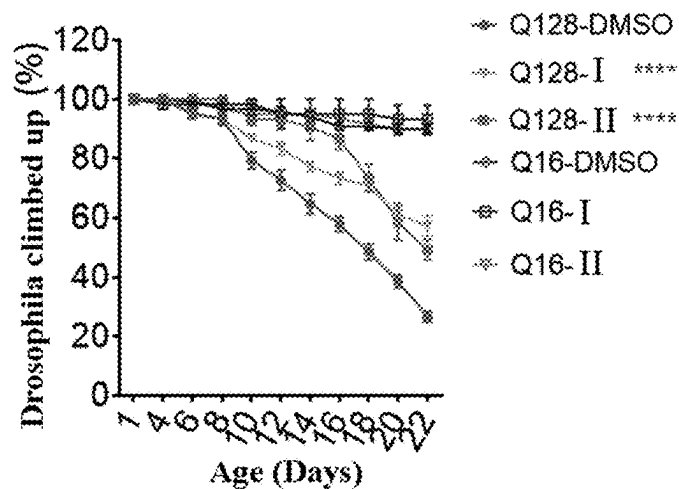


Figure 16

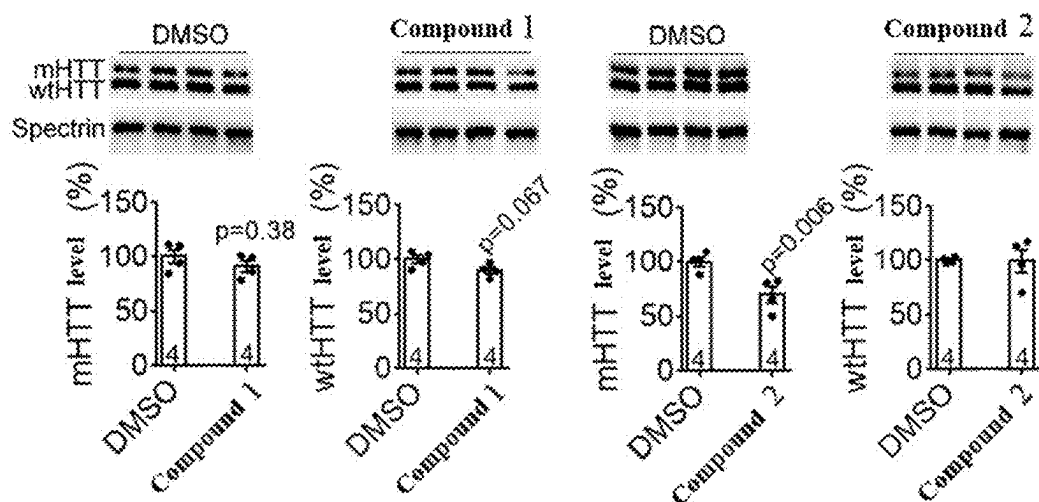


Figure 17

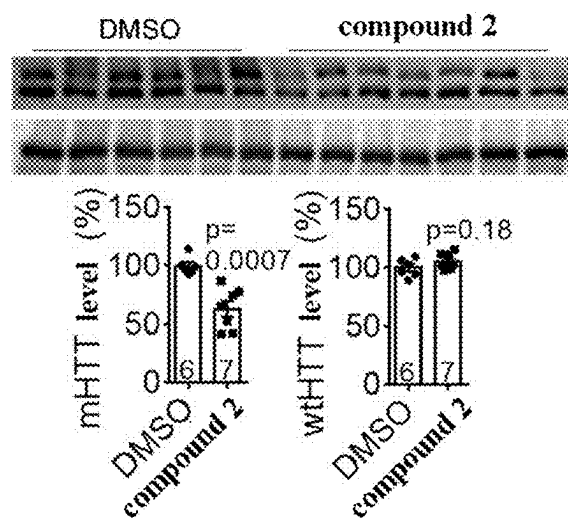


Figure 18

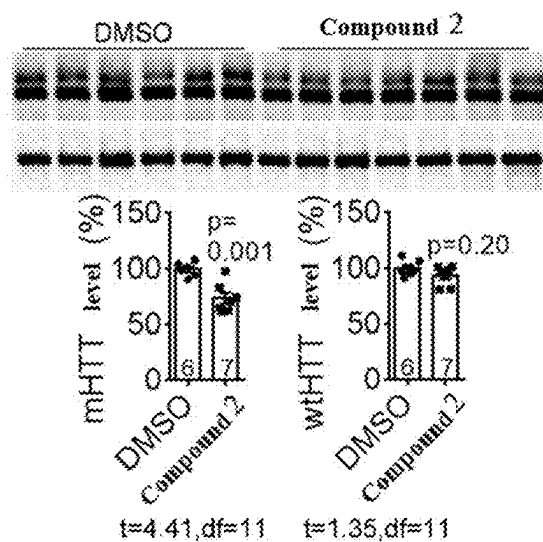


Figure 19

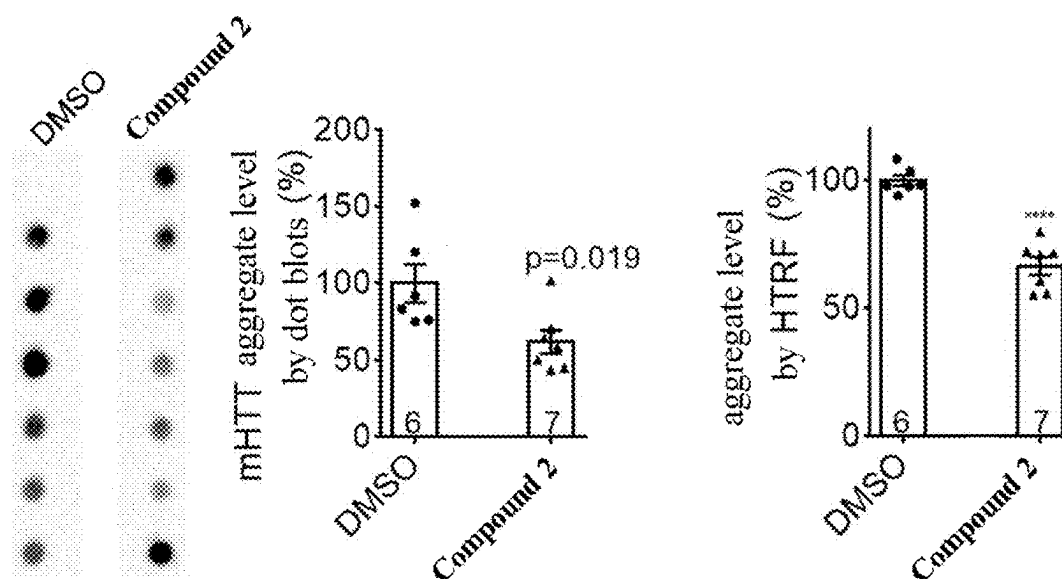


Figure 20

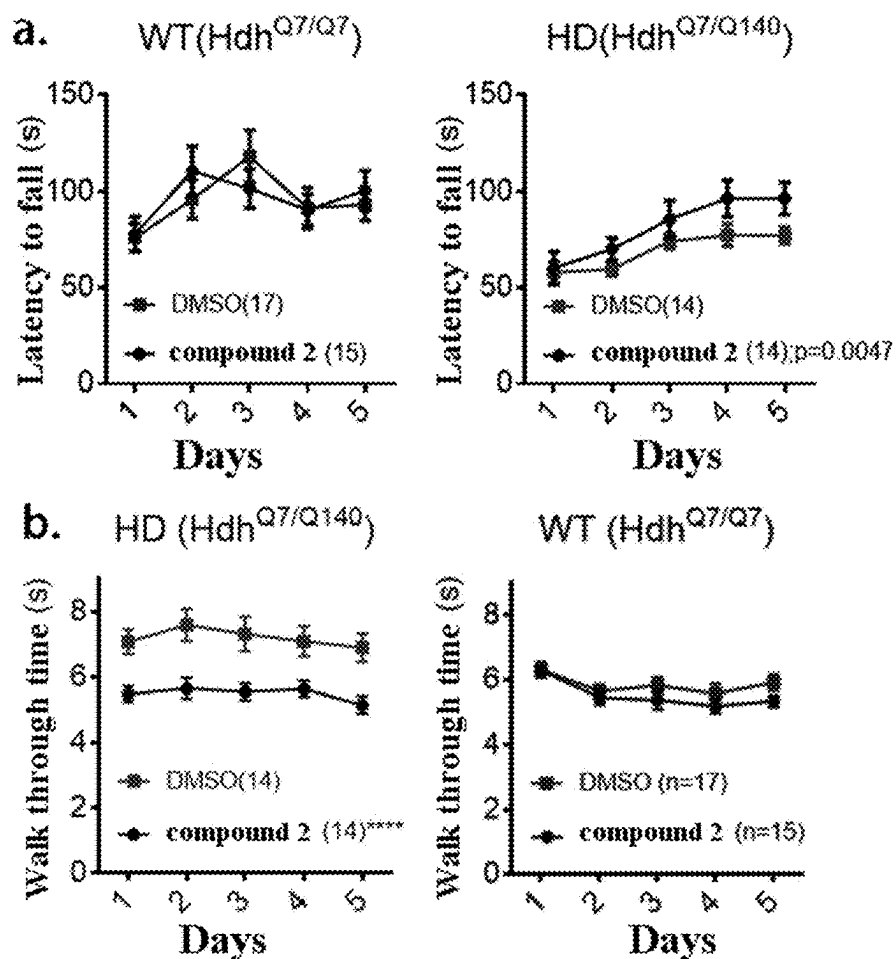


Figure 21

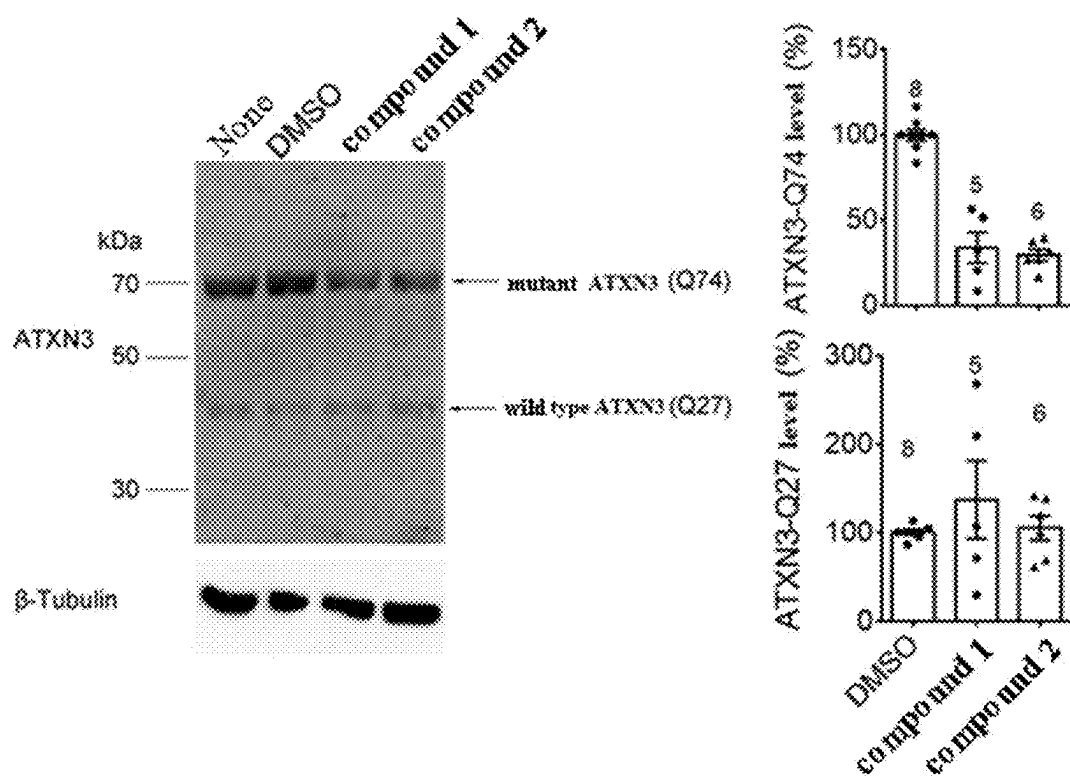


Figure 22

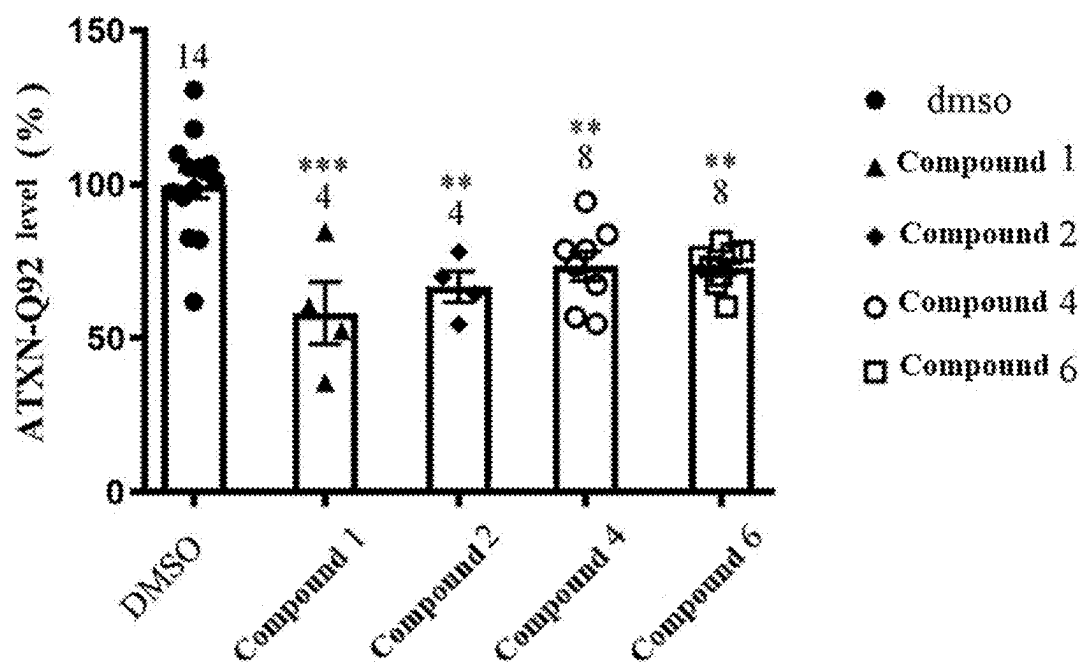


Figure 23

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COMPOUND FOR TREATING NEURODEGENERATIVE DISORDERS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. national stage filing, under 35 U.S.C. § 371(c), of International Application No. PCT/CN2020/078779, filed on Mar. 11, 2020, which claims priority to Chinese Patent Application No. 201910180674.7, filed on Mar. 11, 2019 and Chinese Patent Application No. 201911000198.2, filed Oct. 21, 2019, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to the biopharmaceutical field, and in particular to use of a compound with a substituted bicyclic structure, a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyglutamine (polyQ) related disorder.

BACKGROUND OF THE INVENTION

Neurodegenerative disorder refers to diseases caused by nervous system dysfunction resulted from abnormal death of central neurons. So far, there is no ultimate therapy that can slow down its development. Many neurodegenerative disorders are caused by proteins with unknown activity. Currently, methods that may be used to control protein levels, such as biological tools like RNAi, CRISPR. etc. are difficult to deliver, especially to the nervous system.

A feasible treatment strategy is to use low molecular weight compounds (compounds for short) to control the level of proteins that affect diseases. An emerging approach is to enhance the ubiquitination of disease-causing proteins and target them to the proteasome degradation pathway using proteolysis-targeting chimera (PROTAC) technology, but this method relies on certain E3 ligases, which may not be present in diseased cells. Moreover, the protein degradation ability of the proteasome is limited, and the degradation efficiency of certain large disease proteins or aggregates that cause neurodegenerative disorder is low. As an important protein degradation pathway, autophagy is ubiquitous in eukaryotic cells, with strong protein degradation ability but low selectivity. Some studies increase protein degradation by enhancing autophagy, but this method lacks specificity.

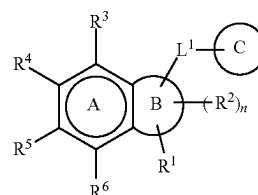
PolyQ-related neurodegenerative disorder is a type of neurodegenerative disorder caused by mutant proteins, which can be effectively treated by reducing the level of mutant proteins. Taking Huntington's disease (HD), the most common neurodegenerative disorder, as an example, it is a monogenetic disorder wherein the mutation in the CAG repeat region of the exon 1 of the HTT gene located in the patient's chromosome 4 results in the expansion of the polyglutamine (polyQ) of the synthesized mutant protein (mHTT). mHTT is susceptible to shearing and aggregating, and causes toxicity which eventually leads to dysfunction and death of specific neuron. The current methods for controlling mHTT levels through low molecular weight compounds lack specificity, and may cause side effects. Furthermore, these methods are nonallele-selective and unable to distinguish between mHTT and wild-type HTT protein (wtHTT), which will lead to a decrease in the level of wtHTT which has important biological functions.

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Similarly, take spinocerebellar ataxia type 3 (SCA3; also known as Machado-Joseph disease, MJD) as an example, which is the most common autosomal dominant spinocerebellar ataxia and is second only to HD as a common polyQ-related disorder in the world. SCA3 is caused by the abnormally expanded polyQ at the C-terminus of the coded protein ATXN3 resulting from the increase in the number of CAG repeats of the Ataxin-3 gene (ATXN3; also known as the MJD1 gene). Some studies have used siRNA, antisense oligonucleotides and other means to act on ATXN3 to reduce the expression of mutant ATXN3 protein and it is confirmed that the reduction in the level of mutant ATXN3 protein can produce therapeutic effects (Wang, Neuroscience, 371, 2018, 138-154). Some studies have focused on controlling the level of mutant ATXN3 protein through low molecular weight compounds. Studies on enhancing autophagy by compounds such as Menzies et al. Brain 2010, 133:93-104; studies on reducing the level of mutant ATXN3 protein by regulating other targets such as Costa M D, Brain, 2016, 139(11)2891-2908. But these studies have not satisfactorily solved the problem of specificity.

DISCLOSURE OF THE INVENTION

In one aspect, the present invention provides use of a compound of formula (I), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder



(I)

wherein:

ring A is a benzene ring;

ring B is a saturated or unsaturated 5- or 6-membered heterocycle, the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

ring C is selected from C₆₋₁₀ aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or C₁₋₆ a hydrocarbon chain;

or ring C is absent, and L¹ is absent;

R¹ is =Y, wherein Y is O or S, or OR⁷;

at each occurrence, R² is each independently selected from hydrogen, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl,

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alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-8} alkyl, C_{2-8} alkenyl, C_{2-8} alkynyl, $=\text{O}$, $=\text{S}$, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{OR}^{a2}$, $-\text{S}(=\text{O})_2\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{NR}^{a2}\text{R}^{b2}$, $-\text{S}(=\text{O})\text{R}^{a2}$ and $-\text{C}(=\text{NR}^{a2})\text{NR}^{b2}\text{R}^{c2}$;

R^3 , R^4 , R^5 , R^6 are each independently selected from H and R^{x2} ;

at each occurrence, R^{x1} and R^{x2} are each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(\text{3- to 7-membered heterocyclyl})$, $-\text{O}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl), $-\text{SH}$, $-\text{S}(\text{C}_{1-6}$ alkyl), $-\text{S}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{S}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{S}(\text{3- to 7-membered heterocyclyl})$, $-\text{S}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{NH}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{N}(\text{C}_{3-6}$ cyclohydrocarbyl) $_2$, $-\text{NH}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{N}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl) $_2$, $-\text{NH}(\text{3- to 7-membered heterocyclyl})$, $-\text{N}(\text{3- to 7-membered heterocyclyl})_2$, $-\text{NH}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl), $-\text{N}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl) $_2$, $=\text{O}$, $-\text{COOH}$ and C_{1-6} alkyl;

at each occurrence, R^7 , R^8 are each independently selected from H, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl and C_{6-10} aryl- C_{1-4} alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{OC}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NHC}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl) and $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl);

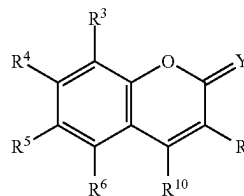
at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, C_{6-10} aryl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl- C_{1-4} alkyl, $-\text{OR}^{x1}$, $-\text{SR}^{x1}$, $-\text{NR}^{x1}\text{R}^{x2}$, $-\text{C}(=\text{O})\text{OR}^{x1}$, $-\text{C}(=\text{O})\text{NR}^{x1}\text{R}^{x2}$, $-\text{C}(=\text{O})\text{R}^{x1}$, $-\text{S}(=\text{O})_2\text{OR}^{x1}$, $-\text{S}(=\text{O})_2\text{R}^{x1}$, $-\text{S}(=\text{O})_2\text{NR}^{x1}\text{R}^{x2}$ and $-\text{S}(=\text{O})\text{R}^{x1}$, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, $=\text{O}$, $=\text{S}$, $-\text{OR}^{x3}$, $-\text{SR}^{x3}$, $-\text{NR}^{x3}\text{R}^{x4}$, $-\text{C}(=\text{O})\text{R}^{x3}$, $-\text{C}(=\text{O})\text{OR}^{x3}$ and $-\text{C}(=\text{O})\text{NR}^{x3}\text{R}^{x4}$;

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at each occurrence, R^{x1} , R^{x2} , R^{x3} , R^{x4} are each independently selected from H, C_{1-8} alkyl, C_{3-10} cyclohydrocarbyl, C_{3-10} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl- C_{1-4} alkyl, C_{6-10} aryl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl- C_{1-4} alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-8} alkyl, C_{2-8} alkenyl, C_{2-8} alkynyl, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$, $=\text{O}$ and $-\text{COOH}$;

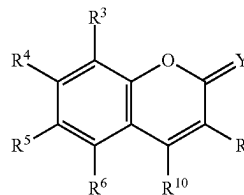
n is 1 or 2;

provided that the compound of formula (I) does not include the following structure:

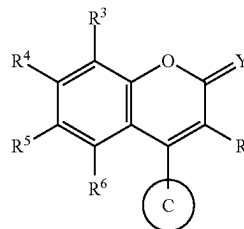


wherein R^3 is selected from $-\text{O}(\text{C}_{1-6}$ alkyl) and $-\text{O}(\text{C}_{1-4}$ alkylene- C_{6-10} aryl); R^{10} is H;

and the compound of formula (I) does not include the following structure:



wherein R^4 is $-\text{O}(\text{C}_{1-6}$ alkyl); R^5 is halogen; R^{10} is CF_3 ; and the compound of formula (I) does not include the following structure:

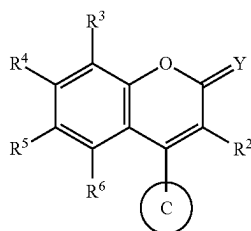


wherein R^4 is selected from $-\text{O}(\text{C}_{1-6}$ alkyl) and $-\text{O}(\text{C}_{1-4}$ alkylene- C_{6-10} aryl);

ring C is C_{6-10} aryl, which is optionally substituted with one or more substituents each independently selected from R^{x1} ;

and the compound of formula (I) does not include the following structure:

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wherein R^4 is selected from $-\text{OH}$ and $-\text{O}(\text{C}_{1-6} \text{ alkyl})$; R^5 is halogen; Ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R^{X1} .

In one embodiment, the neurodegenerative disorder is spinocerebellar ataxia (such as type 1, 2, 3, 6, 7, 12, 17), dentatorubral-pallidoluysian atrophy, Huntington's disease, Huntington's disease-like syndrome-2 or spinal-bulbar muscular atrophy, especially Huntington's disease.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. Effects of compounds on the level of proteins containing polyglutamine in HEK293T cells.

FIG. 2. Affinity binding curves determined by OI-RD of compounds with full-length HTT of different concentrations. The vertical dashed lines indicate the beginning of the association phase and the dissociation phase during the affinity binding. The dashed curve is the result of the global fitting of the Langmuir reaction model.

FIG. 3. Affinity binding curve determined by MST of compound with full-length HTT.

FIG. 4. Effects of compounds on HTT levels in cortical neurons of $\text{Hdh}^{Q140/Q7}$ mice.

FIG. 5. Effects of compound 1 on HTT levels in cortical neurons of $\text{Hdh}^{Q7/Q7}$ mice.

FIG. 6. Detection of N-terminal fragments of mHTT with antibodies MW1 and 3B5H10.

FIG. 7. Cell viability test results of $\text{Hdh}^{Q140/Q7}$ mouse cortical neurons after treatment with the compounds.

FIG. 8. Effects of compounds on mHTT levels in primary fibroblasts of HD patients at the concentration of 100 nM.

FIG. 9. Effects of compounds on mHTT levels in immortalized fibroblasts of HD patients.

FIG. 10. Compound 2 reduces mHTT levels in immortalized fibroblasts of HD patients.

FIG. 11. Effects of compounds on mHTT levels in HD patient iPSC-derived neurons.

FIG. 12. Effects of compounds on the apoptosis of HD patient iPSC-derived neurons. The scale bar is 50 μm .

FIG. 13. Effects of compounds on the apoptosis of HD patient iPSC-derived neurons.

FIG. 14. Effects of compounds on mHTT levels in Huntington's disease Drosophila.

FIG. 15. Effects of compounds on the survival rate of Huntington's disease Drosophila.

FIG. 16. Effects of compounds on the climbing performance of Huntington's disease Drosophila.

FIG. 17. Effects of intracerebroventricular injection of compounds on the levels of mHTT and wtHTT in the cortices of Huntington's disease mice.

FIG. 18. Effects of intraperitoneal injection of compound 2 on the levels of mHTT and wtHTT in the cortices of Huntington's disease mice.

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FIG. 19. Effects of intraperitoneal injection of compound 2 on the levels of mHTT and wtHTT in the striata of Huntington's disease mice.

FIG. 20. Detection of mHTT aggregates in the cortices of Huntington's disease mice after intraperitoneal injection of compound 2.

FIG. 21. Effects of intraperitoneal injection of compound 2 on behavioral defects in Huntington's disease mice.

FIG. 22. Effects of compounds on the level of ATXN3 protein in fibroblasts of patients with spinocerebellar ataxia type 3.

FIG. 23. Effects of compounds on the level of mutant ATXN1 protein in HEK293T cells.

DETAILED DESCRIPTION

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art. If there is a contradiction, the definition provided in this application shall prevail. When a trade name is present herein, it refers to the corresponding commodity or the active ingredient thereof. All patents, published patent applications and publications cited herein are incorporated herein by reference.

General Terminology and Definitions

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art. If there is a contradiction, the definition provided in this application shall prevail. When a trade name is present herein, it refers to the corresponding commodity or the active ingredient thereof. All patents, published patent applications and publications cited herein are incorporated herein by reference.

The terms "including", "comprising", "having", "containing", or "relating to" and other variants thereof, as used herein, are inclusive or open-ended, and not exclusive of other elements or steps of methods that are not enumerated. Those skilled in the art should understand that the above terms such as "include" also cover the meaning of "consist of".

The term "one or more" or the similar expression "at least one" can mean, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more.

When the lower limit and the upper limit of a numerical range are disclosed, any numerical value and any included range falling within the range are specifically disclosed. In particular, each value range of the values disclosed herein should be understood to mean each value and range covered in a wider range. For example, the expression "ATXN1 with polyQ length ≥ 40 " can cover the case where the polyQ length ≥ 41 . For example, "ATXN2 with polyQ length ≥ 33 " can cover the case where the polyQ length ≥ 34 . For example, "ATXN3 with polyQ length ≥ 41 " can cover the case where the polyQ length ≥ 42 and can cover the case where the polyQ length is 74, for example. For example, "ATXN3 with polyQ length ≤ 41 " can cover the case where the polyQ length is 27. For example, "ATXN7 with polyQ length ≥ 19 " can cover the case where the polyQ length ≥ 38 . For example, "TBP with polyQ length ≥ 44 " can cover the case where the polyQ length ≥ 45 . For example, "ATN1 with polyQ length ≥ 39 " can cover the case where the polyQ length ≥ 49 . For example, "HTT with a polyQ length ≥ 36 " can cover the case where the polyQ length is 47, 49, 55, 68, 72, 73, 111, 128, or 140. For another example, "HTT with polyQ length < 36 " can cover the case where the polyQ length is 7, 16, 19, 23, or 25. For example, "AR with polyQ length ≥ 37 " can cover the case where the polyQ length ≥ 38 .

The expression m-n used as used herein refers to the range from m to n and the subrange composed of each point value and each point value. For example, the expression “C₁₋₈” or “C₁₋₈” covers the range of 1-8 carbon atoms and should be understood to also cover any subrange and each point value, such as C₂₋₅, C₃₋₄, C₁₋₂, C₁₋₃, C₁₋₄, C₁₋₅, C₁₋₆, C₁₋₇, etc., and C₁, C₂, C₃, C₄, C₅, C₆, C₇, C₈, etc. For example, the expression “C₃₋₁₀” or “C₃₋₁₀” should also be understood in a similar manner, for example, it can cover any subrange and point value contained therein, such as C₃₋₉, C₆₋₉, C₆₋₈, C₆₋₇, C₇₋₁₀, C₇₋₉, C₇₋₈, C₈₋₉, etc., and C₃, C₄, C₅, C₆, C₇, C₈, C₉, C₁₀, etc. For another example, the expression “3- to 10-membered” should be understood as covering any subrange and each point value, such as 3- to 5-membered, 3- to 6-membered, 3- to 7-membered, 3- to 8-membered, 4- to 5-membered, 4- to 6-membered, 4- to 7-membered, 4- to 8-membered, 5- to 7-membered, 5- to 8-membered, 6- to 7-membered, 6- to 8-membered, 9- to 10-membered, etc., and 3-, 4-, 5-, 6-, 7-, 8-, 9-, 10-membered, etc. Other similar expressions as used herein should also be understood in a similar way.

The term “optional” or “optionally” means that the event or situation described later may or may not occur, and the description includes occurrence of said event or situation and non-occurrence of said event or situation.

The terms “substitute” and “substituted” mean that one or more (e.g. one, two, three, or four) hydrogens are replaced by a selection from the indicated group, provided that not exceeding the normal valence of the designated atom under the circumstance and the substitution forms a stable compound. Combinations of substituents and/or variables are only allowed when the combination forms a stable compound. When describing that a substituent is absent, it should be understood that the substituent can be one or more hydrogen atoms, provided that the structure can result in a stable state of the compound.

If a substituent is described as “optionally substituted”, the substituent may be unsubstituted or may be substituted. If an atom or group is described as being optionally substituted by one or more substituents on a list, one or more hydrogens on the atom or group can be optionally replaced by substituents which are selected independently. When the substituent is oxo (i.e., =O), it means that two hydrogen atoms are replaced.

Unless otherwise specified, as used herein, the point of attachment of a substituent can be from any suitable position of the substituent. When a bond of a substituent is shown to pass through a bond connecting two atoms in a ring, such substituent can bond to any of the ring-forming atoms in the substitutable ring.

When any variable (such as R), as well as variables with superscripts or subscripts (such as R^{X1}, R^{X2}, R⁷, R⁸, R^{a1}, R^{b1}, R^{c1}, R^{a1}, R^{b1}, R^{c2}, etc.) appear more than once in the composition or structure of the compound, its definition in each case is independent at each occurrence. For example, if a group is substituted with 0, 1, 2, 3, or 4 R substituents, the group may optionally be substituted with up to four R substituents, and all the options of R substituent are all independent of each other.

The term “halo” or “halogen” or “halogenated” should be understood to mean fluorine (F), chlorine (Cl), bromine (Br) or iodine (I) atoms, preferably fluorine, chlorine, bromine atoms.

The term “alkyl” refers to a straight or branched saturated aliphatic hydrocarbon group composed of carbon atoms and hydrogen atoms, which is connected to the rest of the molecule by a single bond. “Alkyl” may have 1-8 atoms,

referring to “C₁₋₈ alkyl”, for example, C₁₋₄ alkyl, C₁₋₃ alkyl, C₁₋₂ alkyl, C₃ alkyl, C₄ alkyl, C₁₋₆ alkyl, C₃₋₆ alkyl. Non-limiting examples of alkyl groups include, but are not limited to methyl, ethyl, propyl, butyl, pentyl, hexyl, isopropyl, isobutyl, sec-butyl, tert-butyl, isopentyl, 2-methylbutyl, 1-methylbutyl 1-ethylpropyl, 1,2-dimethylpropyl, neopentyl, 1,1-dimethylpropyl, 4-methylpentyl, 3-methylpentyl, 2-methylpentyl, 1-methylpentyl, 2-ethylbutyl, 1-ethylbutyl, 3,3-dimethylbutyl, 2,2-dimethylbutyl, 1,1-dimethylbutyl, 2,3-dimethylbutyl, 1,3-dimethylbutyl or 1,2-dimethylbutyl, or their isomers.

The term “alkylene”, when used alone or in combination with other groups herein, refers to a straight or branched saturated divalent hydrocarbon group. For example, the term “C₁₋₆ alkylene” refers to an alkylene having 1-6 carbon atoms, for example, methylene, ethylene, propylene, butylene, pentylene, hexylidene, 1-methylethylene, 2-methylethylene, methylpropylene or ethylpropylene, etc.

The term “alkenyl” refers to a straight or branched unsaturated aliphatic hydrocarbon group consisting of carbon atoms and hydrogen atoms with at least one double bond. The alkenyl group may have 2-8 carbon atoms, referring to, “C₂₋₈ alkenyl”, for example, C₂-alkenyl, C₃₋₄ alkenyl. Non-limiting examples of alkenyl groups include, but are not limited to ethenyl, allyl, (E)-2-methylethenyl, (Z)-2-methylethenyl, (E)-but-2-enyl, (Z)-but-2-enyl, (E)-but-1-enyl, (Z)-but-1-enyl, etc.

The term “alkynyl” refers to a straight or branched unsaturated aliphatic hydrocarbon group with at least one triple bond consisting of carbon atoms and hydrogen atoms. The alkynyl group may have 2-8 carbon atoms, referring to “C₂₋₈ alkynyl”, for example, C₂₋₄ alkynyl, C₃₋₄ alkynyl. Non-limiting examples of alkynyl groups include, but are not limited to ethynyl, prop-1-ynyl, prop-2-ynyl, but-1-ynyl, but-2-ynyl, but-3-ynyl, etc.

The term “cyclohydrocarbyl” refers to a saturated or unsaturated non-aromatic cyclic hydrocarbon group composed of carbon atoms and hydrogen atoms, preferably containing 1 or 2 rings. The said cyclohydrocarbyl may be a monocyclic ring, a fused polycyclic ring, a bridged ring or a spiro ring structure. The cyclohydrocarbyl group may have 3-10 carbon atoms, i.e. “C₃₋₁₀ cyclohydrocarbyl”, for example, C₃₋₈ cyclohydrocarbyl, C₅ cyclohydrocarbyl, C₆ cyclohydrocarbyl, C₇ cyclohydrocarbyl. Non-limiting examples of cyclohydrocarbyl include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, bicyclo[2.2. 1]heptyl and spiro[3.3]heptyl, etc. The term also covers situations where the C atom can be substituted by oxo (=O).

The term “heterocyclyl” or “heterocyclic hydrocarbon group” refers to a monocyclic or bicyclic ring system having, for example, 3-10 (suitably 3-8, more suitably 3-7, especially 4-6) ring atoms (3- to 10-membered, 3- to 8-membered, 3- to 7-membered, 3- to 6-membered), in which at least one ring atoms (for example, 1, 2 or 3) are heteroatoms selected from N, O and S, and the remaining ring atoms are C. The ring system can be saturated (also can be understood as the corresponding “heterocycloalkyl”) or unsaturated (i.e., having one or more double bonds and/or triple bonds in the ring). “Heterocyclyl” or “heterocyclic hydrocarbon group” does not possess aromaticity. The term also covers situations where the C atom can be substituted with oxo (=O) and/or the S atom on the ring can be substituted with 1 or 2 oxo (=O).

The heterocyclyl may be, for example, a 4-membered ring, such as azetidiny, oxetanyl, or a 5-membered ring, such as tetrahydrofuranyl, dioxolinyl, pyrrolidinyl, imida-

zolidinyl, pyrazolidinyl, pyrrolinyl, oxopyrrolidinyl, 2-oxoimidazolidin-1-yl; or 6-membered ring, such as tetrahydropyran, piperidinyl, morpholinyl, dithianyl, thiomorpholinyl, piperazinyl, 1,1-dioxo-1,2-thiazinan-2-yl or trithianyl; or a 7-membered ring, such as a diazepine ring. Optionally, the heterocyclyl can be benzo-fused.

The heterocyclyl may be bicyclic without limitation, for example, a 5-membered fused 5-membered ring, such as hexahydrocyclopentane[c]pyrrole-2(1H)-yl; or a 5-membered fused 6-membered bicyclic ring, such as hexahydro-1,2-a]pyrazine-2(1H)-yl.

As mentioned above, the heterocycle may be unsaturated; that is to say, it may contain one or more double bonds without limitation. For example, an unsaturated heterocycle containing a nitrogen atom may be 1,6-dihydropyrimidine, 1,2-dihydropyrimidine, 1,4-dihydropyrimidine, 1,6-dihydropyridine, 1,2-dihydropyridine, 1,4-dihydropyridine, 2,3-dihydro-1H-pyrrole, 3,4-dihydro-1H-pyrrole, 2,5-dihydro-1H-pyrrolyl, 4H-[1,3,4]thiadiazinyl, 4,5-dihydrooxazolyl or 4H-[1,4] thiazinyl ring. An unsaturated heterocycle containing an oxygen atom may be 2H-pyran, 4H-pyran, or 2,3-dihydrofuran, and the unsaturated heterocycle containing a sulfur atom may be 2H-thiopyran or 4H-thiopyran. The heterocycle can be benzo-fused, but not limited thereto, such as dihydroisoquinolinyl ring.

The term “aryl” refers to an all-carbon monocyclic or fused polycyclic (such as bicyclic) aromatic ring group with a conjugated π -electron system. For example, aryl may have 6-14 carbon atoms, suitably 6-10, more suitably 6 or 10. Examples of aryl include, but are not limited to, phenyl, naphthyl, anthracenyl, etc.

The term “heteroaryl” should be understood to preferably mean a monovalent monocyclic, bicyclic, or tricyclic aromatic ring system having 5, 6, 7, 8, 9 or 10 ring atoms (“5- to 10-membered heteroaryl”), especially 5 or 6 or 9 or 10 ring atoms, and at least one (suitably 1-4, more suitably 1, 2 or 3) of the ring atoms are heteroatoms such as oxygen, nitrogen, or sulfur. The heteroatoms may be the same with or different from each other. In addition, the heteroaryl can be benzo-fused in each case. In particular, the heteroaryl is selected from the group consisting of thienyl, furyl, pyrrolyl, oxazolyl, thiazolyl, imidazolyl, pyrazolyl, isoxazolyl, isothiazolyl, oxadiazolyl, triazolyl, thiadiazolyl, etc., and their benzo derivatives, such as benzofuranyl, benzothienyl, benzoxazolyl, benzisoxazolyl, benzimidazolyl, benzotriazolyl, indazole, indolyl, isoindolyl, etc.; or pyridyl, pyridazinyl, pyrimidinyl, pyrazinyl, triazinyl, etc., and their benzo-fused derivatives, such as quinolinyl, quinazolinyl, isoquinolinyl, etc., or azocinyl, indoliziny, purinyl, etc., and their benzo derivatives; or cinnolinyl, phthalazinyl, quinazolinyl, quinolinyl, naphthyridinyl, carbazolyl, acridinyl, etc.

The term “C₁-C₆ hydrocarbon chain” refers to a chain-like group composed of carbon atoms and hydrogen atoms, which may be straight or branched, and contains 1-8 (especially 1-5, such as 1, 2, 3, 4 or 5) carbon atoms. The hydrocarbon chain can be saturated (i.e., C₁-C₆ alkylene) or unsaturated; that is to say, it may contain one or more (preferably one) carbon-carbon double bond or triple bond.

The alkylene may have 1-8 carbon atoms, referring to “C₁₋₆ alkylene”, such as C₁₋₅ alkylene, C₁₋₄ alkylene, C₁₋₃ alkylene, C₁₋₂ alkylene, C₃ alkylene, and C₁ alkylene, i.e., methylene. Non-limiting examples of alkylene include but are not limited to methylene (—CH₂—), 1,1-ethylene (—CH(CH₃)—), 1,2-ethylene (—CH₂CH₂—), 1,1-propylene (—CH(CH₂CH₃)—), 1,2-propylene (—CH₂CH(CH₃)—), 1,3-propylene (—CH₂CH₂CH₂—), 1,4-butylen (—CH₂CH₂CH₂CH₂—) etc.

The term “neurodegenerative disorder” refers to a disease caused by the loss or pathological change of neurons and/or their myelin sheaths. Characteristic pathological structures, such as insoluble aggregates of protein, can be observed in the brain neurons of patients with neurodegenerative disorders. Insoluble aggregates may produce cytotoxicity, further leading to neuron loss and disease.

The term “polyQ” or “polyglutamine” refers to the polyglutamine tract contained by the protein. Glutamine is encoded by cytosine-adenine-guanine (CAG) in the gene. The length of the polyglutamine is related to the number of CAG repeats in gene exons. Therefore, the increase in the number of CAG repeats in gene exons result in polyglutamine expansions in the synthesized protein. Proteins with abnormally expanded polyQ are known to be associated with some neurodegenerative disorder. As used herein, the gene name can use the form of “Q+number” to indicate the number of CAG repeats in exons, such as Q25 or Q72, which respectively indicate 25 repeats or 72 repeats of CAG in exons. In the protein name, the length of the polyglutamine can be expressed in the form of “Q+number” as above, such as Q27 or Q73, which means that the length of the polyglutamine is 27 Q (glutamine) or 73 Q, respectively. The CAG repetitions or glutamine repetitions indicated in the form of “Q+number” herein are all continuous repetitions. Unless otherwise specified, the length of polyQ as used herein refers to the length of the continuous polyglutamine.

The term “polyQ-related neurodegenerative disorder” refers to a neurodegenerative disorder associated with abnormal expansion of polyQ, or a neurodegenerative disorder that responds to levels of proteins containing expanded polyQ. Neurodegenerative disorders are a group of disorders with clinical and genetic heterogeneity.

“Normal polyQ” refers to the polyQ of a protein in a normal physiological state that has a length less than a specific number. Correspondingly, “abnormally expanded” means that the polyQ of the protein has a length longer than the normal length. For diseases or pathological conditions, the length of polyQ is longer. As an example, polyQ-related neurodegenerative disorders include but are not limited to spinocerebellar ataxia (SCA) type 1 (polyQ length ≥ 41), type 2 (polyQ length ≥ 34), type 3 (polyQ length ≥ 62), type 7 (polyQ length ≥ 38), type 12 (polyQ length ≥ 46), type 17 (polyQ length ≥ 45); and dentatorubral-pallidoluysian atrophy (DRPLA, polyQ length ≥ 49), Huntington’s disease (HD, polyQ length ≥ 36) and spinal-bulbar muscular atrophy (SBMA, polyQ length ≥ 38). These diseases are caused by the expansion of CAG repeat regions on ATXN1, ATXN2, ATXN3, ATXN7, ATXN12, TBP, ATN1, HTT and AR genes, respectively (Lesley Jones et al., DNA repair in the trinucleotide repeat disorders, Lancet Neurol. 2017; 16: 88-96). Among them, spinocerebellar ataxia type 3 (SCA3, also known as Machado-Joseph disease, MJD) is the most common autosomal dominant spinocerebellar ataxia and is second only to HD as a common polyQ-related disorder in the world. SCA3 is caused by the abnormally expanded polyQ at the C-terminus of the coded protein ATXN3 resulting from the increase in the number of CAG repeats of the Ataxin-3 gene (ATXN3; also known as the MJD1 gene). Examples of normal polyQ proteins described herein include, but are not limited to, ATXN1 with polyQ length < 40 , ATXN2 with polyQ length < 33 , ATXN3 with polyQ length < 41 , ATXN7 with polyQ length < 19 , ATXN12 with polyQ length < 46 , TBP with polyQ length < 44 , ATN1 with polyQ length < 39 , HTT with polyQ length < 36 , and AR with polyQ length < 37 . Correspondingly, examples of the proteins with abnormally expanded polyQ described herein

include, but are not limited to, ATXN1 with polyQ length ≥ 40 , ATXN2 with polyQ length ≥ 33 , ATXN3 with polyQ length ≥ 41 , ATXN7 with polyQ length ≥ 19 , and ATXN12 with polyQ length ≥ 46 , TBP with polyQ length ≥ 44 , ATN1 with polyQ length ≥ 39 , HTT with polyQ length ≥ 36 , and AR with polyQ length ≥ 37 .

The term "pharmaceutically acceptable" refers to that when contacted with the patient's tissue within the scope of normal medical judgment, no undue toxicity, irritation, allergic reactions, etc. shall arise, having reasonable advantage-disadvantage ratios and effective for the intended use.

The pharmaceutically acceptable salts of the compound of the present invention include acid addition salts and base addition salts thereof. Suitable acid addition salts are formed from acids that form pharmaceutically acceptable salts. Examples include hydrochloride, acetate, aspartate, benzoate, bicarbonate/carbonate, glucoheptonate, gluconate, nitrate, orotate, palmitic acid salt and other similar salt. Suitable base addition salts are formed from bases that form pharmaceutically acceptable salts. Examples include aluminum salts, arginine salts, choline salts, magnesium salts, and other similar salts. The method for preparing the pharmaceutically acceptable salt of the compound of the present invention is known to those skilled in the art.

The compound of the present invention may exist in specific geometric or stereoisomeric forms. The present invention contemplates all such compounds, including cis and trans isomers, (-)- and (+)-enantiomers, (R)- and (S)-enantiomers, diastereomers, (D)-isomers, (L)-isomers, and their racemic mixtures and other mixtures, for example, mixtures enriched in enantiomers or diastereomers, all of which are within the scope of the present invention. There may be other asymmetric carbon atoms in substituents such as alkyl. All these isomers and their mixtures are included in the scope of the present invention. In some embodiments, the preferred compounds are those isomeric compounds that show better biological activity. Purified or partially purified isomers and stereoisomers, or racemic mixtures or diastereomeric mixtures of the compound of the present invention are also included in the scope of the present invention. The purification and separation of such substances can be achieved by standard techniques known in the art.

Optically pure enantiomers can be obtained by resolving racemic mixtures according to conventional methods, for example, by using optically active acids or bases to form diastereomeric salts, or by forming covalent diastereomers. A mixture of diastereomers can be separated into single diastereomers based on their physical and/or chemical differences by methods known in the art (for example, by chromatography or fractional crystallization). Then, release the optically active enantiomeric base or acid from the separated diastereomeric salt. Another method for separating racemic enantiomers can use chiral chromatography (such as a chiral HPLC column). The separated chiral isomers can be subjected to conventional derivatization or non-derivatization before separation, depending on what method can achieve more effective separation of chiral isomers. Enzymatic methods can also be used to separate derivatized or underivatized chiral isomers. Similarly, optically active raw materials can be used to obtain the optically pure compound of the present invention through chiral synthesis.

In addition, the compound of the present invention may exist in the form of tautomers. The present invention includes all possible tautomers of the compound of the present invention, and also includes single tautomers or the form of any mixtures of said tautomers in any ratio.

The compound of the present invention may exist in the form of solvates (preferably hydrates), wherein the compound of the present invention contains a polar solvent as a structural element of the compound crystal lattice, especially for example water, methanol, or ethanol. The amount of polar solvents, especially water, can be present in stoichiometric or non-stoichiometric ratios.

The present invention also covers all possible crystalline forms or polymorphs of the compound of the present invention, which can be a single polymorph or a mixture of more than one polymorph in any ratio.

The present invention also contemplates all pharmaceutically acceptable isotopically-labeled compounds, which are identical to the compounds of the invention except that one or more atoms are replaced by the atom(s) of the same atomic number but having atomic mass or mass number different from the atomic mass or mass number prevailing in nature.

The metabolites of the compound of the present invention are also included within the scope of the present invention, namely the substances formed in the body when the compound of the present invention is administered. The metabolites of compounds can be identified by techniques known in the art, and their activity can be characterized by experimental methods. Such products can be produced, for example, by oxidation, reduction, hydrolysis, amidation, deamidation, esterification, enzymatic hydrolysis, etc. of the administered compound. Therefore, the present invention includes metabolites of the compound of the present invention, including compounds prepared by contacting the compound of the present invention with a mammal for a time sufficient to produce its metabolites.

The present invention further includes within its scope the prodrugs of the compound of the present invention, which are certain derivatives of the compound of the present invention that have less or no pharmacological activity themselves but when administered to or on the body, can be converted into the compound of the present invention with the desired activity by, for example, hydrolytic cleavage. Usually, such prodrugs will be functional group derivatives of the compound, which are easily converted into the desired therapeutically active compound in vivo. For a review of prodrugs and their preparation methods, see, for example, J. Rautio et al. *Nature Reviews Drug Discovery* (2008) 7, 255-270 and *Prodrugs: Challenges and Rewards* (V. Stella et al. ed. Springer, 2007). The prodrugs of the present invention can be prepared, for example, by replacing an appropriate functional group in the compound of the present invention with a certain moiety known to those skilled in the art as "pro-moiety".

The term "polymorphism" or "polymorph" refers to a single polymorph or a mixture of more than one polymorph in any ratio.

The term "crystal form" or "crystal" refers to any solid substance exhibiting a three-dimensional order, as opposed to an amorphous solid substance, which produces a characteristic X-ray powder diffraction pattern with sharp and defined peaks.

The term "amorphous" refers to any solid substance which lacks order in three dimensions.

The term "hydrate" describes a solvate containing a drug and a stoichiometric or non-stoichiometric amount of water.

The term "pharmaceutically acceptable carrier" refers to those substances that have no obvious stimulating effect on the organism and will not damage the biological activity and performance of the active compound. "Pharmaceutically acceptable carriers" include but are not limited to glidants,

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sweeteners, diluents, preservatives, dyes/colorants, flavors, surfactants, wetting agents, dispersants, disintegrants, stabilizer, solvent or emulsifier. Non-limiting examples of the carrier include calcium carbonate, calcium phosphate, various sugars and various starches, cellulose derivatives, gelatin, vegetable oils and polyethylene glycols, etc.

The term "administration" or "administering" or the like refers to a method that enables a compound or composition to be delivered to a desired site of biological action. Such methods comprise but not limited to oral or parenteral (including intracerebroventricular, intravenous, subcutaneous, intraperitoneal, intramuscular, intravascular injection or infusion), local, rectal administration or the like. Especially injection or oral.

As used herein, The terms "treat," "treating" or "treatment," and other grammatical equivalents as used herein, include alleviating, abating or ameliorating a disease or condition, preventing additional symptoms, ameliorating or preventing the underlying metabolic causes of symptoms, inhibiting the disease or condition, e.g., arresting the development of the disease or condition, relieving the disease or condition, causing regression of the disease or condition, relieving a condition caused by the disease or condition, or stopping the symptoms of the disease or condition, and are intended to include prophylaxis. The terms further include achieving a therapeutic benefit and/or a prophylactic benefit. By therapeutic benefit is meant eradication or amelioration of the underlying disorder being treated. Also, a therapeutic benefit is achieved with the eradication or amelioration of one or more of the physiological symptoms associated with the underlying disorder such that an improvement is observed in the patient, notwithstanding that the patient may still be afflicted with the underlying disorder. For prophylactic benefit, the compositions may be administered to a patient at risk of developing a particular disease, or to a patient reporting one or more of the physiological symptoms of a disease, even though a diagnosis of this disease may not have been made.

The terms "active ingredient", "therapeutic agent", "active substance" or "active agent" refer to a chemical entity that can effectively treat or prevent the target disorder, disease, or symptom.

For drugs, drug units or active ingredients, the terms "effective amount", "therapeutically effective amount" or "prophylactically effective amount" refer to an amount of a drug or agent that has acceptable side effects but is sufficient to achieve the desired effect. The effective amount may be determined individually and depends on the age and general condition of the receptor as well as specific active substance. The effective amount in specific case can be determined by a person skilled in the art through conventional test.

As used herein, the term "subject" includes a human or non-human animal. An exemplary human subject includes a human subject having a disease (such as one described herein) (referred to as a patient), or a normal subject. The term "non-human animal" as used herein includes all vertebrates, such as non-mammals (e.g. birds, amphibians, reptiles) and mammals, such as non-human primates, livestock and/or domesticated animals (such as sheep, dog, cat, cow, pig and the like).

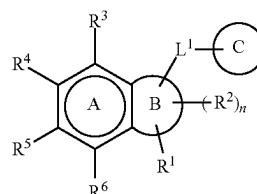
The following detailed description of the invention is intended to illustrate non-limiting embodiments, so that others skilled in the art can more fully understand the technical solution of the present invention, its principles and practical applications, so that others skilled in the art can modify and implement the present invention in various

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manners so that it can be optimally adapted to the requirements of specific applications.

The Compound of the Present Invention and Use Thereof

In one aspect, the present invention provides use of a compound of formula (I), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder



(I)

wherein:

ring A is a benzene ring;

ring B is a saturated or unsaturated 5- or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S.

ring C is selected from C₆₋₁₀ aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R^{X1}.

L¹ is a bond, or a C₁₋₆ hydrocarbon chain;

or ring C is absent, and L¹ is absent;

R¹ is =Y, wherein Y is O or S, or OR⁷;

at each occurrence, R² is each independently selected from hydrogen, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)₂OR^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, C(=S)R^{a1}, P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)₂R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)OR⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)₂R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl),

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—O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, =O, —COOH and C₁₋₆ alkyl;

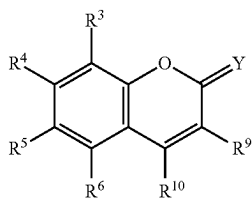
at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₄ alkyl and C₆₋₁₀ aryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl);

at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆-alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR^{r1}, —SR^{r1}, —NR^{r1}R^{r2}, —C(=O)OR^{r1}, —C(=O)NR^{r1}R^{r2}, —C(=O)R^{r1}, —S(=O)₂OR^{r1}, —S(=O)₂R^{r1}, —S(=O)₂NR^{r1}R^{r2} and —S(=O)₂R^{r1}, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, =O, =S, —OR^{r3}, —SR^{r3}, —NR^{r3}R^{r4}, —C(=O)R^{r3}, —C(=O)OR^{r3} and —C(=O)NR^{r3}R^{r4};

at each occurrence, R^{r1}, R^{r2}, R^{r3}, R^{r4} are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —OH, —SH, —NH₂, =O and —COOH;

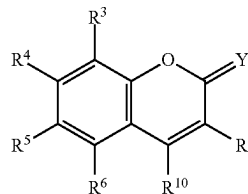
n is 1 or 2;

provided that the compound of formula (I) does not include the following structure:

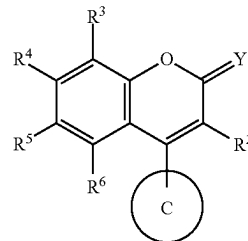


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wherein R³ is selected from —O(C₁₋₆ alkyl) and —O(C₁₋₄ alkylene-C₆₋₁₀ aryl); R¹⁰ is H; and the compound of formula (I) does not include the following structure:



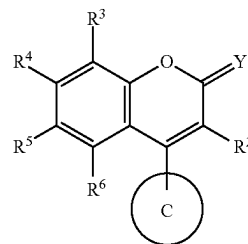
wherein R⁴ is —O(C₁₋₆ alkyl); R⁵ is halogen; R¹⁰ is CF₃; and the compound of formula (I) does not include the following structure:



wherein R⁴ is selected from —O(C₁₋₆ alkyl) and —O(C₁₋₄ alkylene-C₆₋₁₀ aryl);

ring C is C₆₋₁₀ aryl, which is optionally substituted with one or more substituents each independently selected from R^{x1};

and the compound of formula (I) does not include the following structure:

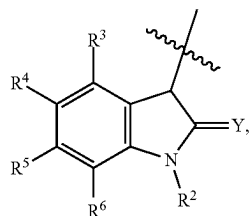


wherein R⁴ is selected from —OH and —O(C₁₋₆ alkyl); R⁵ is halogen; ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R^{x1}.

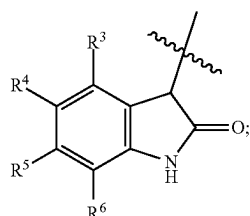
In one embodiment, R¹ is =O. In another embodiment, R¹ is =S. In another embodiment, R¹ is OR⁷.

In one embodiment, ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S. In one embodiment, ring B is a saturated or unsaturated 5-membered heterocycle, wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrrole. In another embodiment, ring B is selected from 2,3-dihydro-1H-pyrrole and 3,4-dihydro-1H-pyrrole, preferably 2,3-dihydro-1H-pyrrole. In yet another embodiment, ring B is pyrrolidine.

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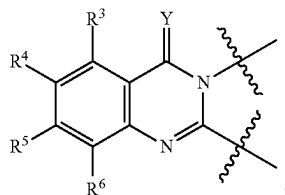


In a more preferred embodiment, the A-B ring system is wherein Y is O or S; and ring C is a 5- to 7-membered heteroaryl, preferably a 5- to 6-membered heteroaryl, especially a 5-membered heteroaryl, wherein the heteroaryl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} . In a particular embodiment, the A-B ring system is



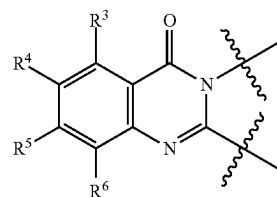
and ring C is a 5- to 7-membered heteroaryl, preferably a 5- to 6-membered heteroaryl, especially a 5-membered heteroaryl, the heteroaryl is optionally substituted by 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} . In another embodiment, ring C contains 1, 2, 3 or 4 heteroatoms, each of which is independently selected from N, O and S, preferably selected from N and O. In yet another embodiment, ring C contains at least one N atom. In one embodiment, ring C is a 5-membered heteroaryl group containing 1 or 2 N atoms, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R^{X1} . In another embodiment, ring C is selected from pyrrole and imidazole.

In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

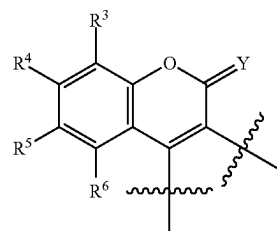


In a more preferred embodiment, the A-B ring system is wherein Y is O or S. In a particular embodiment, the A-B ring system is

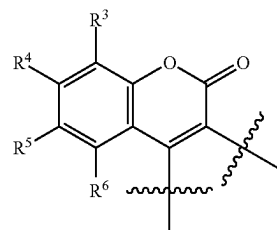
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In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, the A-B ring system is



wherein Y is O or S. In a particular embodiment, the A-B ring system is

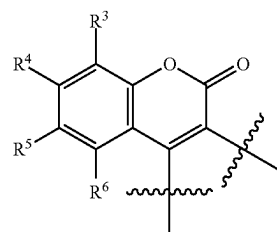


In yet another embodiment, ring C is phenyl, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R^{X1} .

In one embodiment, L^1 is a bond. In another embodiment, L^1 is a C_1 - C_6 hydrocarbon chain.

In a preferred embodiment, L^1 is a C_1 - C_2 hydrocarbon chain.

In one embodiment, the A-B ring system is



ring C is absent, and L^1 is absent.

In another embodiment:

ring A is a benzene ring;

ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

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ring C is selected from C₆₋₁₀ aryl and 5- to 10-membered heteroaryl, wherein aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or a C₁₋₆ hydrocarbon chain;

R¹ is =Y, wherein Y is O or S, or OR⁷;

at each occurrence, R² is each independently selected from hydrogen, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), alkylene-(3- to 7-membered heterocyclyl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), alkylene-3- to 7-membered heterocyclyl)₂, =O, —COOH and C₁₋₆ alkyl;

at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl and C₆₋₁₀ aryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl);

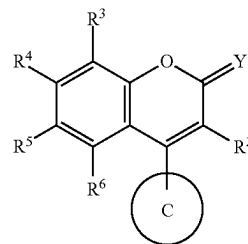
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at each occurrence, R^{a1}, R^{b1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷², —C(=O)R⁷¹, —S(=O)₂OR⁷¹, —S(=O)₂R⁷¹, —S(=O)₂NR⁷¹R⁷² and —S(=O)R⁷¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR⁷³, —SR⁷³, —NR⁷³R⁷⁴, —C(=O)R⁷³, —C(=O)OR⁷³ and —C(=O)NR⁷³R⁷⁴;

at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —OH, —SH, —NH₂, =O and —COOH;

n is 1;

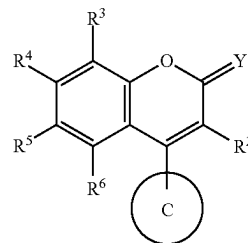
and provided that the compound of formula (I) does not include the following structure:



wherein R⁴ is selected from —O(C₁₋₆ alkyl) and —O(C₁₋₄ alkylene-C₆₋₁₀ aryl);

ring C is C₆₋₁₀ aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1};

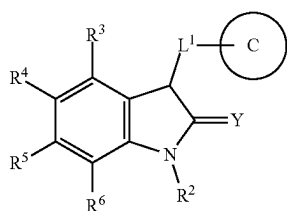
and the compound of formula (I) does not include the following structure:



wherein R⁴ is selected from —OH and —O(C₁₋₆ alkyl); R⁵ is halogen; ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R^{X1}.

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The structure described in the following formula (II) falls within the scope of the structure described in formula (I). In another aspect of the present invention, provided is use of a compound of formula (II), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder



wherein:

Y is O or S;

ring C is 5- to 7-membered heteroaryl, the heteroaryl is optionally substituted with one or more substituents each independently selected from R^{X1} ;

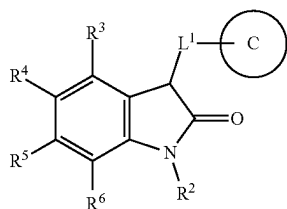
R^2 is selected from H and C_{1-8} alkyl;

L^1 is a bond, or a C_1-C_6 hydrocarbon chain;

R^3, R^4, R^5, R^6 are each independently selected from H and R^{X2} ;

wherein R^{X1}, R^{X2} are as defined in formula (I).

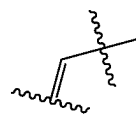
In one embodiment, Y is O. When Y is O, the formula (II) is



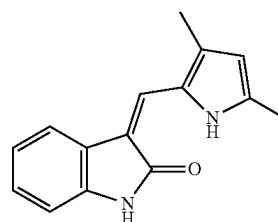
In one embodiment, ring C is a 5- to 6-membered heteroaryl, wherein the heteroaryl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} . In a preferred embodiment, ring C is a 5-membered heteroaryl, which is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} . In another embodiment, ring C contains 1, 2, 3 or 4 heteroatoms, each of which is independently selected from N, O and S, preferably selected from N and O. In another embodiment, ring C contains at least one N atom. In one embodiment, ring C is a 5-membered heteroaryl containing 1 or 2 N atoms, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R^{X1} . In another embodiment, ring C is selected from pyrrole and imidazole.

In one embodiment, L^1 is a bond. In another embodiment, L^1 is a C_1-C_6 hydrocarbon chain. In a preferred embodiment, L^1 is a C_1-C_2 hydrocarbon chain. In one embodiment, L^1 is methylene or methenyl. In a special embodiment, L^1 is methenyl. In another particular embodiment, L^1 is

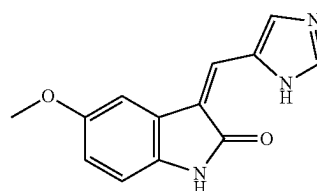
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In another particular embodiment, the compound of formula (II) is selected from:



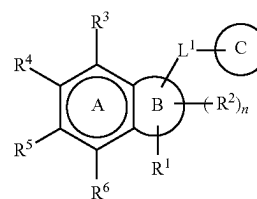
(compound 3)



(compound 4)

The structure described in the following formula (III) falls within the scope of the structure described in formula (I). In another aspect of the present invention, provided is use of a compound of formula (III), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

wherein:



(III)

ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2, or 3 heteroatoms each independently selected from N, O and S;

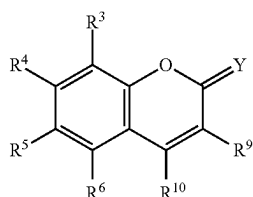
Ring C is a C_{6-10} aryl, optionally substituted with one or more substituents each independently selected from R^{X1} ;

or ring C is absent, and L^1 is absent;

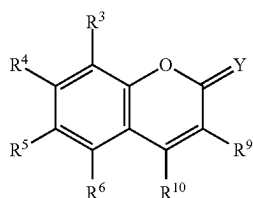
ring A, $L^1, R^1, R^2, R^3, R^4, R^5, R^6, R^{X1}, n$ are as defined in formula (I);

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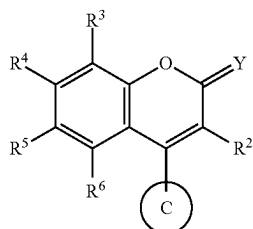
provided that the compound of formula (III) does not include the following structure:



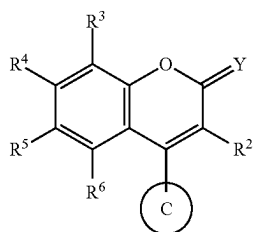
wherein R^3 is selected from $-\text{O}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{O}(\text{C}_{1-4} \text{ alkylene-}\text{C}_{6-10} \text{ aryl})$; R^{10} is H; and the compound of formula (III) does not include the following structure:



wherein R^4 is $-\text{O}(\text{C}_{1-6} \text{ alkyl})$; R^5 is halogen; R^{10} is CF_3 ; and the compound of formula (III) does not include the following structure:



wherein R^4 is selected from $-\text{O}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{O}(\text{C}_{1-4} \text{ alkylene-}\text{C}_{6-10} \text{ aryl})$; ring C is a C_{6-10} aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1} ; and the compound of formula (III) does not include the following structure:

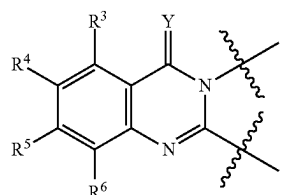


wherein R^4 is selected from $-\text{OH}$ and $-\text{O}(\text{C}_{1-6} \text{ alkyl})$; R^5 is halogen; ring C is a 5- to 10-membered heteroaryl, wherein heteroaryl is substituted with one or more substituents each independently selected from R^{X1} .

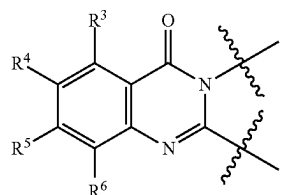
In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein heterocycle contains 1, 2

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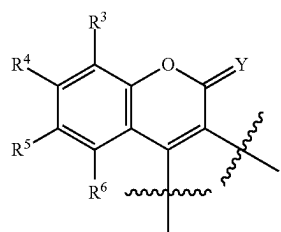
or 3 heteroatoms each independently selected from N, O and S. In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.



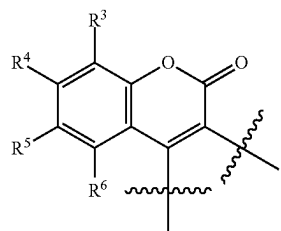
In a more preferred embodiment, the A-B ring system is wherein Y is O or S. In a particular embodiment, the A-B ring system is



In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, the A-B ring system is



wherein Y is O or S. In a particular embodiment, the A-B ring system is



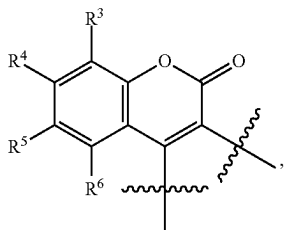
In yet another embodiment, ring C is phenyl, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R^{X1} .

In one embodiment, L^1 is a bond. In another embodiment, L^1 is a $\text{C}_1\text{-C}_6$ hydrocarbon chain.

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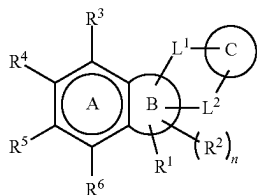
In a preferred embodiment, L^1 is a C_1 - C_2 hydrocarbon chain.

In one embodiment, the A-B ring system is



ring C is absent, and L^1 is absent.

In an alternative embodiment, ring B and ring C of the compound of formula (III) are further connected through L^2 to obtain a variant of the compound of formula (III), which has the following structure of formula (III')



wherein:

ring C is C_{6-10} aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1} ;

R^1 is H, $=O$, or is OR^7 ;

L^1 is a bond, or is a C_1 - C_2 hydrocarbon chain;

L^2 is a bond, or is a C_1 - C_2 hydrocarbon chain;

provided that L^1 and L^2 are not bonds at the same time; R^3 , R^4 , R^5 , R^6 are each independently selected from H and R^{X2} ;

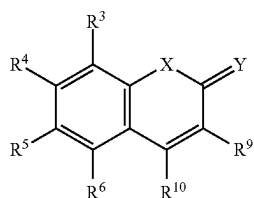
ring A, ring B, R^2 , n, R^{X1} are as defined in formula (III).

In one embodiment, R^1 is H.

In another embodiment, R^2 is $-OH$.

In one embodiment, at each occurrence, R^{X2} is each independently selected from halogen, $-NO_2$, $-CN$, C_{1-6} alkyl, $-OH$ and $-O(C_{1-6}$ alkyl).

The structure described in the following formula (IV) falls within the scope of the structure described in formula (I) and formula (III). In another aspect of the present invention, provided is use of a compound of formula (IV), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder



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wherein:

Y is O or S;

X is O;

R^9 is selected from H, halogen, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $-OR^{a1}$, $-SR^{a1}$, $-NR^{a1}R^{b1}$, $-C(=O)OR^{a1}$, $-C(=O)NR^{a1}R^{b1}$, $-C(=O)R^{a1}$, $-S(=O)_2OR^{a1}$, $-S(=O)_2R^{a1}$, $-S(=O)_2NR^{a1}R^{b1}$ and $-S(=O)R^{a1}$, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-NO_2$, $-CN$, C_{1-6} alkyl, $-OR^{a2}$, $-SR^{a2}$, $-NR^{a2}R^{b2}$, $-C(=O)OR^{a2}$, $-C(=O)NR^{a2}R^{b2}$, $-C(=O)R^{a2}$, $-S(=O)_2OR^{a2}$, $-S(=O)_2R^{a2}$, $-S(=O)_2NR^{a2}R^{b2}$ and $-S(=O)R^{a2}$; wherein R^{a1} , R^{b1} , R^{a2} , R^{b2} are as defined in formula (III);

R^{10} is selected from H, halogen, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl- C_{1-4} alkyl;

R^3 is selected from H, halogen, C_{1-6} alkyl, $-OH$, $-NH_2$, $-NH(C_{1-6}$ alkyl) and $-N(C_{1-6}$ alkyl) $_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-OH$, $-O(C_{1-6}$ alkyl), $-NH_2$, $-NH(C_{1-6}$ alkyl) and $-N(C_{1-6}$ alkyl) $_2$;

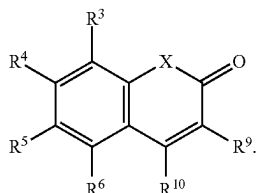
R^4 is selected from H, halogen, $-NO_2$, $-CN$, C_{1-6} alkyl, $-OR^7$, $-SR^7$ and $-NR^7R^8$; at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-NO_2$, $-CN$, $-OH$, $-O(C_{1-6}$ alkyl), $-NH_2$, $-NH(C_{1-6}$ alkyl), $-N(C_{1-6}$ alkyl) $_2$, $-COOH$, $-C(=O)O(C_{1-6}$ alkyl), $-C(=O)NH(C_{1-6}$ alkyl), $-C(=O)N(C_{1-6}$ alkyl) $_2$, $-OC(=O)(C_{1-6}$ alkyl), $-NHC(=O)(C_{1-6}$ alkyl) and $-C(=O)(C_{1-6}$ alkyl); wherein R^7 , R^8 are as defined in formula (III);

R^5 is selected from H, halogen, $-NO_2$, $-CN$, C_{1-6} alkyl, $-OH$, $-O(C_{1-6}$ alkyl), $-NH_2$, $-NH(C_{1-6}$ alkyl), $-N(C_{1-6}$ alkyl) $_2$, $-C(=O)O(C_{1-6}$ alkyl), $-C(=O)NH(C_{1-6}$ alkyl), $-C(=O)N(C_{1-6}$ alkyl) $_2$, $-OC(=O)(C_{1-6}$ alkyl), $-NHC(=O)(C_{1-6}$ alkyl) and $-C(=O)(C_{1-6}$ alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-OH$, $-O(C_{1-6}$ alkyl), $-NH_2$, $-NH(C_{1-6}$ alkyl), $-N(C_{1-6}$ alkyl) $_2$, $-C(=O)O(C_{1-6}$ alkyl), $-C(=O)NH(C_{1-6}$ alkyl), $-C(=O)N(C_{1-6}$ alkyl) $_2$, $-OC(=O)(C_{1-6}$ alkyl), $-NHC(=O)(C_{1-6}$ alkyl) and $-C(=O)(C_{1-6}$ alkyl);

R^6 is selected from H, halogen, $-NO_2$, $-CN$, C_{1-6} alkyl, $-OH$, $-O(C_{1-6}$ alkyl), $-O(benzyl)$, $-SH$, $-S(C_{1-6}$ alkyl), $-S(benzyl)$, $-NH_2$, $-NH(C_{1-6}$ alkyl), $-N(C_{1-6}$ alkyl) $_2$ and $-NH(benzyl)$, wherein the alkyl or benzyl is optionally substituted with one or more substituents selected from halogen, $-OH$, $-O(C_{1-6}$ alkyl), $-NH_2$, $-NH(C_{1-6}$ alkyl) and $-N(C_{1-6}$ alkyl) $_2$; provided that when R^4 is $-O(C_{1-6}$ alkyl) and R^5 is halogen, R^{10} is not CF_3 .

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In one embodiment, Y is O. When Y is O, formula (IV) is



In one embodiment, R⁹ is selected from H, halogen, C₁₋₆ alkyl, —OR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, C(=O)NR^{a1}R^{b1} and —S(=O)₂NR^{a1}R^{b1}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2} and —C(=O)R^{a2}. In one embodiment, R⁹ is selected from H, halogen, —OR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1} and —C(=O)NR^{a1}R^{b1}. In one embodiment, R⁹ is —OR^{a1} or —NR^{a1}R^{b1}. In another embodiment, R⁹ is —C(=O)OR^{a2} or —C(=O)NR^{a2}R^{b2}. In a particular embodiment, R⁹ is —NHC(=O)(C₁₋₆ alkyl). In a particular embodiment, R⁹ is —NHC(=O)CH₃.

In one embodiment, R¹⁰ is selected from H, halogen and methyl. In another embodiment, R¹⁰ is H.

In one embodiment, at each occurrence, R^{a1}, R^{b1} are each independently selected from H, C₁₋₆ alkyl and —C(=O)R^{Y1}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR^{Y3}, —SR^{Y3}, —NR^{Y3}R^{Y4}, —C(=O)R^{Y3}, —C(=O)OR^{Y3} and —C(=O)NR^{Y3}R^{Y4}. Wherein R^{Y1}, R^{Y3}, R^{Y4} are as defined in formula (III). In one embodiment, at each occurrence, R^{a1}R^{b1} are independently selected from H, C₁₋₆ alkyl and —C(=O)(C₁₋₆ alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —C(=O)(C₁₋₆ alkyl), —C(=O)O(C₁₋₆ alkyl), —C(=O)O(C₃₋₆ cyclohydrocarbyl), —C(=O)NH(C₁₋₆ alkyl) and —C(=O)N(C₁₋₆ alkyl)₂. In one embodiment, at each occurrence, R^{a1}, R^{b1} are independently selected from H and C₁₋₆ alkyl.

In one embodiment, at each occurrence, R^{a2}, R^{b2} are each independently selected from H and C₁₋₆ alkyl.

In one embodiment, R³ is selected from H, —OH and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂ and —COOH. In one embodiment, R³ is selected from H, halogen, methyl, —OH, —NH₂ and —NHCH₃, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —NH₂ and —NH(C₁₋₂ alkyl). In one embodiment, R³ is selected from H, halogen, methyl, —OH, —NH₂ and —NHCH₃, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH₃, —NH₂ and —NHCH₃. In one embodiment, R³ is selected from H, halogen, methyl, —OH and —NH₂. In one embodiment, R³ is selected from H, F, Cl, methyl, —OH and —NH₂. In

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another embodiment, R³ is selected from H, F, methyl, —OH and —NH₂. In a particular embodiment, R³ is selected from H, methyl and —OH. In another particular embodiment, R³ is H. In another particular embodiment, R³ is —OH. In another embodiment, R³ is dimethylaminomethyl.

In one embodiment, R⁴ is selected from H, —OR⁷, —SR⁷ and —NR⁷R⁸; at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl) and —C(=O)N(C₁₋₆ alkyl)₂. In one embodiment, R⁴ is C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl) and —C(=O)N(C₁₋₆ alkyl)₂. In a particular embodiment, R⁴ is —CH₂COOH. In another embodiment, R⁴ is selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂. In one embodiment, R⁴ is selected from H, halogen, C₁₋₃ alkyl, —OH, —O(C₁₋₃ alkyl), —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂ and —COOH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₂ alkyl), —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH.

In one embodiment, R⁵ is selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂ and —COOH. In another embodiment, R⁵ is selected from H, halogen, C₁₋₃ alkyl, —OH, —O(C₁₋₃ alkyl), —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂ and —COOH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₂ alkyl), —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH. In one embodiment, R⁵ is selected from H, halogen, C₁₋₃ alkyl, —OH, —O(C₁₋₃ alkyl), —NH₂, —NH(C₁₋₃ alkyl) and —N(C₁₋₃ alkyl)₂, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₂ alkyl), —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂. In one embodiment, R⁵ is dimethylaminomethyl.

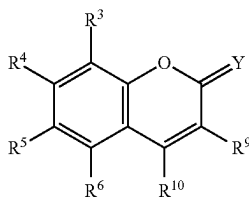
In one embodiment, R⁶ is selected from H, halogen, C₁₋₆ alkyl, —OH and —NH₂, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH₂. In one embodiment, R⁶ is selected from H, F, Cl, Br, C₁₋₆ alkyl, —OH and —NH₂, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH₂. In one embodiment, R⁶ is selected from H, F, Cl, Br, methyl, —OH and —NH₂. In one embodiment, R⁶ is H or —OH.

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In one embodiment, R^3 is methyl, and R^4 is selected from —OH, —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and, C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH. In one embodiment, R^3 is methyl, and R^4 is selected from —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and, C₁₋₃ alkyl substituted with —COOH, wherein substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH.

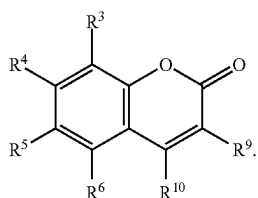
In another embodiment, R^3 is methyl, and R^5 is selected from —OH, —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and, C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH. In another embodiment, R^3 is methyl, and R^5 is selected from —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and, C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH.

The structure described in the following formula (V) falls within the scope of the structure described in formula (I), formula (III) and formula (IV). In another aspect of the present invention, provided is use of a compound of formula (V), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder,



wherein:

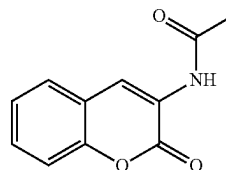
Y , R^9 , R^{10} , R^3 , R^4 , R^5 , R^6 are as defined in formula (IV); in one embodiment, Y is O. When Y is O, formula (V) is



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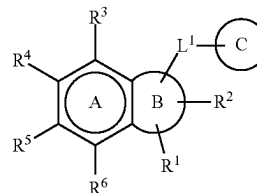
In a particular embodiment, compound of formula (V) is

(compound 5)



The structure described in the following formula (VI) falls within the scope of the structure described in formula (I) and formula (III). In another aspect of the present invention, there is provided use of a compound of formula (VI), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(VI)



wherein:

ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S

ring C is a C₆₋₁₀ aryl, wherein the aryl optionally substituted with one or more substituents each independently selected from R^{X1};

L^1 is a bond, or a C₁-C₆ hydrocarbon chain; R^2 is selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —O, —S, —NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)₂R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —O, —S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)₂R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R^3 , R^4 , R^5 , R^6 are each independently selected from H and R^{X2};

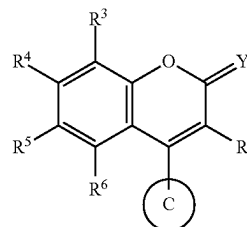
at each occurrence, R^{X1} and R are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷,

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$-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$,
 $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl,
 alkenyl or alkynyl are optionally substituted with one
 or more substituents selected from halogen, $-\text{NO}_2$,
 $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{O}(\text{C}_{3-6} \text{ cyclohydro-}$ 5
 $\text{carbyl})$, $-\text{O}(\text{C}_{1-4} \text{ alkylene-} \text{C}_{3-6} \text{ cyclohydrocarbyl})$,
 $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{O}(\text{C}_{1-4}$
 $\text{alkylene})\text{-(3- to } 7\text{-membered heterocyclyl)}$, $-\text{SH}$,
 $-\text{S}(\text{C}_{1-6} \text{ alkyl})$, $-\text{S}(\text{C}_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(\text{C}_{1-4}$ 10
 $\text{alkylene-} \text{C}_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(3\text{- to } 7\text{-mem-}$
 $\text{bered heterocyclyl})$, $-\text{S}(\text{C}_{1-4} \text{ alkylene})\text{-(3- to } 7\text{-mem-}$
 $\text{bered heterocyclyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$,
 $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{NH}(\text{C}_{3-6} \text{ cyclohydrocarbyl})$, 15
 $-\text{N}(\text{C}_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{NH}(\text{C}_{1-4} \text{ alkylene-}$
 $\text{C}_{3-6} \text{ cyclohydrocarbyl})$, $-\text{N}(\text{C}_{1-4} \text{ alkylene-} \text{C}_{3-6} \text{ cyclo-}$
 $\text{hydrocarbyl})_2$, $-\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$,
 $-\text{N}(3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(\text{C}_{1-4}$ 20
 $\text{alkylene-} 3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{N}(\text{C}_{1-4}$
 $\text{alkylene-} 3\text{- to } 7\text{-membered heterocyclyl})_2$, $=\text{O}$,
 $-\text{COOH}$ and $\text{C}_{1-6} \text{ alkyl}$;
 at each occurrence, R^7 , R^8 are each independently
 selected from H, $\text{C}_{1-6} \text{ alkyl}$, $\text{C}_{2-6} \text{ alkenyl}$, $\text{C}_{2-6} \text{ alkynyl}$, 25
 $\text{C}_{3-6} \text{ cyclohydrocarbyl}$, $\text{C}_{3-6} \text{ cyclohydrocarbyl-} \text{C}_{1-4}$
 alkyl , $3\text{- to } 7\text{-membered heterocyclyl}$, and $3\text{- to } 7\text{-mem-}$
 $\text{bered heterocyclyl-} \text{C}_{1-4} \text{ alkyl}$, wherein the alkyl, alkenyl,
 alkynyl, cyclohydrocarbyl or heterocyclyl are
 optionally substituted with one or more substituents 30
 selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $\text{C}_{1-6} \text{ alkyl}$,
 $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$,
 $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6} \text{ alkyl})$,
 $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, 35
 $-\text{OC}(=\text{O})\text{C}_{1-6} \text{ alkyl}$, $-\text{NHC}(=\text{O})\text{C}_{1-6} \text{ alkyl}$ and
 $-\text{C}(=\text{O})\text{C}_{1-6} \text{ alkyl}$;
 at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each
 independently selected from H, $\text{C}_{1-6}\text{-alkyl}$, $\text{C}_{2-6} \text{ alkenyl}$,
 $\text{C}_{2-6} \text{ alkynyl}$, $\text{C}_{3-6} \text{ cyclohydrocarbyl}$, $\text{C}_{3-6} \text{ cyclo-}$ 40
 $\text{hydrocarbyl-} \text{C}_{1-4} \text{ alkyl}$, $3\text{- to } 7\text{-membered heterocyclyl}$,
 $3\text{- to } 7\text{-membered heterocyclyl-} \text{C}_{1-4} \text{ alkyl}$, $\text{C}_{6-10} \text{ aryl}$,
 $\text{C}_{6-10} \text{ aryl-} \text{C}_{1-4} \text{ alkyl}$, $5\text{- to } 10\text{-membered heteroaryl}$, 5-
 $\text{to } 10\text{-membered heteroaryl-} \text{C}_{1-4} \text{ alkyl}$, $-\text{OR}^{r1}$,
 $-\text{SR}^{r1}$, $-\text{NR}^{r1}\text{R}^{r2}$, $-\text{C}(=\text{O})\text{OR}^{r1}$, $-\text{C}(=\text{O})$ 45
 $\text{NR}^{r1}\text{R}^{r2}$, $-\text{C}(=\text{O})\text{R}^{r1}$, $-\text{S}(=\text{O})_2\text{OR}^{r1}$,
 $-\text{S}(=\text{O})_2\text{R}^{r1}$, $-\text{S}(=\text{O})_2\text{NR}^{r1}\text{R}^{r2}$ and $-\text{S}(=\text{O})\text{R}^{r1}$,
 wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl,
 heterocyclyl, aryl or heteroaryl are optionally substi- 50
 tuted with one or more substituents selected from
 halogen, $=\text{S}$, $-\text{OR}^{r3}$, $-\text{SR}^{r3}$, $-\text{NR}^{r3}\text{R}^{r4}$,
 $-\text{C}(=\text{O})\text{R}^{r3}$, $-\text{C}(=\text{O})\text{OR}^{r3}$ and $-\text{C}(=\text{O})$
 $\text{NR}^{r3}\text{R}^{r4}$;
 at each occurrence, R^{r1} , R^{r2} , R^{r3} , R^{r4} are each indepen- 55
 dently selected from H, $\text{C}_{1-8} \text{ alkyl}$, $\text{C}_{3-10} \text{ cyclohydro-}$
 carbyl , $\text{C}_{3-10} \text{ cyclohydrocarbyl-} \text{C}_{1-4} \text{ alkyl}$, 3- to
 $10\text{-membered heterocyclyl}$, $3\text{- to } 10\text{-membered hetero-}$
 $\text{cyclyl-} \text{C}_{1-4} \text{ alkyl}$, $\text{C}_{6-10} \text{ aryl}$, $\text{C}_{6-10} \text{ aryl-} \text{C}_{1-4} \text{ alkyl}$, 5- to 60
 $10\text{-membered heteroaryl}$, and $5\text{- to } 10\text{-membered hetero-}$
 $\text{aryl-} \text{C}_{1-4} \text{ alkyl}$, wherein the alkyl, alkenyl, alkynyl,
 cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are
 optionally substituted with one or more substituents
 selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $\text{C}_{1-8} \text{ alkyl}$, C_{2-8} 65
 alkenyl , $\text{C}_{2-8} \text{ alkynyl}$, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$, $=\text{O}$ and
 $-\text{COOH}$;

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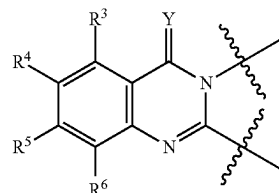
ring A and R^1 are as defined in formula (III);
 provided that the compound of formula (VI) does not
 include the following structure:



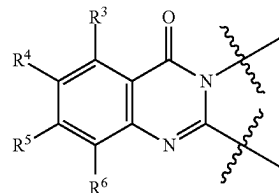
wherein R^4 is selected from $-\text{O}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{O}(\text{C}_{1-4}$
 $\text{alkylene-} \text{C}_{6-10} \text{ aryl})$.

In one embodiment, ring B is a saturated or unsaturated
 6-membered heterocycle, wherein heterocycle contains 1 or
 2 heteroatoms each independently selected from N and O. In
 another embodiment, ring B is dihydropyrimidine. In a
 preferred embodiment, ring B is selected from 1,6-dihydro-
 pyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimi-
 dine.

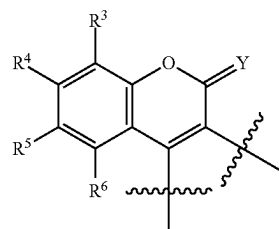
In a more preferred embodiment, the A-B ring system is



wherein Y is O or S. In a particular embodiment, the A-B
 ring system is

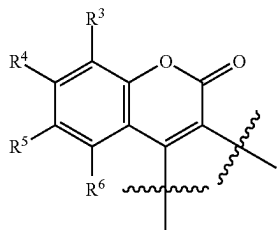


In yet another embodiment, ring B is 2H-pyran or
 4H-pyran. In a preferred embodiment, A-B ring system is



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wherein Y is O or S. In a particular embodiment, A-B ring system is



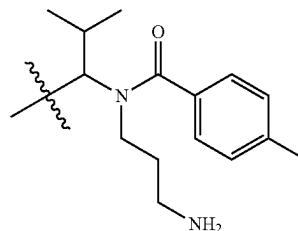
In yet another embodiment, ring C is phenyl, wherein phenyl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} .

In one embodiment, L^1 is a bond. In another embodiment, L^1 is a C_1 - C_6 hydrocarbon chain. In a preferred embodiment, L^1 is C_1 - C_2 hydrocarbon chain.

In yet another embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $=\text{O}$, $=\text{S}$, $-\text{OR}^{a1}$, $-\text{SR}^{a1}$, $-\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{OR}^{a1}$, $-\text{C}(=\text{O})\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{OR}^{a1}$, $-\text{S}(=\text{O})_2\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{NR}^{a1}\text{R}^{b1}$ and $-\text{S}(=\text{O})\text{R}^{a1}$, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{OR}^{a2}$, $-\text{S}(=\text{O})_2\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{NR}^{a2}\text{R}^{b2}$ and $-\text{S}(=\text{O})\text{R}^{a2}$. In another embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, $=\text{O}$, $=\text{S}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $-\text{OH}$, $-\text{O}(C_{1-6} \text{ alkyl})$, $-\text{O}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{O}(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{O}(C=\text{O})(C_{1-6} \text{ alkyl})$, $-\text{O}(C=\text{O})(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(C=\text{O})(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(C=\text{O})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{O}(C=\text{O})(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{SH}$, $-\text{S}(C_{1-6} \text{ alkyl})$, $-\text{S}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{S}(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{NH}_2$, $-\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{NH}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{N}(C_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{NH}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{N}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{NH}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{N}(3- \text{ to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{N}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(C=\text{O})(C_{1-6} \text{ alkyl})$, $-\text{NH}(C=\text{O})(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{NH}(C=\text{O})(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{NH}(C=\text{O})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})(C_{1-6} \text{ alkyl})$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{O}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{C}(=\text{O})\text{O}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{C}(=\text{O})\text{O}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{O}(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{O}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{C}(=\text{O})\text{NH}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{C}(=\text{O})\text{N}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{C}(=\text{O})\text{NH}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{N}(3- \text{ to } 7\text{-membered heterocyclyl})_2$,

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$-\text{C}(=\text{O})\text{NH}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})$, and $-\text{C}(=\text{O})\text{N}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})_2$, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl or are optionally substituted with one or more substituents selected from halogen, nitro, cyano, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$ and $-\text{COOH}$. In yet another embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $-\text{OR}^{a1}$, $-\text{SR}^{a1}$, $-\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{OR}^{a1}$, $-\text{C}(=\text{O})\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{OR}^{a1}$, $-\text{S}(=\text{O})_2\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{NR}^{a1}\text{R}^{b1}$ and $-\text{S}(=\text{O})\text{R}^{a1}$, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$ and $-\text{C}(=\text{O})\text{R}^{a2}$. In a preferred embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a more preferred embodiment, R^2 is selected from H, halogen, C_{1-6} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a further embodiment, R^2 is selected from H, C_{1-4} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from $-\text{NR}^{a2}\text{R}^{b2}$. In a particular embodiment, R^2 is selected from H, C_{1-4} alkyl and $-\text{OH}$, wherein the alkyl is $-\text{CH}_2[\text{CH}(\text{CH}_3)_2]$ and is optionally substituted with one or more substituents selected from $-\text{NR}^{b2}$. In another embodiment, R^2 is an alkyl substituted with $-\text{NR}^{a1}\text{R}^{b1}$. In a particular embodiment, R^2 is



In one embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OW}$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(C_{1-6} \text{ alkyl})$, $-\text{O}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{O}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{O}(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{SH}$, $-\text{S}(C_{1-6} \text{ alkyl})$, $-\text{S}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{S}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{S}(C_{1-4} \text{ alkylene-})(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{NH}_2$, $-\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{NH}(C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{N}(C_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{NH}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})$, $-\text{N}(C_{1-4} \text{ alkylene-}C_{3-6} \text{ cyclohydrocarbyl})_2$, $-\text{NH}(3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{N}(3- \text{ to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})$, $-\text{N}(C_{1-4} \text{ alkylene-}3- \text{ to } 7\text{-membered heterocyclyl})_2$,

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7-membered heterocyclyl)₂, =O and —COOH. In a preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OR⁷, —SR⁷, —NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —OS(=O)₂R⁷ and —NS(=O)₂R⁷R⁸, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), and —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂. In a preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, C₁₋₆ alkyl, —OR⁷ and —NR⁷R⁸, more preferably each independently selected from halogen and —OR⁷, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂. In a more preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from F, Cl, Br, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, and, optionally substituted C₁₋₆ alkyl, wherein the optionally substituted C₁₋₆ alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂. In a particular embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from F, Cl, Br, methyl, —OH and dimethylaminomethyl.

At each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl and C₂₋₆ alkynyl, wherein the alkyl, alkenyl, alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl). In a preferred embodiment, at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl), and —C(=O)(C₁₋₆ alkyl). In a particular embodiment, at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —N(C₁₋₆ alkyl)₂ and —COOH. In another particular embodiment, at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —N(C₁₋₆ alkyl)₂.

In one embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷² and —C(=O)R⁷¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, het-

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erocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR⁷³, —SR⁷³, —NR⁷³R⁷⁴, —C(=O)R⁷³, —C(=O)OR⁷³ and —C(=O)NR⁷³R⁷⁴. In a preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷² and —C(=O)R⁷¹, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR⁷³ and —NR⁷³R⁷⁴. In another preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷² and —C(=O)R⁷¹, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR⁷³ and —NR⁷³R⁷⁴. In a more preferred embodiment, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR⁷¹, —NR⁷¹R⁷² and —C(=O)R⁷¹, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR⁷³ and —NR⁷³R⁷⁴. In a further embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR⁷¹, —NR⁷¹R⁷² and —C(=O)R⁷¹, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NR⁷³R⁷⁴. In a particular embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₃ alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NH₂.

In another embodiment, at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl. In a preferred embodiment, at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₆ alkyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl. In a more preferred embodiment, at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C₁₋₆ alkyl. In a more preferred embodiment, at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl and phenyl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C₁₋₆ alkyl. In a particular embodiment, at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H and p-methylphenyl.

At each occurrence, R^7 , R^8 are each independently selected from H, C_{1-6} alkyl, C_{2-6} alkenyl and C_{2-6} alkynyl, wherein the alkyl, alkenyl, alkynyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(C_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{C}(=\text{O})\text{O}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{OC}(=\text{O})(C_{1-6} \text{ alkyl})$, $-\text{NHC}(=\text{O})(C_{1-6} \text{ alkyl})$ and $-\text{C}(=\text{O})(C_{1-6} \text{ alkyl})$. In a preferred embodiment, at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(C_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{C}(=\text{O})\text{O}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}(C_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(C_{1-6} \text{ alkyl})_2$, $-\text{OC}(=\text{O})(C_{1-6} \text{ alkyl})$, $-\text{NHC}(=\text{O})(C_{1-6} \text{ alkyl})$ and $-\text{C}(=\text{O})(C_{1-6} \text{ alkyl})$. In a particular embodiment, at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and $-\text{N}(C_{1-6} \text{ alkyl})_2$. In another particular embodiment, at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl, wherein alkyl is optionally substituted with one or more substituents selected from halogen and $-\text{N}(C_{1-6} \text{ alkyl})_2$.

In one embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered hetero aryl- C_{1-4} alkyl, $-OR^{j1}$, $-SR^{j1}$, $-NR^{j1}R^{j2}$, $-C(=O)OR^{j1}$, $-C(=O)NR^{j1}R^{j2}$ and $-C(=O)R^{j1}$, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, $-OR^{j3}$, $-SR^{j3}$, $-NR^{j3}R^{j4}$, $-C(=O)R^{j3}$, $-C(=O)OR^{j3}$ and $-C(=O)NR^{j3}R^{j4}$. In a preferred embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered heteroaryl- C_{1-4} alkyl, $-OR^{j1}$, $-SR^{j1}$, $-NR^{j1}R^{j2}$, $-C(=O)OR^{j1}$, $-C(=O)NR^{j1}R^{j2}$ and $-C(=O)R^{j1}$, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, $-OR^3$ and $-NR^{j3}R^{j4}$. In another preferred embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, $-OR^{j1}$, $-NR^{j1}R^{j2}$, $-C(=O)OR^{j1}$, $-C(=O)NR^{j1}R^{j2}$ and $-C(=O)R^{j1}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-OR^3$ and $-NR^{j3}R^{j4}$. In a more preferred embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, $-OR^{j1}$, $-NR^{j1}R^{j2}$ and $-C(=O)R^{j1}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-OR^3$ and $-NR^{j3}R^{j4}$. In a further embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, $-OR^{j1}$, $-NR^{j1}R^{j2}$ and $-C(=O)R^{j1}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and $-NR^{j3}R^{j4}$. In a particular embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-3} alkyl, $-OH$ and p -methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from halogen and $-NH_2$.

In another embodiment, at each occurrence, R¹¹, R¹², R¹³, R¹⁴ are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl,

3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbonyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂ and C₁₋₆ alkyl. In a preferred embodiment, at each occurrence, R^{R1}, R^{R2}, R^{R3}, R^{R4} are each independently selected from H, C₁₋₆ alkyl, C₃₋₆ cyclohydrocarbonyl, C₃₋₆ cyclohydrocarbonyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbonyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂ and C₁₋₆ alkyl. In a more preferred embodiment, at each occurrence, R^{R1}, R^{R2}, R^{R3}, R^{R4} are each independently selected from H, C₁₋₆ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂ and C₁₋₆ alkyl. In a more preferred embodiment, at each occurrence, R^{R1}, R^{R2}, R^{R3}, R^{R4} are each independently selected from H, C₁₋₆ alkyl, phenyl and phenyl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C₁₋₆ alkyl. In a particular embodiment, at each occurrence, R^{R1}, R^{R2}, R^{R3}, R^{R4} are each independently selected from H and p-methylphenyl.

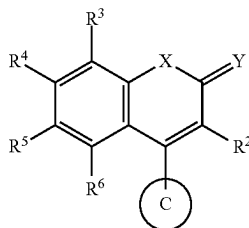
In one embodiment, R^3 , R^6 are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$ and $-\text{O}(\text{C}_{1-6}$ alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, and $-\text{O}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl). In a preferred embodiment, R^3 , R^6 are each independently selected from H, halogen and $-\text{OH}$, more preferably is H or $-\text{OH}$.

In one embodiment, R^4 , R^5 are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl) and $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{O}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{NH}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{NH}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{NH}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl), $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{OC}(=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{NHC}(=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})(\text{C}_{1-6}$ alkyl) and $=\text{O}$. In another embodiment, R^4 , R^5 are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl) and $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6}$ alkyl) $_2$, $-\text{OC}(=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{NHC}(=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{C}(=\text{O})(\text{C}_{1-6}$ alkyl) and $=\text{O}$. In another embodiment, R^4 , R^5 are each independently selected

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from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$ and $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{OC}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$, $-\text{NHC}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$ and $=\text{O}$. In another embodiment, R^4 , R^5 are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{OC}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$, $-\text{NHC}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})(\text{C}_{1-6} \text{ alkyl})$ and $=\text{O}$.

The structure described in the following formula (VII) falls within the scope of the structure described in formula (I), formula (III) and formula (VI). In another aspect of the present invention, provided is use of a compound of formula (VII), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder.

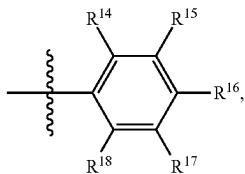


wherein

X is O;

Y, R^3 , R^4 , R^5 , R^6 are as defined in formula (VI);

ring C is

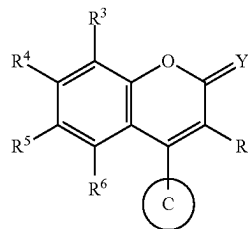


wherein R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$; wherein R^7 , R^8 are as defined in formula (VI); R^2 is selected from H, halogen, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl- C_{1-4} alkyl, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$,

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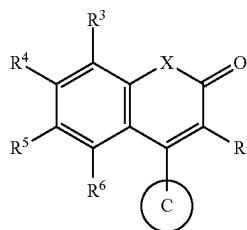
$-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{OR}^{a2}$, $-\text{S}(=\text{O})_2\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{NR}^{a2}\text{R}^{b2}$ and $-\text{S}(=\text{O})\text{R}^{a2}$; wherein R^{a2} , R^{b2} , R^{c2} are as defined in formula (VI);

provided that the compound of formula (VII) does not include the following structure:



wherein R^4 is selected from $-\text{O}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{O}(\text{C}_{1-4} \text{ alkylene-C}_{6-10} \text{ aryl})$.

In one embodiment, Y is O. When Y is O, formula (VII) is



In one embodiment, R^2 is selected from H, halogen and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{OR}^{a2}$, $-\text{S}(=\text{O})_2\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{NR}^{a2}\text{R}^{b2}$ and $-\text{S}(=\text{O})\text{R}^{a2}$. In one embodiment, R^2 is selected from $-\text{OR}^{a1}$, $-\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{OR}^{a1}$ and $-\text{C}(=\text{O})\text{NR}^{a1}\text{R}^{b1}$. In another embodiment, R^2 is selected from H, halogen and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{NH}_2$ and $-\text{COOH}$.

In one embodiment, at each occurrence, R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl- C_{1-4} alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6} \text{ alkyl})_2$. In a preferred embodiment, at each occurrence, R^{a2} , R^{b2} , R^{c2} are each independently selected from H and C_{1-6} -alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$, $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{C}(=\text{O})\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{C}(=\text{O})\text{N}(\text{C}_{1-6} \text{ alkyl})_2$.

In one embodiment, R^3 is selected from H, halogen, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$.

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substituent(s) selected from halogen, —OH, —OCH₃, —NHCH₃ and —N(CH₃)₂. In one embodiment, R³ is selected from H, halogen, —OH, —NH₂ and methyl, wherein the methyl is substituted by —N(CH₃)₂. In yet another embodiment, R³ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂. In one embodiment, R³ is selected from H, halogen, —OH, —OCH₃, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —OCH₃, —NHCH₃ and —N(CH₃)₂. In one embodiment, R³ is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with —N(CH₃)₂. In another embodiment, R³ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —NH₂, —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂. In one embodiment, R³ is selected from H, halogen, —OH and C₁₋₂ alkyl, wherein the alkyl is substituted with one or more substituents selected from halogen, —NH(C₁₋₂ alkyl) and —N(C₁₋₂ alkyl)₂. In one embodiment, R³ is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with substituent(s) selected from selected from halogen, —NH(C₁₋₂ alkyl) and —N(C₁₋₂ alkyl)₂. In one embodiment, R³ is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with substituent(s) selected from halogen, —NHCH₃ and —N(CH₃)₂. In one embodiment, R³ is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with —N(CH₃)₂. In yet another embodiment, R³ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl), and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂ and —NH(C₁₋₄ alkyl). In one embodiment, R³ is selected from H, halogen, —OH, —OCH₃, —NH₂, —NHCH₃, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —O(C₁₋₂ alkyl) and —NH(C₁₋₂ alkyl). In one embodiment, R³ is selected from H, halogen, —OH, —OCH₃, —NH₂, —NHCH₃, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —OCH₃ and —NHCH₃. In one embodiment, R³ is selected from H, halogen, —OH, —NH₂ and methyl, wherein the methyl is substituted by —N(CH₃)₂. In yet another embodiment, R³ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂ and —NH(C₁₋₄ alkyl). In one embodiment, R³ is selected from H, halogen, —OH, —OCH₃, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —O(C₁₋₂ alkyl) and —NH(C₁₋₂ alkyl). In one embodiment, R³ is selected from H, halogen, —OH, —OCH₃, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —OCH₃ and —NHCH₃. In one embodiment, R³ is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with —N(CH₃)₂. In another embodiment, R³ is selected from H, halogen, —OH, —O(C₁₋

In one embodiment, R⁵ is selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl). In one embodiment, R⁵ is selected from H, halogen, C₁₋₄ alkyl, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂. In one embodiment, R⁵ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl), —N(C₁₋₄ alkyl)₂, —C(=O)O(C₁₋₄ alkyl), —C(=O)NH₂, —C(=O)NH(C₁₋₄ alkyl), —C(=O)N(C₁₋₄ alkyl)₂, —OC(=O)(C₁₋₄ alkyl), —NC(=O)(C₁₋₄ alkyl)₂, —C(=O)(C₁₋₄ alkyl), and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl), —N(C₁₋₄ alkyl)₂, —C(=O)O(C₁₋₄ alkyl), —C(=O)NH₂, —C(=O)NH(C₁₋₄ alkyl), —C(=O)N(C₁₋₄ alkyl)₂, —OC(=O)(C₁₋₄ alkyl), —NC(=O)(C₁₋₄ alkyl)₂ and —C(=O)(C₁₋₄ alkyl). In one embodiment, R⁵ is selected from H, halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl), —N(C₁₋₄ alkyl)₂, and substituted C₁₋₄ alkyl, wherein the substituted C₁₋₄ alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₄ alkyl), —NH₂, —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂. In one embodiment, R⁵ is selected from H, halogen, —O(C₁₋₄ alkyl), —NH(C₁₋₄ alkyl), —N(C₁₋₄ alkyl)₂, and substituted C₁₋₂ alkyl, wherein the substituted C₁₋₂ alkyl is substituted with one or more substituents selected from halogen, —O(C₁₋₄ alkyl), —NH(C₁₋₄ alkyl) and —N(C₁₋₄ alkyl)₂. In one embodiment, R⁵ is selected from H, halogen, —O(C₁₋₄ alkyl), —NH(C₁₋₄ alkyl), —N(C₁₋₄ alkyl)₂, and substituted C₁₋₂ alkyl, wherein the substituted C₁₋₂ alkyl is substituted with one or more substituents selected from halogen, —OCH₃, —NHCH₃ and —N(CH₃)₂. In one

embodiment, R⁵ is selected from H, halogen, —O(C₁₋₂ alkyl), —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂, and substituted C₁₋₂ alkyl, wherein the substituted C₁₋₂ alkyl is substituted with one or more substituents selected from halogen, —OCH₃, —NHCH₃ and —N(CH₃)₂).

In one embodiment, R^6 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$. In another embodiment, R^6 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-6} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$ and $-\text{NH}_2$. In one embodiment, R^6 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-2} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-2} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-2} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-2} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$ and $-\text{NH}_2$. In yet another embodiment, R^6 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-4} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-4} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-4} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-4} \text{ alkyl})_2$, wherein the alkyl is optionally substituted with one or more halogen. In another embodiment, R^6 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-2} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-2} \text{ alkyl})$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-4} \text{ alkyl})$ and $-\text{N}(\text{C}_{1-4} \text{ alkyl})_2$. In one embodiment, R^6 is selected from H, F, Cl, Br, C_{1-6} alkyl, $-\text{OH}$ and $-\text{NH}_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$ and $-\text{NH}_2$. In another embodiment, R^6 is selected from H, F, Cl, Br, methyl, $-\text{OH}$ and $-\text{NH}_2$. In a particular embodiment, R^6 is H or $-\text{OH}$, especially H.

In one embodiment, R^3 is methyl, and R^4 is selected from $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-3} \text{ alkyl})$, $-\text{N}(\text{C}_{1-3} \text{ alkyl})_2$, and C_{1-3} alkyl substituted with $-\text{COOH}$, wherein the substituted C_{1-3} alkyl is substituted with one or more substituents selected from $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-2} \text{ alkyl})$, $-\text{N}(\text{C}_{1-2} \text{ alkyl})_2$ and $-\text{COOH}$. In one embodiment, R^3 is methyl, and R^4 is selected from $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-3} \text{ alkyl})$, $-\text{N}(\text{C}_{1-3} \text{ alkyl})_2$, and C_{1-3} alkyl substituted with $-\text{COOH}$, wherein the substituted C_{1-3} alkyl is substituted with one or more substituents selected from $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-2} \text{ alkyl})$, $-\text{N}(\text{C}_{1-2} \text{ alkyl})_2$, and $-\text{COOH}$.

In another embodiment, R³ is methyl and R⁵ is selected from —OH, —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH. In another embodiment, R³ is methyl, and R⁵ is selected from —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂, and —COOH.

In one embodiment, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6} \text{ alkyl})$, $-\text{O}(\text{C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{O}(\text{C}_{1-4} \text{ alkylene-C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{O}(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{O}(\text{C}_{1-4} \text{ alkylene})\text{-}(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{O}(\text{C}=\text{O})(\text{C}_{1-6} \text{ alkyl})$, $-\text{O}(\text{C}=\text{O})(\text{C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4} \text{ alkylene-C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{O}(\text{C}=\text{O})(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4} \text{ alkylene})\text{-}(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{SH}$, $-\text{S}(\text{C}_{1-6} \text{ alkyl})$, $-\text{S}(\text{C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{S}(\text{C}_{1-4} \text{ alkylene-C}_{3-6} \text{ cyclohydrocarb-yl})$, $-\text{S}(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{S}(\text{C}_{1-4} \text{ alkylene})\text{-}(3\text{- to } 7\text{-membered heterocyc-yl})$, $-\text{NH}_2$, $-\text{NH}$

(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, —NH(C=O)(C₁₋₆ alkyl), —NH(C=O)(C₃₋₆ cyclohydrocarbyl), —NH(C=O)(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —NH(C=O)(3- to 7-membered heterocyclyl), —C(=O)(C₁₋₆ alkyl), —C(=O)O(C₁₋₆ alkyl), —C(=O)O(C₃₋₆ cyclohydrocarbyl), —C(=O)O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —C(=O)O(3- to 7-membered heterocyclyl), —C(=O)O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —C(=O)NH₂, —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —C(=O)NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —C(=O)N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —C(=O)NH(3- to 7-membered heterocyclyl), —C(=O)N(3- to 7-membered heterocyclyl)₂, —C(=O)NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), and —C(=O)N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from H, halogen, nitro, cyano, —OH, —SH, —NH₂, =O and —COOH. In one embodiment, R¹⁴, R¹⁵, R¹⁶, R¹⁷, R¹⁸ are each independently selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸ and —C(=O)R⁷. In one embodiment, R¹⁴, R¹⁵, R¹⁶, R¹⁷, R¹⁸ are each independently selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C₁₋₆ alkyl), —O(C=O)(C₃₋₆ cyclohydrocarbyl), —O(C=O)(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, —NH(C=O)(C₁₋₆ alkyl), —NH(C=O)(C₃₋₆ cyclohydrocarbyl), —NH(C=O)(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R¹⁴, R¹⁵, R¹⁷, R¹⁸ are each independently selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C₁₋₆ alkyl), —O(C=O)(C₃₋₆ cyclohydrocarbyl), —O(C=O)(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, —NH(C=O)(C₁₋₆ alkyl), —NH(C=O)(C₃₋₆ cyclohydrocarbyl), —NH(C=O)(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R¹⁴, R¹⁵

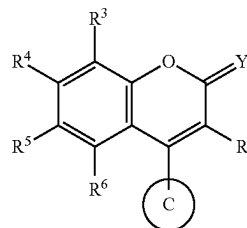
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R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{O}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(\text{C}=\text{O})(3\text{- to } 7\text{-membered heterocyclyl})$, and $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl). In one embodiment, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{O}(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(\text{C}=\text{O})(3\text{- to } 7\text{-membered heterocyclyl})$, and $-\text{O}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene)-(3- to 7-membered heterocyclyl). In one embodiment, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl)₂, $-\text{NH}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{N}(\text{C}_{3-6}$ cyclohydrocarbyl)₂, $-\text{NH}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{N}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl)₂, $-\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{N}(3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl), $-\text{N}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl)₂, $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{NH}(\text{C}=\text{O})(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), and $-\text{NH}(\text{C}=\text{O})(3\text{- to } 7\text{-membered heterocyclyl})$. In one embodiment, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl)₂, $-\text{NH}(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{N}(\text{C}_{3-6}$ cyclohydrocarbyl)₂, $-\text{NH}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{N}(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl)₂, $-\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{N}(3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl), $-\text{N}(\text{C}_{1-4}$ alkylene-3- to 7-membered heterocyclyl)₂, $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl), $-\text{NH}(\text{C}=\text{O})(\text{C}_{3-6}$ cyclohydrocarbyl), $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl) and $-\text{NH}(\text{C}=\text{O})(3\text{- to } 7\text{-membered heterocyclyl})$. In one embodiment, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl)₂ and $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl). In one embodiment, R^{14} , R^{15} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl)₂ and $-\text{NH}(\text{C}=\text{O})(\text{C}_{1-6}$ alkyl). In one embodiment, R^{16} is H. In one embodiment, R^{16} is $-\text{OCH}_3$.

The structure described in the following formula (VIII) falls within the scope of the structure described in formula (I), formula (III) and formula (VII). In another aspect of the present invention, provided is a use of a compound of formula (VIII), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

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(VIII)

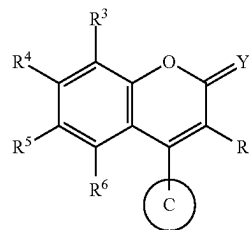


wherein

ring C, Y, R^2 , R^3 , R^4 , R^5 , R^6 are as defined in formula (VII).

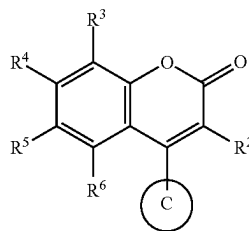
In one embodiment, ring C, Y, R^2 , R^3 , R^4 , R^5 , R^6 in formula (VIII) are as defined in formula (VII);

provided that the compound of formula (VIII) does not include the following structure:



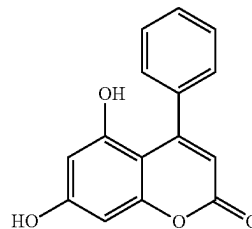
wherein R^4 is selected from $-\text{O}(\text{C}_{1-6}$ alkyl) and $-\text{O}(\text{C}_{1-4}$ alkylene- C_{6-10} aryl).

In one embodiment, Y is O. When Y is O, formula (VIII) is



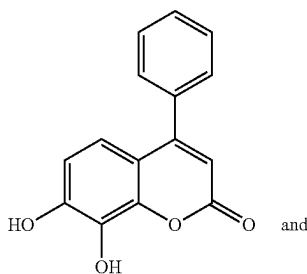
In a particular embodiment, the compound of formula (VIII) is selected from:

(compound 2)

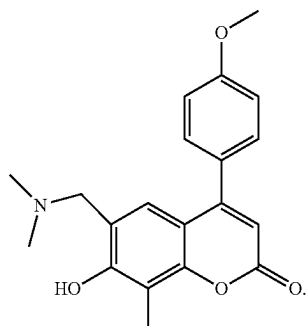


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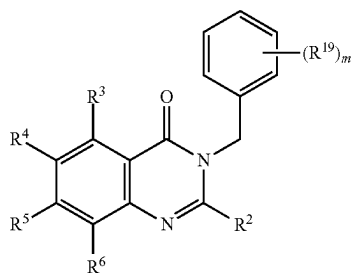


(compound 6)



(compound 7)

The structure described in the following formula (IX) falls within the scope of the structure described in formula (I), formula (III) and formula (VI). In another aspect of the present invention, there is provided use of a compound of formula (IX), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder,



(IX)

wherein:

at each occurrence, R^{19} is each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl), $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$ and $-\text{COOH}$;

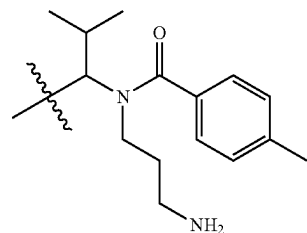
m is 0, 1, 2, 3, 4 or 5;

R^2 , R^3 , R^4 , R^5 , R^6 are as defined in formula (VI).

In one embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyc-

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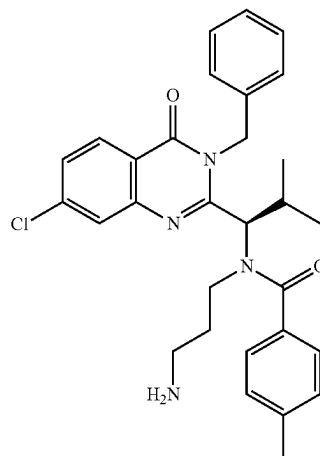
clyl, 3- to 7-, membered heterocyclyl- C_{1-4} alkyl, $-\text{OR}^{a1}$, $-\text{SR}^{a1}$, $-\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{OR}^{a1}$, $-\text{C}(=\text{O})\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{OR}^{a1}$, $-\text{S}(=\text{O})_2\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{NR}^{a1}\text{R}^{b1}$ and $-\text{S}(=\text{O})\text{R}^{a1}$, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^{a2}$, SR^{a2} , $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$ and $-\text{C}(=\text{O})\text{R}^{a2}$. In a preferred embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a more preferred embodiment, R^2 is selected from H, halogen, C_{1-6} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a further embodiment, R^2 is selected from H, C_{1-4} alkyl and $-\text{OH}$, wherein the alkyl is optionally substituted with one or more substituents selected from $-\text{NR}^{a2}\text{R}^{b2}$. In a particular embodiment, R^2 is selected from H, C_{1-4} alkyl and $-\text{OH}$, wherein the alkyl is $-\text{CH}_2[\text{CH}(\text{CH}_3)_2]$, and is optionally substituted with one or more substituents selected from $-\text{NR}^{a2}\text{R}^{b2}$. In another embodiment, R^2 is an alkyl substituted with $-\text{NR}^{a1}\text{R}^{b1}$, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are as defined in formula (VI). In a particular embodiment, R^2 is



In one embodiment, at each occurrence, R^{19} is each independently selected from halogen, C_{1-6} alkyl, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl) and $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{OH}$, $-\text{O}(\text{C}_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(\text{C}_{1-6}$ alkyl) and $-\text{N}(\text{C}_{1-6}$ alkyl) $_2$.

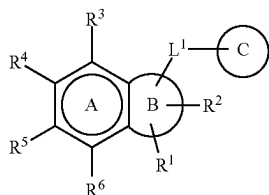
In a particular embodiment, the compound of formula (IX) is

(compound 1)



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In another aspect of the present invention, provided is use of a compound of formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder



wherein:

ring A is a benzene ring;

ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

ring C is C₆₋₁₀ aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or is a C₁-C₆ hydrocarbon chain;

R¹ is =Y, wherein Y is O or S;

R² is selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)NR^{a2}R^{b2}, —S(=O)R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, =O, —COOH and C₁₋₆ alkyl;

at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄

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alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C₁₋₄ alkyl;

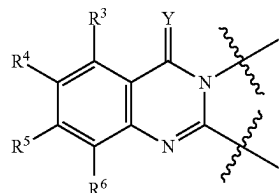
at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷², —C(=O)R⁷¹, —S(=O)₂OR⁷¹, —S(=O)₂R⁷¹, —S(=O)NR⁷¹R⁷² and —S(=O)R⁷¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from =O, =S, —OR⁷³, —SR⁷³, —NR⁷³R⁷⁴, —C(=O)R⁷³, —C(=O)OR⁷³ and —C(=O)NR⁷³R⁷⁴;

at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₈ alkyl, —C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —OH, —SH, —NH₂, =O and —COOH.

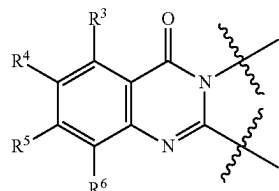
In one embodiment, R¹ is =O. In another embodiment, R¹ is =S.

In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

In a more preferred embodiment, A-B ring system is

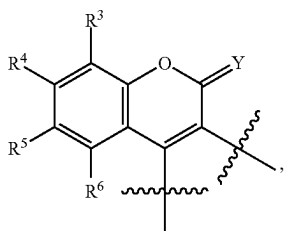


wherein Y is O or S. In a particularly preferred embodiment, A-B ring system is

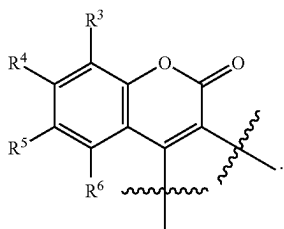


In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, A-B ring system is

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wherein Y is O or S. In a particularly preferred embodiment, A-B ring system is



In yet another embodiment, ring C is phenyl, which is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} .

In one embodiment, L^1 is a bond. In another embodiment, L^1 is a C_{1-6} hydrocarbon chain. In a preferred embodiment, L^1 is a C_{1-2} hydrocarbon chain.

In yet another embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $=\text{O}$, $=\text{S}$, $-\text{OR}^{a1}$, $-\text{SR}^{a1}$, $-\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{OR}^{a1}$, $-\text{C}(=\text{O})\text{NR}^{a1}\text{R}^{b1}$, $-\text{C}(=\text{O})\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{OR}^{a1}$, $-\text{S}(=\text{O})_2\text{R}^{a1}$, $-\text{S}(=\text{O})_2\text{NR}^{a1}\text{R}^{b1}$ and $-\text{S}(=\text{O})\text{R}^{a1}$, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, $=\text{O}$, $=\text{S}$, $-\text{OR}^{a2}$, $-\text{SR}^{a2}$, $-\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{OR}^{a2}$, $-\text{C}(=\text{O})\text{NR}^{a2}\text{R}^{b2}$, $-\text{C}(=\text{O})\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{OR}^{a2}$, $-\text{S}(=\text{O})_2\text{R}^{a2}$, $-\text{S}(=\text{O})_2\text{NR}^{a2}\text{R}^{b2}$ and $-\text{S}(=\text{O})\text{R}^{a2}$. In a preferred embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, $=\text{O}$, $=\text{S}$, $-\text{COOH}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, $-\text{OH}$, $-\text{O}(C_{1-6}$ alkyl), $-\text{O}(C_{3-6}$ cyclohydrocarbyl), $-\text{O}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{O}(C_{1-4}$ alkylene- $(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{SH}$, $-\text{S}(C_{1-6}$ alkyl), $-\text{S}(C_{3-6}$ cyclohydrocarbyl), $-\text{S}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{S}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{S}(C_{1-4}$ alkylene- $(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{NH}_2$, $-\text{NH}(C_{1-6}$ alkyl), $-\text{N}(C_{1-6}$ alkyl) $_2$, $-\text{NH}(C_{3-6}$ cyclohydrocarbyl), $-\text{N}(C_{3-6}$ cyclohydrocarbyl) $_2$, $-\text{NH}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{N}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl) $_2$, $-\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{N}(3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{NH}(C_{1-4}$ alkylene- $3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{N}(C_{1-4}$ alkylene- $3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{C}(=\text{O})(C_{1-6}$ alkyl), $-\text{COOH}$, $-\text{C}(=\text{O})\text{O}(C_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{O}(C_{3-6}$ cyclohydrocarbyl), $-\text{C}(=\text{O})\text{O}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{C}(=\text{O})\text{O}(3\text{- to } 7\text{-membered heterocyclyl})$,

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$-\text{C}(=\text{O})\text{O}(C_{1-4}$ alkylene- $(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{NH}_2$, $-\text{C}(=\text{O})\text{NH}(C_{1-6}$ alkyl), $-\text{C}(=\text{O})\text{N}(C_{1-6}$ alkyl) $_2$, $-\text{C}(=\text{O})\text{NH}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl), $-\text{C}(=\text{O})\text{N}(C_{1-4}$ alkylene- C_{3-6} cyclohydrocarbyl) $_2$, $-\text{C}(=\text{O})\text{NH}(3\text{- to } 7\text{-membered heterocyclyl})$, $-\text{C}(=\text{O})\text{N}(3\text{- to } 7\text{-membered heterocyclyl})_2$, $-\text{C}(=\text{O})\text{NH}(C_{1-4}$ alkylene- $3\text{- to } 7\text{-membered heterocyclyl})$, and $-\text{C}(=\text{O})\text{N}(C_{1-4}$ alkylene- $3\text{- to } 7\text{-membered heterocyclyl})_2$, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, nitro, cyano, $-\text{OH}$, $-\text{SH}$, $-\text{NH}$, $=\text{O}$ and $-\text{COOH}$. In another preferred embodiment, R^2 is selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$ and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a more preferred embodiment, R^2 is selected from H, halogen and C_{1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{1-6} alkyl and $-\text{NR}^{a2}\text{R}^{b2}$. In a further preferred embodiment, R^2 is H or C_{1-4} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from $-\text{NR}^{a2}\text{R}^{b2}$. In a particularly preferred embodiment, R^2 is H or C_{1-4} alkyl, wherein the alkyl is $-\text{CH}[\text{CH}(\text{CH}_3)_2]-$ and is optionally substituted with one or more substituents selected from $-\text{NR}^{a2}\text{R}^{b2}$.

In one embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$, $=\text{O}$, $-\text{COOH}$ and C_{1-6} alkyl. In a preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$ and $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$. In a preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, C_{1-6} alkyl, $-\text{OR}^7$ and $-\text{NR}^7\text{R}^8$. In a more preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, C_{1-6} alkyl, $-\text{OR}^7$ and $-\text{NR}^7\text{R}^8$. In a further preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen and $-\text{OR}^7$. In a more preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from C_{1-6} alkyl, $-\text{OH}$ and $-\text{O}(C_{1-6}$ alkyl). In a particularly preferred embodiment, at each occurrence, R^{X1} and R^{X2} are each independently selected from Cl and $-\text{OH}$.

At each occurrence, R^7 , R^8 are each independently selected from H, C_{1-6} alkyl, C_{2-6} alkenyl and C_{2-6} alkynyl. In a preferred embodiment, at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl. In a particularly preferred embodiment, at each occurrence, R^7 , R^8 are each independently selected from H and C_{1-6} alkyl.

In one embodiment, at each occurrence, R^{a1} , R^{b1} , R^{c1} , R^{a2} , R^{b2} , R^{c2} are each independently selected from H, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, C_{3-6} cyclohydrocarbyl, C_{3-6} cyclohydrocarbyl- C_{1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl- C_{1-4} alkyl, C_{6-10} aryl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl- C_{1-4} alkyl, OR^{r1} , SR^{r1} , $-\text{NR}^{r1}\text{R}^{r2}$, $-\text{C}(=\text{O})\text{OR}^{r1}$, $-\text{C}(=\text{O})\text{NR}^{r1}\text{R}^{r2}$, $-\text{C}(=\text{O})\text{R}^{r1}$, $-\text{S}(=\text{O})_2\text{OR}^{r1}$, $-\text{S}(=\text{O})_2\text{R}^{r1}$, $-\text{S}$

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(=O)₂NR¹R² and —S(=O)R¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from =O, =S, —OR¹³, —SR¹³, —NR¹³R¹⁴, —C(=O)R¹³, —C(=O)OR¹³ and —C(=O)NR¹³R¹⁴. In a preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR¹, —SR¹, —NR¹R², —C(=O)OR¹, —C(=O)NR¹R² and —C(=O)R¹, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from —R¹³ and —NR¹³R¹⁴. In another preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR¹, —NR¹R², —C(=O)OR¹, —C(=O)NR¹R² and —C(=O)R¹, wherein the alkyl is optionally substituted with one or more substituents selected from —R¹³ and —NR¹³R¹⁴. In a more preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR¹, —NR¹R² and —C(=O)R¹, wherein the alkyl is optionally substituted with one or more substituents selected from —NR¹³ and —NR¹³R¹⁴. In a further preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR¹, —NR¹R² and —C(=O)R¹, wherein the alkyl is optionally substituted with one or more substituents selected from —NR¹³R¹⁴. In a particularly preferred embodiment, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₃ alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from —NH₂.

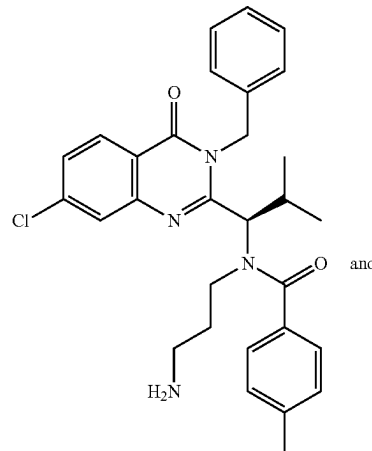
In another embodiment, at each occurrence, R¹, R², R³, R⁴ are each independently selected from H, C₁₋₈ alkyl, —C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl. In a preferred embodiment, at each occurrence, R¹, R², R³, R⁴ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl. In a more preferred embodiment, at each occurrence, R¹, R², R³, R⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl. In a further preferred embodiment, at each occurrence, R¹, R², R³, R⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl and phenyl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C₁₋₆ alkyl. In a particularly preferred

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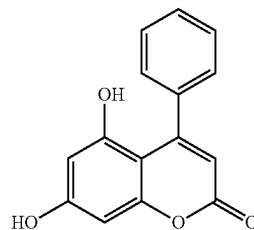
embodiment, at each occurrence, R¹, R², R³, R⁴ are each independently selected from H and p-methylphenyl.

In a particularly preferred embodiment, the compound of formula (I') is selected from:

(compound 1)



(compound 2)



In yet another aspect, the present invention provides use of pharmaceutical composition comprising a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I') in the preparation of a medicament for the prevention or treatment of a polyQ-related neurodegenerative disorder, wherein the pharmaceutical composition comprises a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

In another aspect, the present invention provides a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, or a pharmaceutical composition thereof, which is used to prevent or treat a polyQ-related neurodegenerative disorder.

In a further aspect, the present invention provides a method for preventing or treating a polyQ-related neurodegenerative disorder, which includes administering a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, or a pharmaceutical composition thereof to an individual in need.

The compound of the present invention can be administered to patients orally or parenterally in the form of conventional preparations, for example, capsules, microcapsules, tablets, granules, powders, lozenges, pills, suppositories, injections, suspensions, syrups, patches, creams, lotions, ointments, gels, sprays, solutions, and emulsions. Suitable formulations can use conventional organic or inorganic additives and are prepared by commonly used methods. The organic or inorganic additives are, for example, excipients (for example, sucrose, starch, mannitol, sorbitol, lactose, glucose, cellulose, talc, calcium phosphate or calcium carbonate), binder (for example, cellulose, methylcellulose, hydroxymethylcellulose, polypropylpyrrolidone, polyvinylpyrrolidone, gelatin, gum acacia, polyethylene glycol, sucrose or starch), disintegrants (for example, starch, carboxymethylcellulose, hydroxypropyl starch, low-substituted hydroxypropylcellulose, sodium bicarbonate, calcium phosphate or calcium citrate), lubricants (for example, magnesium stearate, light anhydrous silicic acid, talc or sodium lauryl sulfate), flavoring agents (for example, citric acid, menthol, glycine or tangerine powder), preservatives (for example, sodium benzoate, sodium bisulfite, methyl paraben or propyl paraben), stabilizer (for example, citric acid, sodium citrate or acetic acid), suspending agents (for example, methylcellulose, polyvinylpyrrolidone or aluminum stearate), dispersant (for example, hydroxypropyl methylcellulose), diluent (for example, water) and base wax (for example, cocoa butter, white petrolatum or polyethylene glycol). For example, the effective amount of the compound in the pharmaceutical composition may be an amount that can achieve the desired effect.

Dosage regimens can be adjusted to provide the desired optimal response. For example, when the medicine is administered in the form of injection, it can be administered as a single bolus injection, bolus injection and/or continuous infusion, etc. For example, several divided doses may be administered over time, or the dose may be proportionally reduced or increased as indicated by the urgent need of the treatment situation. It should be noted that the dose value may vary with the type and severity of the condition to be alleviated and may include single or multiple doses. Generally, the dose of treatment varies, depending on the considerations, such as the age, gender, and general health of the patient to be treated; the frequency of treatment and the nature of the desired effect; the degree of tissue damage; the duration of symptoms; and other variables that can be adjusted by individual physicians. It should be further understood that for any particular individual, the specific dosing regimen should be adjusted over time according to the needs of the individual and the professional judgment of the person administering the composition or supervising the administration of the composition. The dosage and regimen of the pharmaceutical composition can be easily determined by a person of ordinary skill in the clinical field. For example, the composition or compound of the present invention may be administered in divided doses from 4 times a day to once every 3 days, and the dosage may be, for example, 0.01 to 1000 mg/time. The required dose can be administered one or more times to obtain the desired result. The pharmaceutical composition according to the present invention can also be provided in unit dosage forms.

In one embodiment, the present invention provides a capsule containing the compound of the present invention without an additional carrier.

The pharmaceutical composition of the present invention may be in the form of tablets, chewable tablets, capsules, solutions, parenteral solutions, lozenges, suppositories, and

suspensions, etc. The composition can be formulated to contain a daily dose or an appropriate portion of the daily dose in a dosage unit, which can be a single tablet or capsule or a liquid of a suitable volume.

In one embodiment, the solution is prepared from a water-soluble salt, such as hydrochloride. Generally, all compositions are prepared according to known methods in pharmaceutical chemistry. Capsules can be prepared by mixing the compound with a suitable carrier or diluent and filling an appropriate amount of the mixture into the capsule. Commonly used carriers and diluents include, but are not limited to, inert powdered substances, such as a variety of different starches, powdered cellulose, especially crystalline and microcrystalline cellulose, sugars like fructose, mannitol and sucrose, cereal flour, and similar edible powder.

Tablets can be prepared by direct compression, wet granulation, or dry granulation. The preparation usually adds diluent, binder, lubricant and disintegrant and the compound. Typical diluents include, for example, various types of starch, lactose, mannitol, kaolin, calcium phosphate or sulfate, inorganic salts (such as sodium chloride) and powdered sugar. Powdered cellulose derivatives are also useful. Typical tablet binders are the following substances, such as starch, gelatin, and sugar (such as lactose, fructose, glucose, etc.). Natural and synthetic gums are also suitable, including gum acacia, alginate, methylcellulose, polyvinylpyrrolidone, etc. Polyethylene glycol, ethylcellulose and wax can also act as binders.

Lubricants can be selected from such slippery solids such as talc, magnesium stearate and calcium stearate, stearic acid and hydrogenated vegetable oils. A tablet disintegrant swells when wet to break the tablet and release the compound. They include starch, clay, cellulose, algin and gum. More specifically, for example, corn and potato starch, methylcellulose, agar, bentonite, lignocellulose, powdered natural sponge, anion exchange resin, alginic acid, guar gum, citrus pomace, carboxymethylcellulose and sodium lauryl sulfate can be used. Tablets can be coated with sugar as a flavoring and sealing agent or coated with a film-forming protective agent to optimize the dissolution performance of the tablet. The composition can also be formulated into chewable tablets, for example, by adding some substances to the formulation, such as mannitol.

When it is desired to be administered as a suppository, a typical base can be used. Cocoa butter is a traditional suppository base, which can be changed by adding wax to slightly increase its melting point. Especially water-miscible suppository bases including polyethylene glycols of various molecular weights are widely used.

The effect of the compound can be delayed or prolonged by a suitable formulation. For example, slowly dissolving pellets of the compound can be prepared and added to tablets or capsules or used as a sustained release implantable device. The technology also includes preparing several pellets with different dissolution rates and filling the capsule with a mixture of pellets. Tablets or capsules can be coated with a film that resists dissolution for a predictable period. Even parenteral preparations can be prepared as a long-acting formulation by dissolving or suspending the compound in an oily or emulsified vehicle that allows it to be slowly dispersed in the serum.

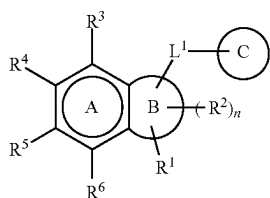
In a preferred embodiment, the polyQ-related neurodegenerative disorder is selected from spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease and spinal-bulbar

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muscular atrophy. In another preferred embodiment, the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12 and spinocerebellar ataxia type 17. In another preferred embodiment, the polyQ-related neurodegenerative disorders are selected from Huntington's disease, spinocerebellar ataxia, type 1 and spinocerebellar ataxia type 3. In a particular embodiment, the polyQ-related neurodegenerative disorder is Huntington's disease. In another particular embodiment, the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 1 or spinocerebellar ataxia type 3.

The embodiments of the present invention can be listed as follows:

[1] Use of a compound of formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder.



wherein

ring A is a benzene ring;

ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

ring C is selected from C₆₋₁₀ aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl are optionally substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or is a C₁₋₆ hydrocarbon chain;

or, ring C is absent, and L¹ is absent;

R¹ is =Y, wherein Y is O or S, or is OR⁷;

at each occurrence, R² is each independently selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocycl-yl, 3- to 10-membered heterocycl-yl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocycl-yl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

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R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocycl-yl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocycl-yl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocycl-yl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocycl-yl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —N(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocycl-yl), —N(3- to 7-membered heterocycl-yl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocycl-yl), —N(C₁₋₄ alkylene-3- to 7-membered heterocycl-yl)₂, =O, —COOH and C₁₋₆ alkyl;

at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocycl-yl, 3- to 7-membered heterocycl-yl-C₁₋₄ alkyl, and C₆₋₁₀ aryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocycl-yl or aryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl);

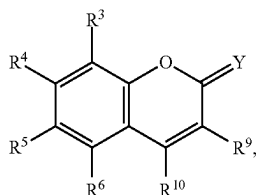
at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocycl-yl, 3- to 7-membered heterocycl-yl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷², —C(=O)R⁷¹, —S(=O)₂OR⁷¹, —S(=O)₂R⁷¹, —S(=O)₂NR⁷¹R⁷² and —S(=O)R⁷¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocycl-yl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR⁷³, —SR⁷³, —NR⁷³R⁷⁴, —C(=O)R⁷³, —C(=O)OR⁷³ and —C(=O)NR⁷³R⁷⁴;

at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocycl-yl, 3- to 10-membered heterocycl-yl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocycl-yl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —OH, —SH, —NH₂, =O and —COOH;

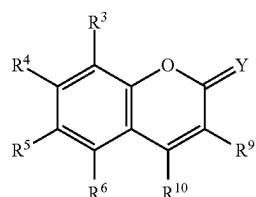
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n is 1 or 2;

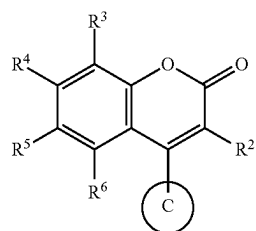
provided that the compound of formula (I'') does not include the following structure:



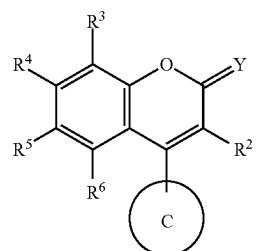
wherein R³ is selected from —O(C₁₋₆ alkyl) and —O(C₁₋₄ alkylene-C₆₋₁₀ aryl); R¹⁰ is H;
and the compound of formula (I'') does not include the following structure:



wherein R⁴ is —O(C₁₋₆ alkyl); R⁵ is halogen; R¹⁰ is CF₃;
and the compound of formula (I'') does not include the following structure:



wherein R⁴ is selected from —O(C₁₋₆ alkyl) and —O(C₁₋₄ alkylene-C₆₋₁₀ aryl);
ring C is C₆₋₁₀ aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1};
and the compound of formula (I'') does not include the following structure:

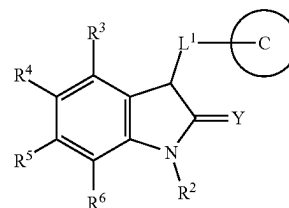


wherein R⁴ is selected from —OH and —O(C₁₋₆ alkyl);
R⁵ is halogen; ring C is 5- to 10-membered heteroaryl,

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5- to 10-membered heteroaryl which is optionally substituted with one or more substituents each independently selected from R^{X1}.

[2] Use of [1] above, wherein the compound of formula (I'') has the following structure of formula (II''):



wherein

Y is O or S;

ring C is 5- to 7-membered heteroaryl, wherein the 5- to 7-membered heteroaryl is optionally substituted with one or more substituents each independently selected from R^{X1};

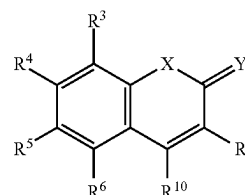
R² is selected from H and C₁₋₈ alkyl;

L¹ is a bond, or is a C₁-C₆ hydrocarbon chain;

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

wherein R^{X1}, R^{X2} are as defined in [1] above.

[3] Use of [1] above, wherein the compound of formula (I'') has the following structure of formula (IV''):



wherein

Y is O or S;

X is O;

R⁹ is selected from H, halogen, C₁₋₆ alkyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1} and —S(=O)R^{a1}, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2} and —S(=O)R^{a2}, wherein R^{a1}, R^{b1}, R^{a2}, R^{b2} are as defined in [1] above;

R¹⁰ is selected from H, halogen, C₁₋₆ alkyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C₁₋₄ alkyl;

R³ is selected from H, halogen, C₁₋₆ alkyl, —OH, —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂, wherein the alkyl is optionally substituted with one or more sub-

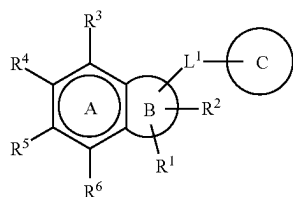
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stituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂; R⁴ is selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OR⁷, —SR⁷ and —NR⁷R⁸; at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl); wherein R⁷, R⁸ are as defined in [1] above;

R⁵ is selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl);

R⁶ is selected from H, halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —O(benzyl), —SH, —S(C₁₋₆ alkyl), —S(benzyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂ and —NH(benzyl), wherein the alkyl or benzyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl) and —N(C₁₋₆ alkyl)₂; provided that when R⁴ is —O(C₁₋₆ alkyl) and R⁵ is halogen, R¹⁰ is not CF₃.

[4] Use of [1] above, wherein the compound of formula (I^m) has the following structure of formula (VI^m):



wherein:

ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

ring C is C₆₋₁₀ aryl, which is substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or is a C₁₋₆ hydrocarbon chain;

R² is selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)₂R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1},

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—OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)₂R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), alkylene-C₃₋₆ cyclohydrocarbyl, —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —NH(C₃₋₆ cyclohydrocarbyl), —N(C₃₋₆ cyclohydrocarbyl)₂, —NH(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), alkylene-C₃₋₆ cyclohydrocarbyl)₂, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl)₂, —NH(C₁₋₄ alkylene-3- to 7-membered heterocyclyl), —N(C₁₋₄ alkylene-3- to 7-membered heterocyclyl)₂, =O, —COOH and C₁₋₆ alkyl;

at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, —OH, —O(C₁₋₆ alkyl), —NH₂, —NH(C₁₋₆ alkyl), —N(C₁₋₆ alkyl)₂, —COOH, —C(=O)O(C₁₋₆ alkyl), —C(=O)NH(C₁₋₆ alkyl), —C(=O)N(C₁₋₆ alkyl)₂, —OC(=O)(C₁₋₆ alkyl), —NHC(=O)(C₁₋₆ alkyl) and —C(=O)(C₁₋₆ alkyl);

at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR^{r1}, —SR^{r1}, —NR^{r1}R^{r2}, —C(=O)OR^{r1}, —C(=O)NR^{r1}R^{r2}, —C(=O)R^{r1}, —S(=O)₂OR^{r1}, —S(=O)₂R^{r1}, —S(=O)₂NR^{r1}R^{r2} and —S(=O)R^{r1}, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR^{r3}, —SR^{r3}, —NR^{r3}R^{r4}, —C(=O)R^{r3}, —C(=O)OR^{r3} and —C(=O)NR^{r3}R^{r4};

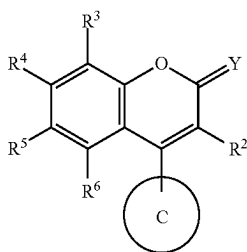
at each occurrence, R^{r1}, R^{r2}, R^{r3}, R^{r4} are each independently selected from H, C₁₋₈ alkyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered hetero-

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cyclyl- C_{1-4} alkyl, C_{6-10} aryl, C_{6-10} aryl- C_{1-4} alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl- C_{1-4} alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-8} alkyl, C_{2-8} alkenyl, C_{2-8} alkynyl, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$, $=\text{O}$ and $-\text{COOH}$;

ring A, R^1 is as defined in [1] above;

provided that the compound of formula (VI'') does not include the following structure:



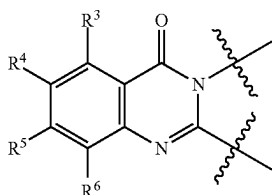
wherein R^4 is selected from $-\text{O}(C_{1-6}$ alkyl) and $-\text{O}(C_{1-4}$ alkylene- C_{6-10} aryl).

[5] Use of [1] above, wherein ring B is dihydropyrimidine.

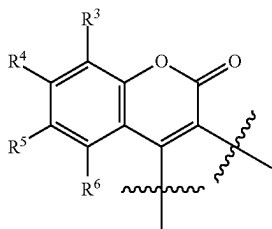
[6] Use of [1] above, wherein ring B is 2H-pyran or 4H-pyran; preferably is 2H-pyran.

[7] Use of any of [1]-[6] above, wherein R^1 is $=\text{O}$.

[8] Use of [5] above, wherein A-B ring system is



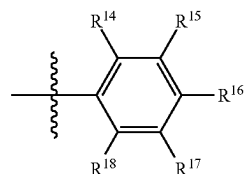
[9] Use of [6] above, wherein A-B ring system is



[10] Use of any of [4]-[9] above, wherein L^1 is a bond, or is a C_1 - C_2 hydrocarbon chain.

[11] Use of any of [4]-[10] above, wherein ring C is phenyl, wherein the phenyl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R^{X1} ; preferably ring C is

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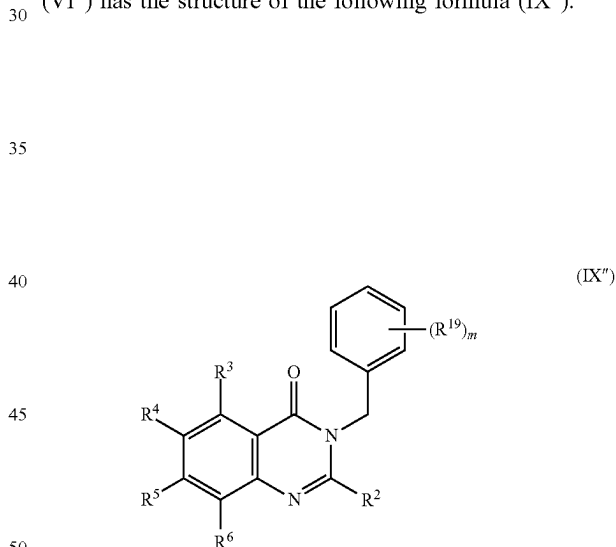


wherein R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, C_{2-6} alkenyl, C_{2-6} alkynyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$;

preferably, R^{14} , R^{15} , R^{16} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$ and $-\text{C}(=\text{O})\text{R}^7$;

more preferably, R^{14} , R^{15} , R^{17} , R^{18} are each independently selected from H, halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NH}(C_{1-6}$ alkyl), $-\text{N}(C_{1-6}$ alkyl) $_2$ and $-\text{NH}(C(=\text{O}))(C_{1-6}$ alkyl); R^{16} is H or $-\text{OCH}_3$.

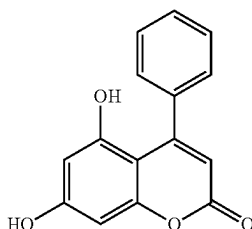
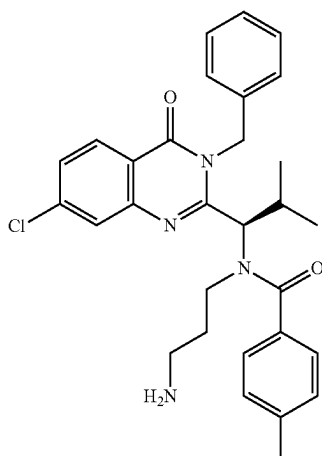
[12] Use of [4] above, wherein the compound of formula (VI'') has the structure of the following formula (IX''):



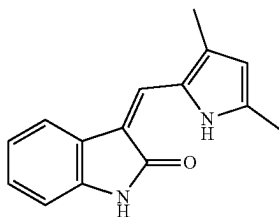
wherein:
at each occurrence, R^{19} are each independently selected from halogen, $-\text{NO}_2$, $-\text{CN}$, C_{1-6} alkyl, $-\text{OR}^7$, $-\text{SR}^7$, $-\text{NR}^7\text{R}^8$, $-\text{C}(=\text{O})\text{OR}^7$, $-\text{C}(=\text{O})\text{NR}^7\text{R}^8$, $-\text{OC}(=\text{O})\text{R}^7$, $-\text{NC}(=\text{O})\text{R}^7\text{R}^8$, $-\text{C}(=\text{O})\text{R}^7$, $-\text{S}(=\text{O})_2\text{OR}^7$, $-\text{S}(=\text{O})_2\text{R}^7$, $-\text{S}(=\text{O})_2\text{NR}^7\text{R}^8$, $-\text{OS}(=\text{O})_2\text{R}^7$, $-\text{NS}(=\text{O})_2\text{R}^7\text{R}^8$ and $-\text{S}(=\text{O})\text{R}^7$, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, $-\text{NO}_2$, $-\text{CN}$, $-\text{OH}$, $-\text{O}(C_{1-6}$ alkyl), $-\text{NH}_2$, $-\text{NH}(C_{1-6}$ alkyl), $-\text{N}(C_{1-6}$ alkyl) $_2$ and $-\text{COOH}$;
 m is 0, 1, 2, 3, 4 or 5;
 R^2 , R^3 , R^4 , R^5 , R^6 are as defined in [4] above.

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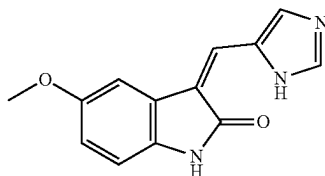
[13] Use of any of [1]-[12] above, wherein the compounds are selected from



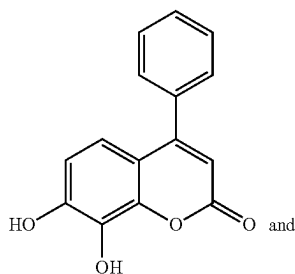
(compound 3)



(compound 4)



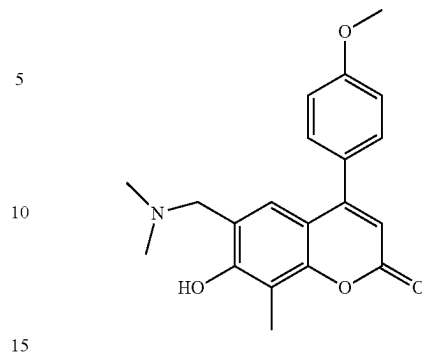
(compound 6)



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-continued

(compound 7)



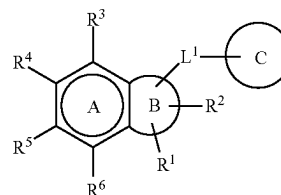
[14] Use of a pharmaceutical composition in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorders; wherein the pharmaceutical composition comprises a compound of formula (I''), formula (II''), formula (III''), formula (III''), formula (IV''), formula (V''), formula (VI''), formula (VII''), formula (VIII''), formula (IX''), formula (X'') or formula (XI'') in any one of [1]-[13] above, or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

[15] Use of any of [1]-[14] above, wherein the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease, and spinal-bulbar muscular atrophy, especially Huntington's disease and spinocerebellar ataxia type 3.

The embodiments of the present invention can also be listed as follows:

<1> Use of a compound of formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(I')



wherein:

ring A is a benzene ring;

ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S;

ring C is C₆₋₁₀ aryl, which is optionally substituted with one or more substituents each independently selected from R^{X1};

L¹ is a bond, or is a C₁-C₆ hydrocarbon chain;

R¹ is =Y, wherein Y is O or S;

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R² is selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2};

R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2};

at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, =O, —COOH and C₁₋₆ alkyl;

at each occurrence, R⁷, R⁸ are each independently selected from H, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C₁₋₄ alkyl;

at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆-alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, C₃₋₆ cyclohydrocarbyl, C₃₋₆ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, —OR⁷¹, —SR⁷¹, —NR⁷¹R⁷², —C(=O)OR⁷¹, —C(=O)NR⁷¹R⁷², —C(=O)R⁷¹, —S(=O)₂OR⁷¹, —S(=O)₂R⁷¹, —S(=O)₂NR⁷¹R⁷² and —S(=O)R⁷¹, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from =O, =S, —OR⁷³, —SR⁷³, —NR⁷³R⁷⁴, —C(=O)R⁷³, —C(=O)OR⁷³ and —C(=O)NR⁷³R⁷⁴;

at each occurrence, R⁷¹, R⁷², R⁷³, R⁷⁴ are each independently selected from H, C₁₋₈ alkyl, —C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, —OH, —SH, —NH₂, =O and —COOH.

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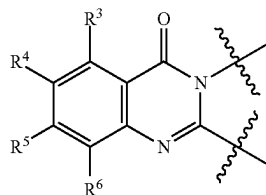
<2> Use of <1> above, wherein ring B is saturated or unsaturated 6-membered heterocycle; wherein the heterocycle contains one or two heteroatoms each independently selected from N and O.

<3> Use of <1> or <2> above, wherein ring B is dihydropyrimidine.

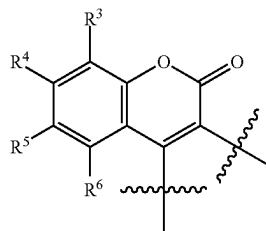
<4> Use of <1> or <2> above, wherein ring B is 2H-pyran or 4H-pyran; preferably is 2H-pyran.

<5> Use of any one of <1> to <4> above, wherein R¹ is =O.

<6> Use of <3> above, wherein A-B ring system is



<7> Use of <4> above, wherein A-B ring system is



<8> Use of any one of <1> to <7> above, wherein L¹ is a bond, or is a C₁₋₂ hydrocarbon chain.

<9> Use of any one of <1> to <8> above, wherein R² is selected from H, halogen, —NO₂, —CN and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₆ alkyl and —NR^{a2}R^{b2}; preferably, R² is selected from H, halogen and C₁₋₆ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C₁₋₆ alkyl and —NR^{a2}R^{b2}, more preferably, R² is H or C₁₋₄ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from —NR^{a2}R^{b2}, especially preferably, R² is H or C₁₋₄ alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from —NR^{a2}R^{b2}, is —CH[CH(CH₃)₂], and is optionally substituted with one or more substituents selected from —NR^{a2}R^{b2}.

<10> Use of <9> above, wherein R² is H.

<11> Use of any one of <1> to <10> above, wherein at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, C₁₋₆ alkyl, —OR⁷ and —NR⁷R⁸; preferably, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen, C₁₋₆ alkyl, —OW and —NR⁷R⁸; more preferably, at each occurrence, R^{X1} and R^{X2} are each independently selected from halogen and —OW; further preferably, at each occurrence, R^{X1} and R^{X2} are each independently selected from C₁, Br, —OH and —O(C₁₋₆ alkyl); especially preferably, at each occurrence, R^{X1} and R^{X2} are each independently selected from C₁ and —OH; wherein at each occurrence, R⁷, R⁸ are each independently selected from H and C₁₋₆ alkyl.

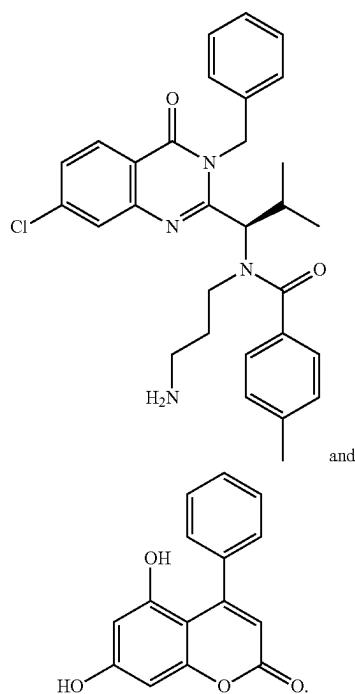
<12> Use of any one of <1> to <11> above, wherein at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each

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independently selected from H, C₁₋₆ alkyl, —OR¹¹, —NR¹¹R¹², —C(=O)OR¹¹, —C(=O)NR¹¹R¹² and —C(=O)R¹¹, wherein the alkyl is optionally substituted with one or more substituents selected from —OR¹³ and —NR¹³R¹⁴; preferably, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR¹¹, —NR¹¹R¹² and —C(=O)R¹¹, wherein the alkyl is optionally substituted with one or more substituents selected from —OR¹³ and —NR¹³R¹⁴; more preferably, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₆ alkyl, —OR¹¹, —NR¹¹R¹² and —C(=O)R¹¹, wherein the alkyl is optionally substituted with one or more substituents selected from —NR¹³R¹⁴; especially preferably, at each occurrence, R^{a1}, R^{b1}, R^{c1}, R^{a2}, R^{b2}, R^{c2} are each independently selected from H, C₁₋₃ alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from —NH₂.

<13> Use of any one of <1> to <12> above, wherein at each occurrence, R¹¹, R¹², R¹³, R¹⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl, phenyl-C₁₋₄ alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —SH, —NH₂, —COOH and C₁₋₆ alkyl; preferably, at each occurrence, R¹¹, R¹², R¹³, R¹⁴ are each independently selected from H, C₁₋₆ alkyl, phenyl and phenyl-C₁₋₄ alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C₁₋₆ alkyl; especially preferably, at each occurrence, R¹¹, R¹², R¹³, R¹⁴ are each independently selected from H and p-methylphenyl.

<14> Use of any one of <1> to <13> above, wherein the compound of formula (I') is selected from:



<15> Use of pharmaceutical composition in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorders; the pharmaceutical

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composition comprises a compound of formula (I') in any one of <1> to <14> above, or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

<16> Use of any one of <1> to <15> above, wherein the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease, and spinal-bulbar muscular atrophy, especially Huntington's disease and spinocerebellar ataxia type 3.

Beneficial Effect

The compound of the present invention has the effect of reducing the level of polyQ-GFP fusion protein with expanded polyQ in cells and does not reduce the level of polyQ-GFP fusion protein with shorter polyQ. Therefore, the compound of the present invention can reduce the level of polyQ protein with expanded polyQ in cells or tissues, thereby having a preventive or therapeutic effect on polyQ protein-related disorders.

The inventors further found that the compound of the present invention have an unexpected regulatory effect on mHTT levels in cells and had good safety. Administration of the compound of the present invention can improve the survival rate and climbing performance of *Drosophila* in HD model. Administration of the compound of the present invention through intraventricular or intraperitoneal injection can allele-selectively reduce the level of mHTT in the cerebral cortices and striata of HD model mice without affecting the level of wtHTT which has important physiological functions, and improved behavioral defects in mice. In the HD patient induced stem cells (iPSC) derived neurons, the compound of the present invention reduces mHTT levels, rescues disease-related phenotypes, and delays neuronal apoptosis. Therefore, the compound of the present invention has good selectivity, therapeutic effects and safety, and is easy to pass through the BBB, potentially facilitating oral administration.

In addition, the compound of the present invention can significantly reduce the level of mutant ATXN3 protein in cells, so it can be used for the treatment or prevention of spinocerebellar ataxia type 3.

Therefore, the compound of the present invention can be used to treat polyQ protein-related disorders and has the advantages of high selectivity and good safety, as well as a good prospect for development of an oral drug.

EXAMPLES

Unless otherwise specified, the instruments and reagents used herein are all commercially available.

In all statistical analyses in this article, * represents p<0.05; ** represents p<0.01; *** represents p<0.001, **** represents p<0.0001. For the comparison between the two groups, the statistical analysis method used is the two-tailed unpaired t test. For comparisons between three or more groups, two-tailed one-way ANOVA was applied when there is only one variable, and two-tailed two-way ANOVA was applied when there are two variables.

Abbreviation

ATXN	ataxin	HTT	Huntingtin
TBP	TATA binding protein	ATN	atrophin
fl	full-length		
BBB	blood-brain barrier		
AR	androgen receptor		
BDNF	brain derived neurotrophic factor		

Experimental Materials, Reagents and Method Steps
Compounds

Compound 1: ispinesib, PubChem CID: 6851740, can be purchased from Selleck, catalog number S1452;

Compound 2: PubChem CID 5398649, can be purchased from ChemDiv, catalog number D715-2435;

Compound 3: Semaxanib, can be purchased from Selleck, CAS No. 194413-58-6;

Compound 4: Su9516, can be purchased from Selleck, CAS No. 377090-84-1;

Compound 5: can be purchased from Sinopharm Chemical Reagent Co. Ltd, CAS No. 779-30-6;

Compound 6: can be purchased from TargetMol, CAS No. 842-01-3; Compound 7: can be purchased from Chem-Div.

Antibody

The antibodies used for Western-blot, HTRF and/or immunofluorescence/immunohistochemistry are as follows: HTT antibody 2B7 (Weiss et al. Anal Biochem 2009, 395, 8-15), 4C9, ab1 (Sapp et al. J Biol Chem 2012, 287, 13487-13499) and MW1 (Ko et al. Brain research bulletin 2001, 56, 319-329) were prepared by the method of the prior art; the antibody S830 used for immunostaining to detect HTT aggregates was gifted from Dr. Gillian Bates; other antibodies were purchased from companies such as Millipore and Sigma.

Preparation and Verification of Recombinant Human Full-Length HTT Protein

(1) The human HTT gene (GenBank: NM_002111.8) with (CAG)₂₃ or (CAG)₇₃ was synthesized de novo by Genewiz Inc. The human HTT gene was cloned into the modified pCAG vector (from Addgene) with N-terminal protein A tag.

(2) The plasmid was transfected into human embryonic kidney E293 cells for expression using polyethyleneimine (PEI, from Polysciences, 23966). The protein is purified with IgG monoclonal antibody-agarose (Smart-lifesciences, SA030010), digested with TEV protease to remove the protein A tag, and further purified with Mono Q and Superose 6 (5/150 GL) columns from GE healthcare. Validation is conducted using Coomassie blue staining and Western-blot. Cells Used for Tests

Primary cultured cortical neurons: The brains of Hdh^{Q140/Q7} and Hdh^{Q7/Q7} newborn mice (P0) were dissected, digested, dissociated, and cultured.

Some primary patient fibroblasts and wild-type cells are from HD patients (Q47, Q55) of the Mongolian Huntington's disease family and healthy controls (WT, Q19). SCA3 cell line was from patient (Q74). The HD Q68 fibroblast cell line was from Coriell Cell Repositories (Camden, N.J., USA). Immortalized fibroblasts and induced stem cells (iPSCs) are prepared from primary fibroblasts. HEK293T cells were from ATCC.

Animals Used for Tests

Huntington's Disease *Drosophila*

Nervous system driver line elav-GAL4 (c155), HTT expressing lines UAS-fl-HTT-Q16 and UAS-fl-HTT-Q128

were from Bloomington *Drosophila* Stock Center (<http://flystocks.bio.indiana.edu/>) and kept in a 25° C. incubator.

Crosses were set up between virgin female *Drosophila* carrying elav-GAL4 driver and the UAS-fl-HTT-Q16 or UAS-fl-HTT-Q128 male *Drosophila* to generate a transgenic *Drosophila* driven by elav-GAL4 to express human HTT full-length protein (Q16) or (Q128) in the nervous system.

Huntington's Disease Mice

Mice expressing wild-type HTT gene (Hdh^{Q7/Q7}) were from the Marian Difiglia laboratory of Massachusetts General Hospital, Harvard University. The Q140 gene knock-in heterozygous mouse (Hdh^{Q140/Q7}) was prepared according to the method of the prior art (Menalled et al., J Comp Neurol, 2003, 465: 11-26).

Protein Analysis

Homogeneous time-resolved fluorescence (HTRF) analysis: the cell or tissue lysate was diluted with the original lysis buffer PBS+1% (v/v) Triton X-100+1×Complete™ protease inhibitor to lyse the sample, then HTRF assay buffer (50 mM NaH₂PO₄, 400 mM NaF, 0.1% BSA, 0.05% (v/v) Tween-20, 1% (v/v) Triton X-100, pH 7.4) was used to dilute the specified antibody pair for detection. In HTRF buffer, the donor antibody concentration was 0.023 ng/μL and the acceptor antibody concentration was 1.4 ng/μL.

Determination of the amount of protein: the amount of protein was determined by the above method. Background correction was performed using blank sample. The protein concentration for all samples were determined to correct the loadings. Different protein concentrations or cell numbers per well were measured to ensure that the signals are within the linear range.

Cell Analysis

Immunofluorescence: cells were washed, fixed, permeabilized and blocked, then incubated with primary antibody overnight at 4° C., then washed three times with blocking buffer, and incubated with secondary antibody for 1 hour at room temperature, stained with DAPI, and mounted. Images were taken by Zeiss Axio Vert A1 confocal microscope. TUBB3 or co-localization was analyzed using ImageJ.

Example 1 Effects of the Compounds on the Levels of Proteins Containing Expanded Polyglutamine

The inventors used HEK293T cells exogenously expressing polyQ-GFP fusion proteins (Q72-GFP or Q25-GFP) to detect compounds that can control the level of proteins containing expanded polyglutamine in the cells. The method was as follows:

(1) PolyQ-GFP cDNA sequence (expressing Met-polyQ-GFP, wherein polyQ was Q72 or Q25) was synthesized de novo and subcloned into pcDNA vector. It was transfected into HEK293T cells (ATCC, CRL-3216) through forward transfection to obtain Q72-GFP expressing cells and Q25-GFP expressing cells, respectively. And the same method was used to obtain cells expressing Q53-GFP, cells expressing Q46-GFP, and cells expressing Q38-GFP.

(2) The polyQ-GFP proteins expressed by (1) were purified separately to obtain Q72-GFP, Q53-GFP, Q46-GFP, Q38-GFP and Q25-GFP (SEQ ID NO: 1). Among them, Q72-GFP, Q53-GFP, Q46-GFP, Q38-GFP differed from the amino acid sequence of SEQ ID NO: 1 in the length of the expanded polyglutamine.

(3) The Q72-GFP expressing cells and Q25-GFP expressing cells obtained in step (1) were used. The cells were treated with 100 nM compound 1, or 50 nM compound 2 for 2 days. Then the level of polyQ-GFP was measured by

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Incucyte fluorescence counting (FIG. 1). It was observed that compound 1 and compound 2 effectively reduced the level of the protein containing expanded polyglutamine (Q72-GFP) in HEK293T cells, but did not reduce the level of the protein containing the shorter polyglutamine (Q25-GFP). This result indicated that the compound of the present invention can selectively reduce the level of abnormally amplified polyQ protein in cells.

Example 2 Interaction of the Compounds with Proteins Containing Expanded Polyglutamine

2.1 OI-RD Detection of the Affinity of Compounds to Proteins Containing Expanded Polyglutamine

Compound chips were prepared using a contact microarray printer (SmartArrayer 136, CapitalBio Corporation) according to the prior art (Zhu et al., Sensors (Basel) 2016, 16(3), 378; Fei et al., J Biomed Opt 2010, 15(1), 016018). Each compound was printed in triplicate on the microarray. OI-RD was used to measure the affinity reaction parameters between the compounds and GFP or the polyQ-GFP protein obtained in step (2) of Example 1. These compounds did not bind to Q25-GFP or GFP, but binded to proteins with expanded polyglutamine. For example, compound 2 showed certain binding affinity to proteins with 38 or more glutamine repeats. The affinity reaction parameters are shown in Table 1.

TABLE 1

The association rate constants, dissociation rate constants, and equilibrium dissociation constants of the affinity reactions of compound 2 with Q72-GFP, Q53-GFP, Q46-GFP, and Q38-GFP			
	K_{on} (min · nM) ⁻¹	K_{off} (min) ⁻¹	K_d (nM)
Q72-GFP	6.20×10^{-5}	1.92×10^{-2}	309.7
Q53-GFP	1.08×10^{-4}	4.65×10^{-3}	43.1
Q46-GFP	7.65×10^{-6}	2.57×10^{-2}	3359.5
Q38-GFP	2.99×10^{-6}	1.5×10^{-2}	5016.7

2.2 MST Detection of the Affinity of Compounds to Proteins Containing Expanded Polyglutamine

Furthermore, the affinity of compound 2 with proteins containing expanded polyglutamine was detected with micro thermophoresis (MST) using Monolith NT.115 instrument from NanoTemper Technologies. The reaction buffer was 20 mM HEPES, pH 7.4, 150 mM NaCl, and protein concentration was 500 nM. Compound 2 showed no binding affinity with Q25-GFP, and its K_d with Q72-GFP is 2.80 μ M.

In summary, the results of OI-RD detection and MST detection indicated that the compound of the present invention selectively binded to proteins containing expanded polyglutamine.

Example 3 Effect of the Compounds with Full-Length mHTT

3.1 OI-RD Detection of Affinity Between Compounds and Full-Length HTT

Compound chips were prepared using a contact microarray printer (SmartArrayer 136, CapitalBio Corporation) according to the prior art (Zhu et al., Sensors (Basel) 2016, 16(3), 378; Fei et al., J Biomed Opt 2010, 15(1), 016018). Each compound was printed in triplicate on the microarray. OI-RD was used to measure the affinity reaction parameters of the compounds with full-length mHTT (fHTT-Q73) and full-length HTT-Q23 (fHTT-Q23, SEQ ID NO: 2) (FIG. 2). The compound showed no binding affinity with fHTT-Q23,

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and the association rate constants, dissociation rate constants and equilibrium dissociation constants of the affinity reactions with fHTT-Q73 are shown in Table 2.

TABLE 2

The association rate constants, dissociation rate constants and equilibrium dissociation constants of the affinity reactions of the compounds with fHTT-Q73			
	K_{on} (min · nM) ⁻¹	K_{off} (min) ⁻¹	K_d (nM)
compound 1	6.43×10^{-5}	2.07×10^{-2}	321.9
compound 2	4.81×10^{-4}	1.19×10^{-2}	24.7

3.2 MST Detection of the Affinity of the Compound with Full-Length HTT

The affinity of the compounds with full-length HTT was verified with micro thermophoresis (MST) using Monolith NT.115 instrument from NanoTemper Technologies (FIG. 3). The reaction buffer was 20 mM HEPES, pH 7.4, 150 mM NaCl. Protein concentration was 500 nM. The compound of the present invention showed no binding affinity with fHTT-Q23. The K_d of compound 1 and fHTT-Q73 was 0.13 μ M; the K_d of compound 2 and fHTT-Q73 was 2.82 μ M.

In summary, the results of OI-RD measurement and MST assay both showed that the compound of the present invention selectively binded with fHTT-Q73. This conclusion was inherently consistent with the phenomenon that the compound selectively reduced protein levels containing expanded polyglutamine in HEK293T cells, indicating that the compound's effect on the above-mentioned protein levels may be achieved by selectively binding with proteins containing expanded polyglutamine.

Example 4 Effects of the Compounds on Cortical Neurons in HD Model Mice

4.1 Effects on mHTT and wtHTT Levels

(1) The primary cultured Hdh^{Q140/Q7} and Hdh^{Q7/Q7} mouse cortical neuron cells were treated with the compound for 2 days. Then the levels of mHTT and wtHTT were detected by Western-blot (antibody 2166) (FIG. 4 and FIG. 5). Compound 1 and compound 2 reduced the level of mHTT in cortical neurons of Q140 knock-in heterozygous mice (Hdh^{Q140/Q7}), but hardly affected the level of wtHTT. Compound 1 did not reduce wtHTT levels in cortical neurons of Hdh^{Q7/Q7} mice.

(2) mHTT was detected using anti-polyQ antibodies MW1 and 3B5H10, and observed for bands of proteins with smaller molecular weights. As a result, no increase in the N-terminal fragments of mHTT was observed (FIG. 6). The detected decrease in mHTT level was not due to an increase in site-specific cleavage.

4.2 Cytotoxicity Test

CellTiter-glo (Promega, G7570) was used according to the protocol provided in the kit to determine the viability of Hdh^{Q140/Q7} mouse cortical neurons after treatment with the specified compound (FIG. 7).

The compound of the present invention showed no cytotoxicity to Hdh^{Q140/Q7} mouse cortical neurons at the test concentration described in 4.1. The detected decrease in mHTT level was not due to neuronal cell loss.

In summary, the compound of the present invention allele-selectively reduced the level of mHTT in cells, and had no cytotoxicity, and had good safety.

Example 5 Effects of the Compounds on mHTT
and wtHTT Levels of Huntington's Disease Patient
Fibroblasts

5.1 Effect on the Levels of mHTT and wtHTT in HD Patient
Primary Fibroblasts

Preliminary experiments were conducted. As a result, the compound exhibited the best mHTT reduction effect at a concentration of 100 nM.

Experiment method: The fibroblasts (Q49, Q55, Q68) of HD patients were treated with 100 nM compound for 2 days. Then mHTT (antibody pair: 2B7/MW1) and total HTT (antibody pair: 2B7/2166) were detected by HTRF. The results are shown in FIG. 8. Reduced mHTT levels were observed in primary fibroblasts (Q49, Q55, Q68) of HD patients. No reduction in HTT levels was observed in wild-type primary fibroblasts.

5.2 Effects on the Level of mHTT in HD Patient Immortalized Fibroblasts

A test method similar to that described in 5.1, HTRF (antibody pair: 2B7/MW1) was used to detect the effects of the compound of the present invention on the level of mHTT in HD patient immortalized fibroblasts (FIG. 9). Decreased mHTT levels were observed on HD patient immortalized fibroblasts. FIG. 10 shows that compound 2 reduced the level of mHTT in HD patient immortalized fibroblasts at different doses.

Example 6 Effects of the Compounds on mHTT
Level and Neuronal Apoptosis of HD Patient
iPSC-Derived Neurons

6.1 Effect on mHTT and wtHTT Levels
Experimental Materials:

A test method similar to that described in 5.1, HTRF (antibody pair: 2B7/MW1) was used to detect the effects of compound 1 and compound 2 on the level of mHTT in HD patient iPSC-derived neurons (Q47) (FIG. 11). Decreased mHTT levels were observed in HD patient iPSC-derived neurons.

Using the same method, the following compound 3, compound 4, compound 5, compound 6, and compound 7 were observed to reduce the mHTT level in the HD patient iPSC-derived neurons; wherein compound 3 reduced mHTT levels by approximately 12.2%, compound 4 reduced mHTT levels by approximately 24.9%, compound 5 reduced mHTT levels by approximately 26.8%, compound 6 reduced mHTT levels by approximately 24.2%, and compound 7 reduced mHTT levels by approximately 19.7%.

The compound of the present invention rescued disease-related phenotypes in HD patient iPSC-derived neurons through autophagy.

6.2 Effects on Neuronal Apoptosis

(1) Immunostaining

HD patient iPSC-derived neurons (Q47) were treated using 100 nM compound 1 or 50 nM compound 2 for 1 day. Then the cells were stressed (by BDNF removal).

Neuronal specific tubulin marker TUBB3 was stained with DAPI. The TUBB3 signal covered area was normalized to the cell nuclei counts, and the data was normalized to the wild type controls to analyze neuronal apoptosis (FIG. 12).

(2) Caspase-3 Activity Detection

After removing BDNF, loss of processes and shrinkage of neurons were observed in HD neurons.

NucView 488 (Biotium, 30029) was used to detect active caspase-3. After removing BDNF, Incucyte (Essen Bioscience, IncuCyte FLR) was used to capture images every 3

hours in the incubator, which were analyzed with Incucyte 2011A software. Three batches were tested and the results were consistent (FIG. 13). Compound 1 and compound 2 significantly improved the loss of processes and shrinkage of HD neurons after removal of BDNF.

Example 7 Effects of the Compounds on
Huntington's Disease Drosophilae

7.1 Effects on mHTT Level

Experimental Method:

Q128 Drosophilae and Q16 Drosophilae were randomized into a negative control group and a positive drug group (compound 1, compound 2), with 75 drosophilae in each group. The negative control group was given the corresponding solvent DMSO, and the positive drug groups were given the corresponding positive drug.

Drosophilae were kept in standard food at 25° C. The newly hatched Drosophilae were transferred to vials with food containing the positive drug (10 µM in 400 µL DMSO) or DMSO for control. The corn food was changed every other day.

After fed for 6 consecutive days, the *Drosophila* head protein was extracted on the 7th day. The mHTT level was measured by HTRF (antibody pair: 2B7/MW1), where each sample included head proteins from five *Drosophila* (FIG. 14). Compound 1 and compound 2 reduced mHTT levels in transgenic Drosophilae expressing human HTT full-length protein (Q128).

7.2 Effects of Compound on Survival Rate

75 age-matched virgin Drosophilae were put into empty plastic vials containing standard food, and the survival rate of each vial was recorded daily to measure the lifespan (FIG. 15). The survival rate of the Q128 *Drosophila* positive drug group was improved compared to the control group.

7.3 Effects on Climbing Performance

15 age-matched virgin Drosophilae were put into empty vials and tapped down so that they were at the bottom of the vials. The percentage of Drosophilae that had climbed past a 7-cm-high line after 15 seconds was recorded. The mean of five observations for each vial was plotted every day, and the data from multiple vials containing different batches of Drosophilae was plotted and analyzed.

The results are shown in FIG. 16. The number in brackets indicates the number of tested vials. The Q128 *Drosophila* positive drug group showed improved climbing performance compared to the control group.

In the above experiments in 7.2 and 7.3, no compound was observed to have an effect on Q16 Drosophilae.

Example 8 Effects of the Compounds on HD
Model Mice

Experimental animal: The mice were grouped and housed in individually vented cages with a 12-hour light/dark cycle, with a maximum of 5 adult mice per cage.

8.1 Effects of Intracerebroventricular Injection of Compounds on the Levels of mHTT and wtHTT in the Cortices of HD Mice

Experimental animals: Hd^{Q140/Q27} mice (3 months old), 4 in each group.

Experimental method: one icv-injection was conducted per day using 2 µL artificial cerebrospinal fluid (ACSF: 1 mM glucose, 119 mM NaCl, 2.5 mM KCl, 1.3 mM MgSO₄, 2.5 mM CaCl₂, 26.2 mM, NaHCO₃, 1 mM NaH₂PO₄)

containing a compound at a concentration of 25 μ M. 2 μ L ACSF containing the same amount of DMSO was used as a control.

After 10 days of injection, the brain cortical neuron proteins of the mice were extracted, and the levels of mHTT and wtHTT were detected by Western-blot (antibody 2166). Each test includes three repetitions for each sample, and the mean values was calculated (FIG. 17). Compound 2 significantly reduced mHTT level in Hdh^{Q140/Q7} mouse cortices and showed mHTT selectivity relative to wtHTT.

8.2 Compound 2 Administered by Intraperitoneal Injection

Experimental method: The compound or control DMSO was diluted with 0.9% NaCl intravenous infusion solution to 0.05 μ g/ μ L. One ip-injection (0.5 mg/kg) was conducted per day. After 14 days of injection, tissue extraction or behavioral experiment were conducted.

In vivo compound detection in brain tissues of ip-injected mice: 2 hours after ip-injection of DMSO or compound in mice, the brain was dissected, weighed, and the compound in the brain tissue was extracted, analyzed with UPLC-MS (Acquity Ultra Performance Liquid Chromatography System, Acquity UPLC BEH C18 (1.7 μ m, 2.1 $\times$ 50 mm) column and Xevo TQ-S mass spectrometer, Waters Corporation, Milford, MA, USA) for LC-MS/MS analysis. The level of compound 2 reached the brain tissue was 9.48 ng/g. No mass spectrum signal of compound was observed for the control (DMSO) group. The compound of the present invention can be delivered to the brain of mice through the BBB smoothly, which potentially facilitates oral administration of the compound.

8.3 Effects of Intraperitoneal Injection of Compounds on the Levels of mHTT and wtHTT in the Cortices and Striata of HD Mice

(1) 13 Hdh^{Q140/Q7} mice (5 months old) were randomized into 2 groups.

Administer compound 2 as described in 8.2. One ip-injection was conducted per day. After 14 days of injection, the protein was extracted and the levels of mHTT and wtHTT in the cortices of HD mice were detected by Western-blot (FIG. 18).

(2) A total of 13 mice Hdh^{Q140/Q7} mice (10 months old) were used, 6-7 mice in each group. According to the above method, the mouse brain striatum neuronal protein was extracted and the levels of mHTT and wtHTT were detected by Western-blot (FIG. 19).

(3) mHTT aggregates in the cortices of Hdh^{Q140/Q7} mice were detected by dot-blot experiment (antibody: 4C9, bar plot showing the results of two repetitions) and HTRF (antibody pair: 4C9/4C₉). Sampling was repeated for each mouse two to three times on average (FIG. 20). No increase in mHTT aggregates was observed. The decrease in mHTT levels in the compound 2 treated group was not due to changes in mHTT solubility.

In summary, intraperitoneal injection of compound 2 reduced the levels of mHTT in the cortices and striata of Hdh^{Q140/Q7} mice, and showed mHTT selectivity relative to wtHTT. Therefore, compound 2 had the prospect of being developed to an oral drug.

8.4 Effects of Intraperitoneal Injection of Compound 2 on Behavioral Defects in HD Mice

Experimental animals: 32 Hdh^{Q140/Q7} mice were randomized into 2 groups; 28 (Hdh^{Q7/Q7}) mice were randomized into 2 groups.

Experimental method: administer compound 2 according to the procedure described in 8.2. One ip-injection was conducted per day, and behavioral experiments were conducted after 14 days of injection.

All behavioral experiments were carried out during the light phase. Before starting the experiments, all mice were kept in the behavioral test room under dim red light for one hour.

Rotarod test: The mice were pre-trained for 3 consecutive days (on the rotarod rotating at 4 rpm for 2 minutes). The result of each experiment was recorded as time on the rod (time on the rotating rod), until falling from the rod or until the end of the task. Each test include three repetitions with an inter-trial interval of 60 min in order to reduce stress and fatigue. The means of three trials were analyzed for each mouse (FIG. 21a).

Balance beam test: a 2 cm thick meter stick with a total length of 100 cm was suspended on the platform on both sides. There was a bright light at the starting point and a dark box with food at the endpoint. The total time for each mouse to walk through the balance beam was recorded (FIG. 21b).

The compound of the present invention improved the Huntington's disease related behavioral defects in HD model mice, and had no effect on wild-type mice.

Example 9 Effects of the Compounds and the ATXN3 with Abnormally Expanded polyQ

9.1 Preparation of Recombinant Human MBP-ATXN3

(1) His8 tag and TEV protease cleavage site is added in pMal-C2x plasmid (from New England Biolabs) to prepare prokaryotic expression vector pMBP. The ATXN3 gene (GenBank: NM_001127696.2) is edited to control the length of the expanded polyglutamine, the gene is amplified and cloned into the prepared pMBP to obtain the pMBP-ATXN3-Q28 and pMBP-ATXN3-Q68 plasmids with polyQ repeated 28 and 68 times, respectively.

(2) The expression plasmids pMBP-ATXN3-Q28 and pMBP-ATXN3-Q68 were introduced into *Escherichia coli* Rosetta(DE3)pLysS for expression. Preliminary purification was conducted with HisTrap HP column (GE Healthcare, 17524701). The product was concentrated by ultrafiltration and then further purified with Superose 6 Increase 10/300 GL size exclusion column (GE Healthcare).

(3) As verified by SDS-PAGE, the purity of the prepared human MBP-ATXN3-Q28 and MBP-ATXN3-Q68 exceeds 98%.

9.2 MST Detection of the Affinity of the Compound to ATXN3 with Abnormally Expanded polyQ

Following the method in 2.2, the affinity of compound 2 with MBP-ATXN3-Q28 and MBP-ATXN3-Q68 protein was verified by microscale thermophoresis.

The results showed that compound 2 did not bind with normal ATXN3 protein (MBP-ATXN3-Q28), and the K_d of compound 2 with abnormal ATXN3 protein (MBP-ATXN3-Q68) was 2.77 μ M. The compound of the present invention selectively binded to ATXN3 with abnormally expanded polyQ.

Example 10 Effects of the Compounds on ATXN3 Level in Fibroblasts of Patients with Spinocerebellar Ataxia Type 3

SCA3 patient fibroblasts (Q74), wild-type cells (Q27) were treated with 100 nM compound 1 or 50 nM compound 2 for 2 days. Then the levels of mutant ATXN3 protein (ATXN3-Q74, SEQ ID NO: 3) and wild-type ATXN3 protein (ATXN3-Q27, SEQ ID NO: 4) were determined by Western-blot (FIG. 22). A decrease in mutant ATXN3 protein level was observed on SCA3 patient fibroblasts

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(Q74), but no decrease in wild-type ATXN3 protein level was observed. Therefore, the compound can be used to treat SCA3.

Example 11 Effects of the Compounds on the Level of ATXN1 with Abnormally Expanded polyQ in Cells

(1) PolyQ-ATXN1 cDNA (where polyQ is Q92) was synthesized de novo and subcloned into pcDNA vector. It

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transfected into HEK293T cells (ATCC, CRL-3216) to obtain cells expressing His-ATXN1-Q92.

(2) Cells prepared in (1) were treated with 100 nM compound (compound 1, or compound 2, or compound 4, or compound 6) for 2 days. Then the level of mutant ATXN1 protein (His-ATXN1-Q92, SEQ ID NO: 5 with His tag at the N-terminal) was detected by HTRF (antibody pair: anti-His-Tb/MW1-D2, FIG. 23). Decreased levels of mutant ATXN1 were observed on fibroblasts (Q92) of SCA1 patients. Therefore, the compound of the present invention can be used to treat or prevent spinocerebellar ataxia type 1.

Sequence Listing

Q25-GFP:

[SEQ ID NO: 1]

MQQQQQQQQQQQQQQQQQQQQQQQQQQQQMSKGEELFTGVVPILVELDGDVNGH

KFSVRGEGEGDATNGKLTLLKFICTTGKLPVPWPTLVTTLTYGVCFSRYPDHMKRHD

FFKSAMPEGYVQERTISFKDDGTYKTRAEVKFEGDTLVNRIELKGIDFKEDGNILGHK

LEYNFNFSHNVIYITADKQKNGIKANFKIRHNVEDGSVQLADHYQQNTPIGDGPVLLPD

NHYLSTQSVLSKDPNEKRDHMLVLEFVTAAGITHGGSG

HTT-Q23:

[SEQ ID NO: 2]

MATLEKLMKAFESLSFQQQQQQQQQQQQQQQQQQQQPPPPPPPPPPPPPPPPPP

PQAQPLLPQPQPPPPPPPPPPGPAVAEEPLHRPKKELSATKKDRVNHCLTICENIVAQSV

RNSPEFQKLLGIAMELFLLCSDDAESDVRMVADECLNKVIKALMDSNLPRLQLELYK

EIKKNGAPRSLRAALWRFAELAHLPVQKCRPYLVNLLPCLTRTSKRPEESVQETLAA

AVPKIMASFGNFANDNEIKVLLKAFIANLKSSSPTIRRTAAGSAVSIQHSRRTQYFYS

WLLNVLLGLLVPVEDEHSTLLILGVLLTLRYLVPLLQQQVKDTSLKGSFGVTRKEMEY

SPSAEQLVQVYELTLHHTQHGDHNVVTGALELLQQLFRTPPPELLQTLTAVGGIGQLT

AAKEESGGRSRSGSIVELIAGGGSSCPVLSRKQKGVLLGEEEALEDDSESRSVSSS

ALTASVKDEISGELAASSGVSTPGSAGHDIITEQPRSQHTLQADSVDLASCDLTSSATD

GDEEDILSHSSSQSAVSPDPAMDLDNGTQASSPISDSSQTTEGPDSAVTPSDSSEIVL

DGTDNQYLGQLIGQPQDEDEEATGILPDEASEAFRNSSMALQQAHLKNNMSHCQPS

DSSVDKFLRDEATEPGDQENKPCRIKGDIGQSTDDDSAPLVHCVRLLSASFLLTGGK

NVLVPDRDVRVSVKALALSCVGAVALHPESFFSKLYKVPLDTTEYPPEQYVSDILNY

IDHGDPQVRGATAILCGTLICSLRSRPFVGDWMTIRTTLTGNTFSLADCIPLLRKTLK

DESSVTCKLACTAVRNCVMSLCSYSELGLQLIIDVLTNRNSSFYWLVRTELLETAEI

DFRLVSFLEAKAENLHRGAHHYTGLLKLQERVLNNVVIHLLGDEDPVRHVAAASLI

RLVPKLFYKCDQQAQADPVVAVARDQSSVYLKLLMHETQPPSHFVSVTITRIYRGYNLL

PSITDVTMNNLSRVIAAVSHELITSTTRALTFCCEALCLLSTAPPVCIWSLGHWCYV

PPLSASDESRSCTVGMATMILTLLSSAWFPLDLSAQDALILAGNLLAASAPKSLRSS

WASEEEANPAATQEEVWPALGDRALVPMVEQLFSLLLKVINICAHVLDVAPGPAIK

AALPSLTNPPSLSPIRRKGKEKEPGEQASVPLSPKKGSEASAASRQSDTSGPVTTSKSSS

LGSFYHLPSYKLDHDLKATHANYKVTLDLQNSTEKGFGFLRSALDVLSQLIELATLQ

DIGKCVIEILGYLKSCFSREPMMATVCVQQLKTLFGTNLASQFDGLSSNPSKSQGRA

QRLGSSSVRPGLYHYCFMAYTHFTQALADASLRNMVQAEQENDTSGWFDVLQKVS

TQLKTNLTSTVTKNRADKNAIHNHIRLFEPLVIKALKQYTTTTCVQLKQKQVLDLLAQLV

QLRVNYCLDSDQVFIGFVLKQFEYIEVGQFRESEAIIPNIFFFLVLLSYERYHSKQIIGIP

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KIIQLCDGIMASGRKAVTHAIPALQPIVHDLFVLRGTNKADAGKELETQKEVVVSMML
RLIQYHQVLEMPILVLQQCHKENEDKWKRLSRQIADIILPMLAKQQMHIDSHEALGV
LNTLFEILAPSSLRPVDMLLRSMFVTPNTMASVSTVQLWISGILAILRVLISQSTEDIVL
SRIQELSFSPYLISCTVINRLRDGDSTSTLEEHSSEKQIKNLPETFSRFLQLVGIILEDI
VTQQLKVMESQQTFTYQCQELGTLMLCLIHIFKSGMFRRITAAATRLFRSDGCGGSFY
TLDSLNLRRASMITTHPALVLLWCQIILLVNHTDYRWAEVQQTPKRHSLSSTKLLSP
QMSGEEEDSDLAALKGMCNREIVRRGALILFCDYVCQNLHDSHETWLIIVNHIQDLIS
LSHEPPVQDFISAVHRNSAASGLFIQAIQSRCENLSTPTMLKKTLCLEGIHLSQSGAVL
TLYVDRLLCTPFRVLARMVDILACRRVEMLLAANLQSSMAQLPMEELNRIQEYQLQSS
GLAQRHQRLYSLLDRFRLSTMQDSLSPSPVSSHPLDGDGHVSLETVSPDKDWYVHL
VKSQCWTRSDSALLEGAELVNRI PAEDMNAFMNSEFNLSLLAPCLSLGMSEISGGQ
KSALFEAAREVTLARVSGTVQQLPAVHHVFQPELPAEPAAYWSKLNDLFGDAALYQS
LPTLARALAQYLVVVSKLP SHLHLPPEKEKDIVKFVVATLEALSWHLIHEQIPLSLDLQ
AGLDCCCLALQLPGLWSVVSSTEFVTHACSLIYCVHFIL EAVAVQPGEQLLSPERRTNT
PKAISEEEEEVDPTQNPKYITAACEMVAEMVESLQSVLALGHKRN SGVPAFLTPLLR
NIIISLARLPLVNSYTRVPLVWKLGWSPKPGD FGTAFFEIPVEFLQEKEVFKEFIYRIN
TLGWTSRQTQEETWATLLGVLTQPLVMEQEE SPPEEDTERTQINVLAVQAITSVLVSA
MTVPVAGNPAVSCLEQQPRNKPLKALDTRFGRKLSIIRGIVEQEIQAMVSKRENIATHH
LYQAWDPVPSLSPATTGALISHEKLLLQINPERELGSM SYKLGQVSIH SVWLGNSITPL
REEEWDEEEEEADAPAPSPPTSPVNSRKHRAGVDI HSCSQFLLELYSRWILPSSAR
RTPAILISEVVRSLLVSDLFTERNQFELMYVTLELRRVHSEDEILAQYLV PATCKAA
AVLGMDKAVAEPVSRLLSTLRSSHLP SRVGALHGVLYVLECDLDDTAKQLIPVISD
YLLSNLKGIAHCVNIHSQQHVLMCATAFYLIENYPLDVGPEFSASIIQMGVMLSGS
EESTPSIIYH CALRGLERLLLSEQLSRLDAESLVKLSVDRNVHSPHRAMAALGLMLT
CMYTGKEKVSPGRTS DPNPAAPDSESVIVAMERVS VLFDRIRKGPPEARVVARILPQF
LDDFFPPQDIMNKVIGEFLSNQQPYQPMATV VYKVFQTLHSTGQSSMVRDWMVLSL
SNFTQRAPVAMATWSLSCFFVSASTSPWVAAILPHVISRMGKLEQVDVNLFLCLVATDF
YRHQIEEELDRRAFQSVLEVVAAPGSPYHRLLTCLRN VHKVTTT

ATXN3-Q74:

[SEQ ID NO: 3]

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YRTFLQQPSGNMDDSGFFSIQVISNALKVWGLELILFNSPEYQRLRIDP INERSFICNYK
EHWFTVRKLGKQWPNLNSLLTGPELISDTYLALFLAQLQEGYSIFVVKGDL PDCEA
DQLQMIRVQQMHRPKLIGEELAQLKEQRVHKTD LERMLEANDGSGMLDEDEEDLQ
RALALSRQEIDMEDEEADLRRAIQLSMQSSRNISQDMTQTSGTNLTSEELRKREAY
FEKQQKQKQQ
QQQQQQQQQQQQQQQQQQQQQQQQQQQQQ RDLSGQSSHP CERPATSSGALGSDLG
KACSPFIMFATFTLYLT

ATXN3-Q27:

[SEQ ID NO: 4]

MESIFHEKQEGSLCAQHCLNLLQGEYFSPVELSSIAHQLDEEERMRMAEGVTS
YRTFLQQPSGNMDDSGFFSIQVISNALKVWGLELILFNSPEYQRLRIDP INERSFICNYK
EHWFTVRKLGKQWPNLNSLLTGPELISDTYLALFLAQLQEGYSIFVVKGDL PDCEA

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DQLLMIRVQQMHRPKLIGEELAQLKEQVRHKTDLERMLEANDGSGMLDEDEEDLQ
 RALALSRQEIDMEDEEADLRRAIQLSMQSSRNISQDMTQTSGTNLTSEELRKRREAY
 FEKQQQKQQQQQQQQQQQQQQQQQQQQQQQQQDLGQSSHPCERPATSSGAL
 GSDLGKACSPFIMFATFTLYLT
 ATXN1-Q92:
 [SEQ ID NO: 5]
 MKNQERSNECLPPKKREIPATSRSSSEKAPTLPNDHRVEGTAWLPGNPGGRGH
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 VSPVQYAHLPHTFQFIGSSQYSGTYASFIPSQLIPPTANPVTSAVASAAGATTPSQRSQLE
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 QQQ
 QQQQQQQHQHQQQQQQQQQQQQQQHLRAPGLITPGSPPPAQONQYVHISSSPQNT
 GRTASPPAIPVHLHPHQTMIPTLTGLPPSQVVMQYADSGSHFVPREATKKAESSRLQQ
 AIQAKEVLNGEMEKSRRYGAPSSADLGLGKAGGKSVPHPYESRHVVVHPSPSDYSSR
 DPGSVRASVMVLPNSTPAADLEVQQATHREASPSTLNDKSLHLGKPGHRSYALSP
 HTVIQTTHSASEPLPVGLPATAFYAGTQPPVIGYLSGQQQAITYAGSLPQHLVIPGTQPL
 LIPVGSTDMESGAAPAIVTSSPQFAAVPHTFVTALPKSENFNEALVTQAAYPAMVQ
 AQIHLFPVVQSVASPAAPPTLPPYFMKGSIIQLANGELKKVEDLKTDFIQSAEISNDL
 KIDSSTVERIEDSHSPGVAVIQFAVGEHRAQVSVEVLVEYPFFVFGQGWSGCCPERTSQ
 LFDLPCSKLSVGDVCISLTLKNLKNKSVKKGQVPDPASVLLKHKADGLAGSRHRYA
 EQENGINGSAQMLSENGELKFPPEKMGLPAAPFLTKEIPSKPAATRKRWSAPESRKL
 EKSEDEPPLTLPKPSLIPQEVKICIEGRSNVGK

 SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 5

<210> SEQ ID NO 1

<211> LENGTH: 261

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<220> FEATURE:

<223> OTHER INFORMATION: Q25-GFP

<400> SEQUENCE: 1

Met	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
1				5					10					15	
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Met	Ser	Lys	Gly	Glu	Glu	
				20					25				30		
Leu	Phe	Thr	Gly	Val	Val	Pro	Ile	Leu	Val	Glu	Leu	Asp	Gly	Asp	Val
				35				40				45			
Asn	Gly	His	Lys	Phe	Ser	Val	Arg	Gly	Glu	Gly	Glu	Gly	Asp	Ala	Thr
				50			55				60				
Asn	Gly	Lys	Leu	Thr	Leu	Lys	Phe	Ile	Cys	Thr	Thr	Gly	Lys	Leu	Pro
				65			70			75				80	
Val	Pro	Trp	Pro	Thr	Leu	Val	Thr	Thr	Leu	Thr	Tyr	Gly	Val	Gln	Cys
				85				90						95	
Phe	Ser	Arg	Tyr	Pro	Asp	His	Met	Lys	Arg	His	Asp	Phe	Phe	Lys	Ser
				100				105						110	

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Ala Met Pro Glu Gly Tyr Val Gln Glu Arg Thr Ile Ser Phe Lys Asp
 115 120 125

Asp Gly Thr Tyr Lys Thr Arg Ala Glu Val Lys Phe Glu Gly Asp Thr
 130 135 140

Leu Val Asn Arg Ile Glu Leu Lys Gly Ile Asp Phe Lys Glu Asp Gly
 145 150 155 160

Asn Ile Leu Gly His Lys Leu Glu Tyr Asn Phe Asn Ser His Asn Val
 165 170 175

Tyr Ile Thr Ala Asp Lys Gln Lys Asn Gly Ile Lys Ala Asn Phe Lys
 180 185 190

Ile Arg His Asn Val Glu Asp Gly Ser Val Gln Leu Ala Asp His Tyr
 195 200 205

Gln Gln Asn Thr Pro Ile Gly Asp Gly Pro Val Leu Leu Pro Asp Asn
 210 215 220

His Tyr Leu Ser Thr Gln Ser Val Leu Ser Lys Asp Pro Asn Glu Lys
 225 230 235 240

Arg Asp His Met Val Leu Leu Glu Phe Val Thr Ala Ala Gly Ile Thr
 245 250 255

His Gly Gly Ser Gly
 260

<210> SEQ ID NO 2
 <211> LENGTH: 3144
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <220> FEATURE:
 <223> OTHER INFORMATION: HTT-Q23

<400> SEQUENCE: 2

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 20 25 30

Gln Gln Gln Gln Gln Gln Gln Gln Pro Pro Pro Pro Pro Pro Pro
 35 40 45

Pro Pro Pro Gln Leu Pro Gln Pro Pro Pro Gln Ala Gln Pro Leu Leu
 50 55 60

Pro Gln Pro Gln Pro Pro Pro Pro Pro Pro Pro Pro Pro Gly Pro
 65 70 75 80

Ala Val Ala Glu Glu Pro Leu His Arg Pro Lys Lys Glu Leu Ser Ala
 85 90 95

Thr Lys Lys Asp Arg Val Asn His Cys Leu Thr Ile Cys Glu Asn Ile
 100 105 110

Val Ala Gln Ser Val Arg Asn Ser Pro Glu Phe Gln Lys Leu Leu Gly
 115 120 125

Ile Ala Met Glu Leu Phe Leu Leu Cys Ser Asp Asp Ala Glu Ser Asp
 130 135 140

Val Arg Met Val Ala Asp Glu Cys Leu Asn Lys Val Ile Lys Ala Leu
 145 150 155 160

Met Asp Ser Asn Leu Pro Arg Leu Gln Leu Glu Leu Tyr Lys Glu Ile
 165 170 175

Lys Lys Asn Gly Ala Pro Arg Ser Leu Arg Ala Ala Leu Trp Arg Phe
 180 185 190

Ala Glu Leu Ala His Leu Val Arg Pro Gln Lys Cys Arg Pro Tyr Leu
 195 200 205

Val 210	Asn	Leu	Leu	Pro	Cys	Leu 215	Thr	Arg	Thr	Ser	Lys 220	Arg	Pro	Glu	Glu
Ser 225	Val	Gln	Glu	Thr	Leu 230	Ala	Ala	Ala	Val	Pro 235	Lys	Ile	Met	Ala	Ser 240
Phe	Gly	Asn	Phe	Ala 245	Asn	Asp	Asn	Glu	Ile 250	Lys	Val	Leu	Leu	Lys 255	Ala
Phe	Ile	Ala	Asn 260	Leu	Lys	Ser	Ser	Ser	Pro 265	Thr	Ile	Arg	Arg	Thr	Ala
Ala	Gly	Ser 275	Ala	Val	Ser	Ile	Cys 280	Gln	His	Ser	Arg	Arg 285	Thr	Gln	Tyr
Phe	Tyr 290	Ser	Trp	Leu	Leu	Asn 295	Val	Leu	Leu	Gly	Leu 300	Leu	Val	Pro	Val
Glu 305	Asp	Glu	His	Ser	Thr 310	Leu	Leu	Ile	Leu	Gly 315	Val	Leu	Leu	Thr	Leu 320
Arg	Tyr	Leu	Val	Pro 325	Leu	Leu	Gln	Gln	Gln 330	Val	Lys	Asp	Thr	Ser 335	Leu
Lys	Gly	Ser	Phe 340	Gly	Val	Thr	Arg	Lys 345	Glu	Met	Glu	Val	Ser 350	Pro	Ser
Ala	Glu	Gln 355	Leu	Val	Gln	Val	Tyr 360	Glu	Leu	Thr	Leu	His 365	His	Thr	Gln
His 370	Gln	Asp	His	Asn	Val	Val 375	Thr	Gly	Ala	Leu	Glu 380	Leu	Leu	Gln	Gln
Leu 385	Phe	Arg	Thr	Pro 390	Pro	Pro	Glu	Leu	Leu	Gln 395	Thr	Leu	Thr	Ala	Val 400
Gly	Gly	Ile	Gly 405	Gln	Leu	Thr	Ala	Ala	Lys 410	Glu	Glu	Ser	Gly	Gly 415	Arg
Ser	Arg	Ser	Gly 420	Ser	Ile	Val	Glu	Leu	Ile 425	Ala	Gly	Gly	Gly 430	Ser	Ser
Cys	Ser	Pro 435	Val	Leu	Ser	Arg	Lys 440	Gln	Lys	Gly	Lys 445	Val	Leu	Leu	Gly
Glu 450	Glu	Glu	Ala	Leu	Glu	Asp 455	Asp	Ser	Glu	Ser	Arg 460	Ser	Asp	Val	Ser
Ser 465	Ser	Ala	Leu	Thr 470	Ala	Ser	Val	Lys	Asp 475	Glu	Ile	Ser	Gly	Glu	Leu 480
Ala	Ala	Ser	Ser	Gly 485	Val	Ser	Thr	Pro	Gly 490	Ser	Ala	Gly	His	Asp 495	Ile
Ile	Thr	Glu	Gln 500	Pro	Arg	Ser	Gln	His 505	Thr	Leu	Gln	Ala	Asp 510	Ser	Val
Asp	Leu 515	Ala	Ser	Cys	Asp	Leu 520	Thr	Ser	Ser	Ala	Thr 525	Asp	Gly	Asp	Glu
Glu 530	Asp	Ile	Leu	Ser	His 535	Ser	Ser	Ser	Gln	Val	Ser 540	Ala	Val	Pro	Ser
Asp 545	Pro	Ala	Met	Asp 550	Leu	Asn	Asp	Gly	Thr 555	Gln	Ala	Ser	Ser	Pro	Ile 560
Ser	Asp	Ser	Ser	Gln 565	Thr	Thr	Thr	Glu	Gly 570	Pro	Asp	Ser	Ala	Val	Thr 575
Pro	Ser	Asp	Ser 580	Ser	Glu	Ile	Val	Leu 585	Asp	Gly	Thr	Asp	Asn	Gln	Tyr 590
Leu	Gly 595	Leu	Gln	Ile	Gly	Gln 600	Pro	Gln	Asp	Glu	Asp 605	Glu	Glu	Ala	Thr
Gly 610	Ile	Leu	Pro	Asp 615	Glu	Ala	Ser	Glu	Ala	Phe	Arg 620	Asn	Ser	Ser	Met
Ala	Leu	Gln	Gln	Ala	His	Leu	Leu	Lys	Asn	Met	Ser	His	Cys	Arg	Glu

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625	630	635	640
Pro Ser Asp Ser Ser Val Asp Lys Phe Val Leu Arg Asp Glu Ala Thr	645	650	655
Glu Pro Gly Asp Gln Glu Asn Lys Pro Cys Arg Ile Lys Gly Asp Ile	660	665	670
Gly Gln Ser Thr Asp Asp Asp Ser Ala Pro Leu Val His Cys Val Arg	675	680	685
Leu Leu Ser Ala Ser Phe Leu Leu Thr Gly Gly Lys Asn Val Leu Val	690	695	700
Pro Asp Arg Asp Val Arg Val Ser Val Lys Ala Leu Ala Leu Ser Cys	705	710	715
Val Gly Ala Ala Val Ala Leu His Pro Glu Ser Phe Phe Ser Lys Leu	725	730	735
Tyr Lys Val Pro Leu Asp Thr Thr Glu Tyr Pro Glu Glu Gln Tyr Val	740	745	750
Ser Asp Ile Leu Asn Tyr Ile Asp His Gly Asp Pro Gln Val Arg Gly	755	760	765
Ala Thr Ala Ile Leu Cys Gly Thr Leu Ile Cys Ser Ile Leu Ser Arg	770	775	780
Ser Arg Phe His Val Gly Asp Trp Met Gly Thr Ile Arg Thr Leu Thr	785	790	795
Gly Asn Thr Phe Ser Leu Ala Asp Cys Ile Pro Leu Leu Arg Lys Thr	805	810	815
Leu Lys Asp Glu Ser Ser Val Thr Cys Lys Leu Ala Cys Thr Ala Val	820	825	830
Arg Asn Cys Val Met Ser Leu Cys Ser Ser Ser Tyr Ser Glu Leu Gly	835	840	845
Leu Gln Leu Ile Ile Asp Val Leu Thr Leu Arg Asn Ser Ser Tyr Trp	850	855	860
Leu Val Arg Thr Glu Leu Leu Glu Thr Leu Ala Glu Ile Asp Phe Arg	865	870	875
Leu Val Ser Phe Leu Glu Ala Lys Ala Glu Asn Leu His Arg Gly Ala	885	890	895
His His Tyr Thr Gly Leu Leu Lys Leu Gln Glu Arg Val Leu Asn Asn	900	905	910
Val Val Ile His Leu Leu Gly Asp Glu Asp Pro Arg Val Arg His Val	915	920	925
Ala Ala Ala Ser Leu Ile Arg Leu Val Pro Lys Leu Phe Tyr Lys Cys	930	935	940
Asp Gln Gly Gln Ala Asp Pro Val Val Ala Val Ala Arg Asp Gln Ser	945	950	955
Ser Val Tyr Leu Lys Leu Leu Met His Glu Thr Gln Pro Pro Ser His	965	970	975
Phe Ser Val Ser Thr Ile Thr Arg Ile Tyr Arg Gly Tyr Asn Leu Leu	980	985	990
Pro Ser Ile Thr Asp Val Thr Met Glu Asn Asn Leu Ser Arg Val Ile	995	1000	1005
Ala Ala Val Ser His Glu Leu Ile Thr Ser Thr Thr Arg Ala Leu	1010	1015	1020
Thr Phe Gly Cys Cys Glu Ala Leu Cys Leu Leu Ser Thr Ala Phe	1025	1030	1035
Pro Val Cys Ile Trp Ser Leu Gly Trp His Cys Gly Val Pro Pro	1040	1045	1050

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Leu Ser	Ala Ser Asp Glu Ser	Arg Lys Ser Cys Thr	Val Gly Met
1055	1060	1065	
Ala Thr	Met Ile Leu Thr Leu	Leu Ser Ser Ala Trp	Phe Pro Leu
1070	1075	1080	
Asp Leu	Ser Ala His Gln Asp	Ala Leu Ile Leu Ala	Gly Asn Leu
1085	1090	1095	
Leu Ala	Ala Ser Ala Pro Lys	Ser Leu Arg Ser Ser	Trp Ala Ser
1100	1105	1110	
Glu Glu	Glu Ala Asn Pro Ala	Ala Thr Lys Gln Glu	Glu Val Trp
1115	1120	1125	
Pro Ala	Leu Gly Asp Arg Ala	Leu Val Pro Met Val	Glu Gln Leu
1130	1135	1140	
Phe Ser	His Leu Leu Lys Val	Ile Asn Ile Cys Ala	His Val Leu
1145	1150	1155	
Asp Asp	Val Ala Pro Gly Pro	Ala Ile Lys Ala Ala	Leu Pro Ser
1160	1165	1170	
Leu Thr	Asn Pro Pro Ser Leu	Ser Pro Ile Arg Arg	Lys Gly Lys
1175	1180	1185	
Glu Lys	Glu Pro Gly Glu Gln	Ala Ser Val Pro Leu	Ser Pro Lys
1190	1195	1200	
Lys Gly	Ser Glu Ala Ser Ala	Ala Ser Arg Gln Ser	Asp Thr Ser
1205	1210	1215	
Gly Pro	Val Thr Thr Ser Lys	Ser Ser Ser Leu Gly	Ser Phe Tyr
1220	1225	1230	
His Leu	Pro Ser Tyr Leu Lys	Leu His Asp Val Leu	Lys Ala Thr
1235	1240	1245	
His Ala	Asn Tyr Lys Val Thr	Leu Asp Leu Gln Asn	Ser Thr Glu
1250	1255	1260	
Lys Phe	Gly Gly Phe Leu Arg	Ser Ala Leu Asp Val	Leu Ser Gln
1265	1270	1275	
Ile Leu	Glu Leu Ala Thr Leu	Gln Asp Ile Gly Lys	Cys Val Glu
1280	1285	1290	
Glu Ile	Leu Gly Tyr Leu Lys	Ser Cys Phe Ser Arg	Glu Pro Met
1295	1300	1305	
Met Ala	Thr Val Cys Val Gln	Gln Leu Leu Lys Thr	Leu Phe Gly
1310	1315	1320	
Thr Asn	Leu Ala Ser Gln Phe	Asp Gly Leu Ser Ser	Asn Pro Ser
1325	1330	1335	
Lys Ser	Gln Gly Arg Ala Gln	Arg Leu Gly Ser Ser	Ser Val Arg
1340	1345	1350	
Pro Gly	Leu Tyr His Tyr Cys	Phe Met Ala Pro Tyr	Thr His Phe
1355	1360	1365	
Thr Gln	Ala Leu Ala Asp Ala	Ser Leu Arg Asn Met	Val Gln Ala
1370	1375	1380	
Glu Gln	Glu Asn Asp Thr Ser	Gly Trp Phe Asp Val	Leu Gln Lys
1385	1390	1395	
Val Ser	Thr Gln Leu Lys Thr	Asn Leu Thr Ser Val	Thr Lys Asn
1400	1405	1410	
Arg Ala	Asp Lys Asn Ala Ile	His Asn His Ile Arg	Leu Phe Glu
1415	1420	1425	
Pro Leu	Val Ile Lys Ala Leu	Lys Gln Tyr Thr Thr	Thr Thr Cys
1430	1435	1440	

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Val	Gln	Leu	Gln	Lys	Gln	Val	Leu	Asp	Leu	Leu	Ala	Gln	Leu	Val
1445						1450					1455			
Gln	Leu	Arg	Val	Asn	Tyr	Cys	Leu	Leu	Asp	Ser	Asp	Gln	Val	Phe
1460						1465					1470			
Ile	Gly	Phe	Val	Leu	Lys	Gln	Phe	Glu	Tyr	Ile	Glu	Val	Gly	Gln
1475						1480					1485			
Phe	Arg	Glu	Ser	Glu	Ala	Ile	Ile	Pro	Asn	Ile	Phe	Phe	Phe	Leu
1490						1495					1500			
Val	Leu	Leu	Ser	Tyr	Glu	Arg	Tyr	His	Ser	Lys	Gln	Ile	Ile	Gly
1505						1510					1515			
Ile	Pro	Lys	Ile	Ile	Gln	Leu	Cys	Asp	Gly	Ile	Met	Ala	Ser	Gly
1520						1525					1530			
Arg	Lys	Ala	Val	Thr	His	Ala	Ile	Pro	Ala	Leu	Gln	Pro	Ile	Val
1535						1540					1545			
His	Asp	Leu	Phe	Val	Leu	Arg	Gly	Thr	Asn	Lys	Ala	Asp	Ala	Gly
1550						1555					1560			
Lys	Glu	Leu	Glu	Thr	Gln	Lys	Glu	Val	Val	Val	Ser	Met	Leu	Leu
1565						1570					1575			
Arg	Leu	Ile	Gln	Tyr	His	Gln	Val	Leu	Glu	Met	Phe	Ile	Leu	Val
1580						1585					1590			
Leu	Gln	Gln	Cys	His	Lys	Glu	Asn	Glu	Asp	Lys	Trp	Lys	Arg	Leu
1595						1600					1605			
Ser	Arg	Gln	Ile	Ala	Asp	Ile	Ile	Leu	Pro	Met	Leu	Ala	Lys	Gln
1610						1615					1620			
Gln	Met	His	Ile	Asp	Ser	His	Glu	Ala	Leu	Gly	Val	Leu	Asn	Thr
1625						1630					1635			
Leu	Phe	Glu	Ile	Leu	Ala	Pro	Ser	Ser	Leu	Arg	Pro	Val	Asp	Met
1640						1645					1650			
Leu	Leu	Arg	Ser	Met	Phe	Val	Thr	Pro	Asn	Thr	Met	Ala	Ser	Val
1655						1660					1665			
Ser	Thr	Val	Gln	Leu	Trp	Ile	Ser	Gly	Ile	Leu	Ala	Ile	Leu	Arg
1670						1675					1680			
Val	Leu	Ile	Ser	Gln	Ser	Thr	Glu	Asp	Ile	Val	Leu	Ser	Arg	Ile
1685						1690					1695			
Gln	Glu	Leu	Ser	Phe	Ser	Pro	Tyr	Leu	Ile	Ser	Cys	Thr	Val	Ile
1700						1705					1710			
Asn	Arg	Leu	Arg	Asp	Gly	Asp	Ser	Thr	Ser	Thr	Leu	Glu	Glu	His
1715						1720					1725			
Ser	Glu	Gly	Lys	Gln	Ile	Lys	Asn	Leu	Pro	Glu	Glu	Thr	Phe	Ser
1730						1735					1740			
Arg	Phe	Leu	Leu	Gln	Leu	Val	Gly	Ile	Leu	Leu	Glu	Asp	Ile	Val
1745						1750					1755			
Thr	Lys	Gln	Leu	Lys	Val	Glu	Met	Ser	Glu	Gln	Gln	His	Thr	Phe
1760						1765					1770			
Tyr	Cys	Gln	Glu	Leu	Gly	Thr	Leu	Leu	Met	Cys	Leu	Ile	His	Ile
1775						1780					1785			
Phe	Lys	Ser	Gly	Met	Phe	Arg	Arg	Ile	Thr	Ala	Ala	Ala	Thr	Arg
1790						1795					1800			
Leu	Phe	Arg	Ser	Asp	Gly	Cys	Gly	Gly	Ser	Phe	Tyr	Thr	Leu	Asp
1805						1810					1815			
Ser	Leu	Asn	Leu	Arg	Ala	Arg	Ser	Met	Ile	Thr	Thr	His	Pro	Ala
1820						1825					1830			
Leu	Val	Leu	Leu	Trp	Cys	Gln	Ile	Leu	Leu	Leu	Val	Asn	His	Thr

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1835	1840	1845
Asp Tyr Arg Trp Trp Ala Glu Val Gln Gln Thr Pro Lys Arg His		
1850	1855	1860
Ser Leu Ser Ser Thr Lys Leu Leu Ser Pro Gln Met Ser Gly Glu		
1865	1870	1875
Glu Glu Asp Ser Asp Leu Ala Ala Lys Leu Gly Met Cys Asn Arg		
1880	1885	1890
Glu Ile Val Arg Arg Gly Ala Leu Ile Leu Phe Cys Asp Tyr Val		
1895	1900	1905
Cys Gln Asn Leu His Asp Ser Glu His Leu Thr Trp Leu Ile Val		
1910	1915	1920
Asn His Ile Gln Asp Leu Ile Ser Leu Ser His Glu Pro Pro Val		
1925	1930	1935
Gln Asp Phe Ile Ser Ala Val His Arg Asn Ser Ala Ala Ser Gly		
1940	1945	1950
Leu Phe Ile Gln Ala Ile Gln Ser Arg Cys Glu Asn Leu Ser Thr		
1955	1960	1965
Pro Thr Met Leu Lys Lys Thr Leu Gln Cys Leu Glu Gly Ile His		
1970	1975	1980
Leu Ser Gln Ser Gly Ala Val Leu Thr Leu Tyr Val Asp Arg Leu		
1985	1990	1995
Leu Cys Thr Pro Phe Arg Val Leu Ala Arg Met Val Asp Ile Leu		
2000	2005	2010
Ala Cys Arg Arg Val Glu Met Leu Leu Ala Ala Asn Leu Gln Ser		
2015	2020	2025
Ser Met Ala Gln Leu Pro Met Glu Glu Leu Asn Arg Ile Gln Glu		
2030	2035	2040
Tyr Leu Gln Ser Ser Gly Leu Ala Gln Arg His Gln Arg Leu Tyr		
2045	2050	2055
Ser Leu Leu Asp Arg Phe Arg Leu Ser Thr Met Gln Asp Ser Leu		
2060	2065	2070
Ser Pro Ser Pro Pro Val Ser Ser His Pro Leu Asp Gly Asp Gly		
2075	2080	2085
His Val Ser Leu Glu Thr Val Ser Pro Asp Lys Asp Trp Tyr Val		
2090	2095	2100
His Leu Val Lys Ser Gln Cys Trp Thr Arg Ser Asp Ser Ala Leu		
2105	2110	2115
Leu Glu Gly Ala Glu Leu Val Asn Arg Ile Pro Ala Glu Asp Met		
2120	2125	2130
Asn Ala Phe Met Met Asn Ser Glu Phe Asn Leu Ser Leu Leu Ala		
2135	2140	2145
Pro Cys Leu Ser Leu Gly Met Ser Glu Ile Ser Gly Gly Gln Lys		
2150	2155	2160
Ser Ala Leu Phe Glu Ala Ala Arg Glu Val Thr Leu Ala Arg Val		
2165	2170	2175
Ser Gly Thr Val Gln Gln Leu Pro Ala Val His His Val Phe Gln		
2180	2185	2190
Pro Glu Leu Pro Ala Glu Pro Ala Ala Tyr Trp Ser Lys Leu Asn		
2195	2200	2205
Asp Leu Phe Gly Asp Ala Ala Leu Tyr Gln Ser Leu Pro Thr Leu		
2210	2215	2220
Ala Arg Ala Leu Ala Gln Tyr Leu Val Val Val Ser Lys Leu Pro		
2225	2230	2235

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Ser His	Leu His	Leu Pro	Pro	Glu Lys	Glu Lys	Asp	Ile Val	Lys	
2240			2245			2250			
Phe Val	Val Ala	Thr Leu	Glu	Ala Leu	Ser Trp	His	Leu Ile	His	
2255			2260			2265			
Glu Gln	Ile Pro	Leu Ser	Leu	Asp Leu	Gln Ala	Gly	Leu Asp	Cys	
2270			2275			2280			
Cys Cys	Leu Ala	Leu Gln	Leu	Pro Gly	Leu Trp	Ser	Val Val	Ser	
2285			2290			2295			
Ser Thr	Glu Phe	Val Thr	His	Ala Cys	Ser Leu	Ile	Tyr Cys	Val	
2300			2305			2310			
His Phe	Ile Leu	Glu Ala	Val	Ala Val	Gln Pro	Gly	Glu Gln	Leu	
2315			2320			2325			
Leu Ser	Pro Glu	Arg Arg	Thr	Asn Thr	Pro Lys	Ala	Ile Ser	Glu	
2330			2335			2340			
Glu Glu	Glu Glu	Val Asp	Pro	Asn Thr	Gln Asn	Pro	Lys Tyr	Ile	
2345			2350			2355			
Thr Ala	Ala Cys	Glu Met	Val	Ala Glu	Met Val	Glu	Ser Leu	Gln	
2360			2365			2370			
Ser Val	Leu Ala	Leu Gly	His	Lys Arg	Asn Ser	Gly	Val Pro	Ala	
2375			2380			2385			
Phe Leu	Thr Pro	Leu Leu	Arg	Asn Ile	Ile Ile	Ser	Leu Ala	Arg	
2390			2395			2400			
Leu Pro	Leu Val	Asn Ser	Tyr	Thr Arg	Val Pro	Pro	Leu Val	Trp	
2405			2410			2415			
Lys Leu	Gly Trp	Ser Pro	Lys	Pro Gly	Gly Asp	Phe	Gly Thr	Ala	
2420			2425			2430			
Phe Pro	Glu Ile	Pro Val	Glu	Phe Leu	Gln Glu	Lys	Glu Val	Phe	
2435			2440			2445			
Lys Glu	Phe Ile	Tyr Arg	Ile	Asn Thr	Leu Gly	Trp	Thr Ser	Arg	
2450			2455			2460			
Thr Gln	Phe Glu	Glu Thr	Trp	Ala Thr	Leu Leu	Gly	Val Leu	Val	
2465			2470			2475			
Thr Gln	Pro Leu	Val Met	Glu	Gln Glu	Glu Ser	Pro	Pro Glu	Glu	
2480			2485			2490			
Asp Thr	Glu Arg	Thr Gln	Ile	Asn Val	Leu Ala	Val	Gln Ala	Ile	
2495			2500			2505			
Thr Ser	Leu Val	Leu Ser	Ala	Met Thr	Val Pro	Val	Ala Gly	Asn	
2510			2515			2520			
Pro Ala	Val Ser	Cys Leu	Glu	Gln Gln	Pro Arg	Asn	Lys Pro	Leu	
2525			2530			2535			
Lys Ala	Leu Asp	Thr Arg	Phe	Gly Arg	Lys Leu	Ser	Ile Ile	Arg	
2540			2545			2550			
Gly Ile	Val Glu	Gln Glu	Ile	Gln Ala	Met Val	Ser	Lys Arg	Glu	
2555			2560			2565			
Asn Ile	Ala Thr	His His	Leu	Tyr Gln	Ala Trp	Asp	Pro Val	Pro	
2570			2575			2580			
Ser Leu	Ser Pro	Ala Thr	Thr	Gly Ala	Leu Ile	Ser	His Glu	Lys	
2585			2590			2595			
Leu Leu	Leu Gln	Ile Asn	Pro	Glu Arg	Glu Leu	Gly	Ser Met	Ser	
2600			2605			2610			
Tyr Lys	Leu Gly	Gln Val	Ser	Ile His	Ser Val	Trp	Leu Gly	Asn	
2615			2620			2625			

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Ser Ile 2630	Thr Pro Leu Arg	Glu 2635	Glu Glu Trp Asp	Glu 2640	Glu Glu Glu
Glu Glu 2645	Ala Asp Ala Pro	Ala 2650	Pro Ser Ser Pro	Pro 2655	Thr Ser Pro
Val Asn 2660	Ser Arg Lys His	Arg 2665	Ala Gly Val Asp	Ile 2670	His Ser Cys
Ser Gln 2675	Phe Leu Leu Glu	Leu 2680	Tyr Ser Arg Trp	Ile 2685	Leu Pro Ser
Ser Ser 2690	Ala Arg Arg Thr	Pro 2695	Ala Ile Leu Ile	Ser 2700	Glu Val Val
Arg Ser 2705	Leu Leu Val Val	Ser 2710	Asp Leu Phe Thr	Glu 2715	Arg Asn Gln
Phe Glu 2720	Leu Met Tyr Val	Thr 2725	Leu Thr Glu Leu	Arg 2730	Arg Val His
Pro Ser 2735	Glu Asp Glu Ile	Leu 2740	Ala Gln Tyr Leu	Val 2745	Pro Ala Thr
Cys Lys 2750	Ala Ala Ala Val	Leu 2755	Gly Met Asp Lys	Ala 2760	Val Ala Glu
Pro Val 2765	Ser Arg Leu Leu	Glu 2770	Ser Thr Leu Arg	Ser 2775	Ser His Leu
Pro Ser 2780	Arg Val Gly Ala	Leu 2785	His Gly Val Leu	Tyr 2790	Val Leu Glu
Cys Asp 2795	Leu Leu Asp Asp	Thr 2800	Ala Lys Gln Leu	Ile 2805	Pro Val Ile
Ser Asp 2810	Tyr Leu Leu Ser	Asn 2815	Leu Lys Gly Ile	Ala 2820	His Cys Val
Asn Ile 2825	His Ser Gln Gln	His 2830	Val Leu Val Met	Cys 2835	Ala Thr Ala
Phe Tyr 2840	Leu Ile Glu Asn	Tyr 2845	Pro Leu Asp Val	Gly 2850	Pro Glu Phe
Ser Ala 2855	Ser Ile Ile Gln	Met 2860	Cys Gly Val Met	Leu 2865	Ser Gly Ser
Glu Glu 2870	Ser Thr Pro Ser	Ile 2875	Ile Tyr His Cys	Ala 2880	Leu Arg Gly
Leu Glu 2885	Arg Leu Leu Leu	Ser 2890	Glu Gln Leu Ser	Arg 2895	Leu Asp Ala
Glu Ser 2900	Leu Val Lys Leu	Ser 2905	Val Asp Arg Val	Asn 2910	Val His Ser
Pro His 2915	Arg Ala Met Ala	Ala 2920	Leu Gly Leu Met	Leu 2925	Thr Cys Met
Tyr Thr 2930	Gly Lys Glu Lys	Val 2935	Ser Pro Gly Arg	Thr 2940	Ser Asp Pro
Asn Pro 2945	Ala Ala Pro Asp	Ser 2950	Glu Ser Val Ile	Val 2955	Ala Met Glu
Arg Val 2960	Ser Val Leu Phe	Asp 2965	Arg Ile Arg Lys	Gly 2970	Phe Pro Cys
Glu Ala 2975	Arg Val Val Ala	Arg 2980	Ile Leu Pro Gln	Phe 2985	Leu Asp Asp
Phe Phe 2990	Pro Pro Gln Asp	Ile 2995	Met Asn Lys Val	Ile 3000	Gly Glu Phe
Leu Ser 3005	Asn Gln Gln Pro	Tyr 3010	Pro Gln Phe Met	Ala 3015	Thr Val Val
Tyr Lys	Val Phe Gln Thr	Leu	His Ser Thr Gly	Gln	Ser Ser Met

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3020	3025	3030
Val Arg Asp Trp Val Met Leu Ser Leu Ser Asn Phe Thr Gln Arg		
3035	3040	3045
Ala Pro Val Ala Met Ala Thr Trp Ser Leu Ser Cys Phe Phe Val		
3050	3055	3060
Ser Ala Ser Thr Ser Pro Trp Val Ala Ala Ile Leu Pro His Val		
3065	3070	3075
Ile Ser Arg Met Gly Lys Leu Glu Gln Val Asp Val Asn Leu Phe		
3080	3085	3090
Cys Leu Val Ala Thr Asp Phe Tyr Arg His Gln Ile Glu Glu Glu		
3095	3100	3105
Leu Asp Arg Arg Ala Phe Gln Ser Val Leu Glu Val Val Ala Ala		
3110	3115	3120
Pro Gly Ser Pro Tyr His Arg Leu Leu Thr Cys Leu Arg Asn Val		
3125	3130	3135
His Lys Val Thr Thr Cys		
3140		

<210> SEQ ID NO 3

<211> LENGTH: 412

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<220> FEATURE:

<223> OTHER INFORMATION: ATXN3-Q74

<400> SEQUENCE: 3

Met Glu Ser Ile Phe His Glu Lys Gln Glu Gly Ser Leu Cys Ala Gln		
1	5	10
His Cys Leu Asn Asn Leu Leu Gln Gly Glu Tyr Phe Ser Pro Val Glu		
20	25	30
Leu Ser Ser Ile Ala His Gln Leu Asp Glu Glu Glu Arg Met Arg Met		
35	40	45
Ala Glu Gly Gly Val Thr Ser Glu Asp Tyr Arg Thr Phe Leu Gln Gln		
50	55	60
Pro Ser Gly Asn Met Asp Asp Ser Gly Phe Phe Ser Ile Gln Val Ile		
65	70	75
Ser Asn Ala Leu Lys Val Trp Gly Leu Glu Leu Ile Leu Phe Asn Ser		
85	90	95
Pro Glu Tyr Gln Arg Leu Arg Ile Asp Pro Ile Asn Glu Arg Ser Phe		
100	105	110
Ile Cys Asn Tyr Lys Glu His Trp Phe Thr Val Arg Lys Leu Gly Lys		
115	120	125
Gln Trp Phe Asn Leu Asn Ser Leu Leu Thr Gly Pro Glu Leu Ile Ser		
130	135	140
Asp Thr Tyr Leu Ala Leu Phe Leu Ala Gln Leu Gln Gln Glu Gly Tyr		
145	150	155
Ser Ile Phe Val Val Lys Gly Asp Leu Pro Asp Cys Glu Ala Asp Gln		
165	170	175
Leu Leu Gln Met Ile Arg Val Gln Gln Met His Arg Pro Lys Leu Ile		
180	185	190
Gly Glu Glu Leu Ala Gln Leu Lys Glu Gln Arg Val His Lys Thr Asp		
195	200	205
Leu Glu Arg Met Leu Glu Ala Asn Asp Gly Ser Gly Met Leu Asp Glu		
210	215	220
Asp Glu Glu Asp Leu Gln Arg Ala Leu Ala Leu Ser Arg Gln Glu Ile		

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225	230	235	240
Asp Met Glu Asp Glu Glu Ala Asp Leu Arg Arg Ala Ile Gln Leu Ser	245	250	255
Met Gln Gly Ser Ser Arg Asn Ile Ser Gln Asp Met Thr Gln Thr Ser	260	265	270
Gly Thr Asn Leu Thr Ser Glu Glu Leu Arg Lys Arg Arg Glu Ala Tyr	275	280	285
Phe Glu Lys Gln Gln Gln Lys Gln Gln Gln Gln Gln Gln Gln Gln	290	295	300
Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln	305	310	315
Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln	320	325	330
Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln	335	340	345
Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln	350	355	360
Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln Gln	365	370	375
Gln Arg Asp Leu Ser Gly Gln Ser Ser His Pro Cys Glu Arg Pro Ala	380	385	390
Thr Ser Ser Gly Ala Leu Gly Ser Asp Leu Gly Lys Ala Cys Ser Pro	395	400	405
Phe Ile Met Phe Ala Thr Phe Thr Leu Tyr Leu Thr	410		

<210> SEQ ID NO 4

<211> LENGTH: 365

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<220> FEATURE:

<223> OTHER INFORMATION: ATXN3-Q27

<400> SEQUENCE: 4

Met Glu Ser Ile Phe His Glu Lys Gln Glu Gly Ser Leu Cys Ala Gln	1	5	10	15
His Cys Leu Asn Asn Leu Leu Gln Gly Glu Tyr Phe Ser Pro Val Glu	20	25	30	
Leu Ser Ser Ile Ala His Gln Leu Asp Glu Glu Glu Arg Met Arg Met	35	40	45	
Ala Glu Gly Gly Val Thr Ser Glu Asp Tyr Arg Thr Phe Leu Gln Gln	50	55	60	
Pro Ser Gly Asn Met Asp Asp Ser Gly Phe Phe Ser Ile Gln Val Ile	65	70	75	80
Ser Asn Ala Leu Lys Val Trp Gly Leu Glu Leu Ile Leu Phe Asn Ser	85	90	95	
Pro Glu Tyr Gln Arg Leu Arg Ile Asp Pro Ile Asn Glu Arg Ser Phe	100	105	110	
Ile Cys Asn Tyr Lys Glu His Trp Phe Thr Val Arg Lys Leu Gly Lys	115	120	125	
Gln Trp Phe Asn Leu Asn Ser Leu Leu Thr Gly Pro Glu Leu Ile Ser	130	135	140	
Asp Thr Tyr Leu Ala Leu Phe Leu Ala Gln Leu Gln Gln Glu Gly Tyr	145	150	155	160
Ser Ile Phe Val Val Lys Gly Asp Leu Pro Asp Cys Glu Ala Asp Gln	165	170	175	
Leu Leu Gln Met Ile Arg Val Gln Gln Met His Arg Pro Lys Leu Ile				

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180					185					190					
Gly	Glu	Glu	Leu	Ala	Gln	Leu	Lys	Glu	Gln	Arg	Val	His	Lys	Thr	Asp
	195						200					205			
Leu	Glu	Arg	Met	Leu	Glu	Ala	Asn	Asp	Gly	Ser	Gly	Met	Leu	Asp	Glu
	210					215					220				
Asp	Glu	Glu	Asp	Leu	Gln	Arg	Ala	Leu	Ala	Leu	Ser	Arg	Gln	Glu	Ile
	225					230					235				240
Asp	Met	Glu	Asp	Glu	Glu	Ala	Asp	Leu	Arg	Arg	Ala	Ile	Gln	Leu	Ser
			245						250					255	
Met	Gln	Gly	Ser	Ser	Arg	Asn	Ile	Ser	Gln	Asp	Met	Thr	Gln	Thr	Ser
			260					265					270		
Gly	Thr	Asn	Leu	Thr	Ser	Glu	Glu	Leu	Arg	Lys	Arg	Arg	Glu	Ala	Tyr
		275					280					285			
Phe	Glu	Lys	Gln	Gln	Gln	Lys	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	
	290					295					300				
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	
	305					310					315				320
Gln	Gln	Arg	Asp	Leu	Ser	Gly	Gln	Ser	Ser	His	Pro	Cys	Glu	Arg	Pro
			325						330					335	
Ala	Thr	Ser	Ser	Gly	Ala	Leu	Gly	Ser	Asp	Leu	Gly	Lys	Ala	Cys	Ser
			340					345					350		
Pro	Phe	Ile	Met	Phe	Ala	Thr	Phe	Thr	Leu	Tyr	Leu	Thr			
		355					360					365			

<210> SEQ ID NO 5

<211> LENGTH: 895

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<220> FEATURE:

<223> OTHER INFORMATION: ATXN1-Q92

<400> SEQUENCE: 5

Met	Lys	Ser	Asn	Gln	Glu	Arg	Ser	Asn	Glu	Cys	Leu	Pro	Pro	Lys	Lys
1			5						10					15	
Arg	Glu	Ile	Pro	Ala	Thr	Ser	Arg	Ser	Ser	Glu	Glu	Lys	Ala	Pro	Thr
		20						25					30		
Leu	Pro	Ser	Asp	Asn	His	Arg	Val	Glu	Gly	Thr	Ala	Trp	Leu	Pro	Gly
		35					40					45			
Asn	Pro	Gly	Gly	Arg	Gly	His	Gly	Gly	Gly	Arg	His	Gly	Pro	Ala	Gly
	50					55					60				
Thr	Ser	Val	Glu	Leu	Gly	Leu	Gln	Gln	Gly	Ile	Gly	Leu	His	Lys	Ala
	65				70					75				80	
Leu	Ser	Thr	Gly	Leu	Asp	Tyr	Ser	Pro	Pro	Ser	Ala	Pro	Arg	Ser	Val
			85						90					95	
Pro	Val	Ala	Thr	Thr	Leu	Pro	Ala	Ala	Tyr	Ala	Thr	Pro	Gln	Pro	Gly
			100					105					110		
Thr	Pro	Val	Ser	Pro	Val	Gln	Tyr	Ala	His	Leu	Pro	His	Thr	Phe	Gln
		115					120					125			
Phe	Ile	Gly	Ser	Ser	Gln	Tyr	Ser	Gly	Thr	Tyr	Ala	Ser	Phe	Ile	Pro
	130					135					140				
Ser	Gln	Leu	Ile	Pro	Pro	Thr	Ala	Asn	Pro	Val	Thr	Ser	Ala	Val	Ala
	145				150					155				160	
Ser	Ala	Ala	Gly	Ala	Thr	Thr	Pro	Ser	Gln	Arg	Ser	Gln	Leu	Glu	Ala
			165						170					175	
Tyr	Ser	Thr	Leu	Leu	Ala	Asn	Met	Gly	Ser	Leu	Ser	Gln	Thr	Pro	Gly

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180					185					190				
His	Lys	Ala	Glu	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
	195					200					205			
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
	210					215					220			
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
	225					230					235			240
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
					245					250				255
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
					260					265				270
Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
					275					280				285
His	Gln	His	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln	Gln
	290					295					300			
Gln	His	Leu	Ser	Arg	Ala	Pro	Gly	Leu	Ile	Thr	Pro	Gly	Ser	Pro
	305				310					315				320
Pro	Ala	Gln	Gln	Asn	Gln	Tyr	Val	His	Ile	Ser	Ser	Ser	Pro	Gln
				325					330					335
Thr	Gly	Arg	Thr	Ala	Ser	Pro	Pro	Ala	Ile	Pro	Val	His	Leu	His
			340					345					350	Pro
His	Gln	Thr	Met	Ile	Pro	His	Thr	Leu	Thr	Leu	Gly	Pro	Pro	Ser
		355					360					365		Gln
Val	Val	Met	Gln	Tyr	Ala	Asp	Ser	Gly	Ser	His	Phe	Val	Pro	Arg
	370					375					380			Glu
Ala	Thr	Lys	Lys	Ala	Glu	Ser	Ser	Arg	Leu	Gln	Gln	Ala	Ile	Gln
	385				390					395				400
Lys	Glu	Val	Leu	Asn	Gly	Glu	Met	Glu	Lys	Ser	Arg	Arg	Tyr	Gly
			405						410					415
Pro	Ser	Ser	Ala	Asp	Leu	Gly	Leu	Gly	Lys	Ala	Gly	Gly	Lys	Ser
			420				425						430	Val
Pro	His	Pro	Tyr	Glu	Ser	Arg	His	Val	Val	Val	His	Pro	Ser	Pro
		435					440					445		Ser
Asp	Tyr	Ser	Ser	Arg	Asp	Pro	Ser	Gly	Val	Arg	Ala	Ser	Val	Met
	450					455					460			Val
Leu	Pro	Asn	Ser	Asn	Thr	Pro	Ala	Ala	Asp	Leu	Glu	Val	Gln	Gln
	465				470					475				480
Thr	His	Arg	Glu	Ala	Ser	Pro	Ser	Thr	Leu	Asn	Asp	Lys	Ser	Gly
			485						490					495
His	Leu	Gly	Lys	Pro	Gly	His	Arg	Ser	Tyr	Ala	Leu	Ser	Pro	His
		500						505					510	Thr
Val	Ile	Gln	Thr	Thr	His	Ser	Ala	Ser	Glu	Pro	Leu	Pro	Val	Gly
		515					520					525		Leu
Pro	Ala	Thr	Ala	Phe	Tyr	Ala	Gly	Thr	Gln	Pro	Pro	Val	Ile	Gly
	530					535					540			Tyr
Leu	Ser	Gly	Gln	Gln	Gln	Ala	Ile	Thr	Tyr	Ala	Gly	Ser	Leu	Pro
	545					550				555				560
His	Leu	Val	Ile	Pro	Gly	Thr	Gln	Pro	Leu	Leu	Ile	Pro	Val	Gly
			565						570					575
Thr	Asp	Met	Glu	Ala	Ser	Gly	Ala	Ala	Pro	Ala	Ile	Val	Thr	Ser
			580				585						590	Ser
Pro	Gln	Phe	Ala	Ala	Val	Pro	His	Thr	Phe	Val	Thr	Thr	Ala	Leu
		595					600						605	Pro

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Lys Ser Glu Asn Phe Asn Pro Glu Ala Leu Val Thr Gln Ala Ala Tyr			
610	615	620	
Pro Ala Met Val Gln Ala Gln Ile His Leu Pro Val Val Gln Ser Val			
625	630	635	640
Ala Ser Pro Ala Ala Ala Pro Pro Thr Leu Pro Pro Tyr Phe Met Lys			
	645	650	655
Gly Ser Ile Ile Gln Leu Ala Asn Gly Glu Leu Lys Lys Val Glu Asp			
	660	665	670
Leu Lys Thr Glu Asp Phe Ile Gln Ser Ala Glu Ile Ser Asn Asp Leu			
	675	680	685
Lys Ile Asp Ser Ser Thr Val Glu Arg Ile Glu Asp Ser His Ser Pro			
	690	695	700
Gly Val Ala Val Ile Gln Phe Ala Val Gly Glu His Arg Ala Gln Val			
705	710	715	720
Ser Val Glu Val Leu Val Glu Tyr Pro Phe Phe Val Phe Gly Gln Gly			
	725	730	735
Trp Ser Ser Cys Cys Pro Glu Arg Thr Ser Gln Leu Phe Asp Leu Pro			
	740	745	750
Cys Ser Lys Leu Ser Val Gly Asp Val Cys Ile Ser Leu Thr Leu Lys			
	755	760	765
Asn Leu Lys Asn Gly Ser Val Lys Lys Gly Gln Pro Val Asp Pro Ala			
	770	775	780
Ser Val Leu Leu Lys His Ser Lys Ala Asp Gly Leu Ala Gly Ser Arg			
785	790	795	800
His Arg Tyr Ala Glu Gln Glu Asn Gly Ile Asn Gln Gly Ser Ala Gln			
	805	810	815
Met Leu Ser Glu Asn Gly Glu Leu Lys Phe Pro Glu Lys Met Gly Leu			
	820	825	830
Pro Ala Ala Pro Phe Leu Thr Lys Ile Glu Pro Ser Lys Pro Ala Ala			
	835	840	845
Thr Arg Lys Arg Arg Trp Ser Ala Pro Glu Ser Arg Lys Leu Glu Lys			
	850	855	860
Ser Glu Asp Glu Pro Pro Leu Thr Leu Pro Lys Pro Ser Leu Ile Pro			
865	870	875	880
Gln Glu Val Lys Ile Cys Ile Glu Gly Arg Ser Asn Val Gly Lys			
	885	890	895

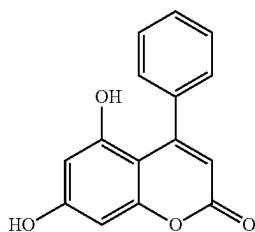
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What is claimed is:

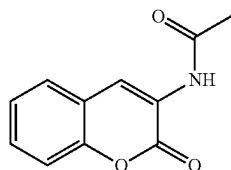
1. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising administering an effective amount of a compound or a pharmaceutically acceptable salt thereof to the subject, wherein

the polyQ-related neurodegenerative disorder is Huntington's disease; and

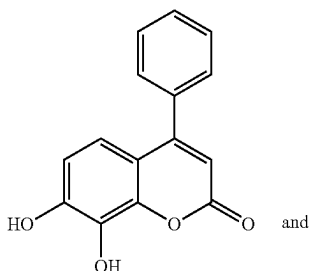
the compound is selected from:



(compound 2)

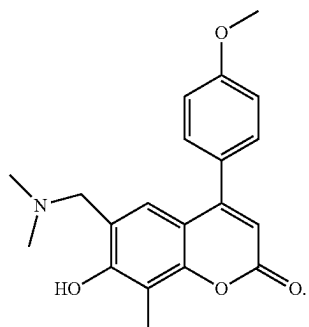


(compound 5)



(compound 6)

(compound 7)



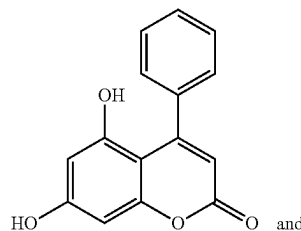
2. The method of claim 1, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.

3. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising adminis-

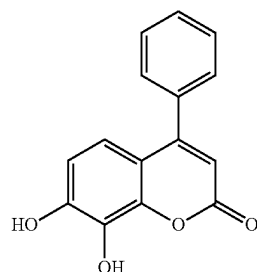
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tering an effective amount of a compound, or a pharmaceutically acceptable salt thereof, to the subject, wherein the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 1; and the compound is selected from:

(compound 2)



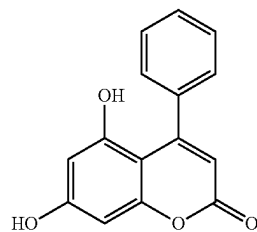
(compound 6)



4. The method of claim 3, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.

5. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising administering an effective amount of a compound, or a pharmaceutically acceptable salt thereof, to the subject, wherein the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 3; and the compound is:

(compound 2)



6. The method of claim 5, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.

* * * * *